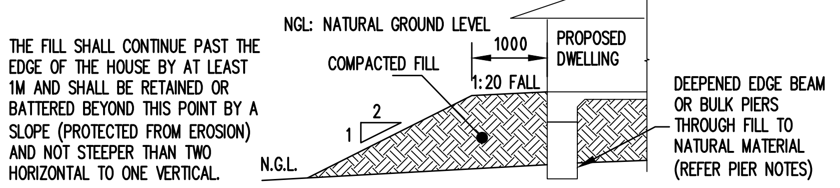
PROPOSED RESIDENTIAL DEVELOPMENT
AT LOT 515 (NO. 18) THORPE WAY, BOX HILL, NSW

IMPORTANT NOTE:
"EFFECTS OF TREES + 'ABNORMAL' FOUNDATION MOISTURE CONDITIONS"
THIS SLAB LAYOUT DESIGN DOES NOT INCLUDE THE EFFECTS OF PREVIOUS, EXISTING OR FUTURE TREES, THEREFORE, IF ANY TREES WERE REMOVED OR ARE PRESENT WITHIN THE VICINITY OF THIS PROPOSED RESIDENCE OR ARE CONSIDERED FOR FUTURE PLANTING, THE DESIGN ENGINEER MUST BE ADVISED FOR RE-CONSIDERATION OF THIS DESIGN. ALSO, THIS SLAB LAYOUT DESIGN DOES NOT ACCOMMODATE GROUND MOVEMENTS GENERATED BY "ABNORMAL" FOUNDATION MOISTURE CONDITIONS. FOR A BETTER UNDERSTANDING OF THESE REQUIREMENTS, THE BUILDERS/OWNERS ATTENTION IS DRAWN TO:
1. APPENDIX CH OF AS2870-2011 "GUIDE TO DESIGN OF FOOTINGS FOR TREES.
2. APPENDIX B OF AS2870-2011 "PERFORMANCE REQUIREMENTS AND FOUNDATION MAINTENANCE"
3. CSIRO PAMPHLET "GUIDE TO HOME OWNERS ON FOUNDATION MAINTENANCE AND FOOTING PERFORMANCE".

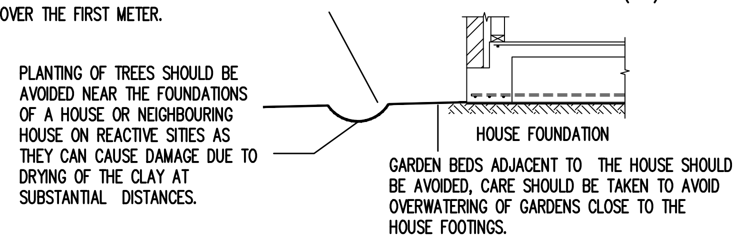
GENERAL

G1. THESE DRAWING SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANTS DRAWING AND SPECIFICATIONS AND WITH SUCH OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT. ALL DISCREPANCIES SHALL BE REFEREED TO THE ENGINEER FOR DECISION BEFORE PROCEEDING WITH THE WORK.
G2. ALL DIMENSION SHOWN SHALL BE VERIFIED ON SITE. ENGINEERS DRAWINGS MUST NOT BE SCALED. (REFER ARCHITECTURAL DRAWINGS).
G3. THE STRUCTURAL ENGINEER SHALL PREPARE, SIGN AND ISSUE STRUCTURAL DRAWINGS FOR BUILDERS COMPULSORY IMPLEMENTATION DURING THE FULL OF THE CONSTRUCTION CONTRACT AND/OR ANY MAINTENANCE PERIOD CONTRACTS.
G4. DURING CONSTRUCTION THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING THE STRUCTURE IN A STABLE CONDITION AND ENSURE NO PART SHALL BE OVERSTRESSED UNDER CONSTRUCTION ACTIVITIES.
G5. ALL MATERIALS & WORKMANSHIP SHALL BE IN ACCORDANCE WITH CURRENT AUSTRALIAN STANDARDS AND SPECIFICATIONS.
G6. DURING THE CONSTRUCTION STAGE AND/OR ANY MAINTENANCE PERIOD, PROPER AND ADEQUATE CONTRACT WORKS INSURANCE AND PUBLIC LIABILITY INSURANCES IS COMPULSORY CLAIMS OR DAMAGE TO ANY ADJACENT PROPERTY OR BUILDING SHALL NOT BE RESPONSIBLE OF THE ENGINEER.
G7. THESE DRAWINGS ARE SIGNED SUBJECT TO A CERTIFICATE OF INSPECTION BEING ISSUED ISSUED BY ENGINEER. ALL PIER HOLES AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.
G8. SHOULD ANY STRUCTURAL ELEMENTS PRESENT DIFFICULTY WITH RESPECT TO CONTRACTIBILITY, SAFETY OR POTENTIAL HAZARDS, THE MATTERS SHALL BE REFERRED TO THE STRUCTURAL ENGINEER FOR RESOLUTION BEFORE PROCEEDING WITH THE WORK.
G9. TREATMENT OF CUT AND FILL SLOPING SITES:



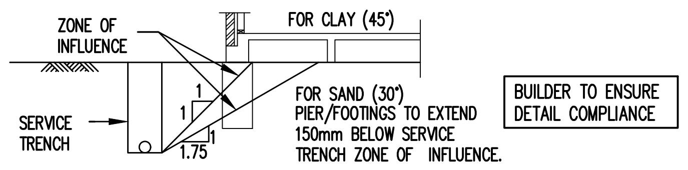
FOUNDATION AND FOOTING

F1. STRIP TOPSOIL CONTAINING ORGANIC MATTER. PROOF ROLL FILL SUB GRADE AND REMOVE ANY SOFT ZONES.
F2. WHERE ADDITIONAL FILL IS REQUIRED TO UNDERSIDE OF SLABS ON GROUND, NON COHESIVE MATERIALS SUCH AS SAND AND GRAVEL DUST SHALL BE PLACED BY "CONTROLLED" COMPACTION IN HORIZONTAL LAYERS OF 200mm (LOOSE) MAXIMUM DEPTH. THIS FILL SHALL BE COMPACTED TO AT LEAST 95% OF STANDARD MAXIMUM DRY DENSITY (SMDD) AND COMPACTED IN ACCORDANCE WITH AS 2870.
F3. FOR SLAB ON GROUND, SAND 50mm APPROXIMATE THICKNESS IS TO BE SPREAD AS A LEVELING LAYER AND WELL WATERED DOWN.
F4. DAMP-PROOFING MEMBRANE UNPUNCTURED AND TAPED AT LAPS, IS TO BE PLACED OVER THE SAND, SUFFICIENT MEMBRANE BEING PROVIDED AT EDGES TO RETURN UNDER BRICKWORK. WHERE NO BRICKWORK, TAPE MEMBRANE TO SIDE OF FOOTING BELOW GROUND.
F5. FOOTINGS ARE NOT TO BE CONSTRUCTED IN FILL MATERIAL, ALL STRUCTURAL FILL TO BE APPROVED BY GEOTECHNICAL ENGINEER.
F6. FOOTING HAVE BEEN DESIGNED FOR ALLOWABLE EQUAL UNIFORM BEARING CAPACITY OF 150KPa ON FIRM NATURAL GROUND. ALL FOUNDATION MATERIAL SHALL BE APPROVED FOR THIS BEARING CAPACITY BEFORE PLACING REINFORCEMENT.
F7. EXCAVATION SHALL CONTINUE UNTIL THE REQUIRED BEARING CAPACITY IS FOUND. THE OVER-EXCAVATION SHALL BE BACK-FILLED WITH A MASS CONCRETE MIX TO THE APPROVAL OF THE ENGINEER.
F8. REFER JOB SPECIFICATION DETAILS (PAGE 3) FOR SITE CLASSIFICATION. WE DISCLOSE THAT WE HAVE NOT VERIFIED THIS REPORT AND THAT WE RELY ON ITS FINDINGS. THIS OFFICE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS.
F9. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING EXCAVATIONS IN A STABLE CONDITION WITHOUT AFFECTING ADJACENT PROPERTIES OR SERVICES. WHERE REQUIRED, TEMPORARY SHORING SHALL BE PROVIDED TO THE SIDE OF FOOTING EXCAVATIONS.
F10. THE OWNERS ATTENTION SHOULD BE DRAWN TO APPENDIX B OF AS2870 "PERFORMANCE REQUIREMENTS AND FOUNDATION MAINTENANCE" ON COMPLETION OF THE JOB.
F11. THE SITE SHALL BE GRADED OR DRAINED SO THAT WATER CANNOT POND AGAINST OR NEAR THE HOUSE. THE GROUND IMMEDIATELY ADJACENT TO THE HOUSE SHALL BE GRADED TO A UNIFORM FALL OF 50mm (min) AWAY FROM THE HOUSE OVER THE FIRST METER.



FOUNDATION PIERS

P1. THE CONTRACTOR SHALL INVESTIGATE THE PRESENCE OF ANY EXISTING SERVICES IN THE GROUND LIKELY TO BE AFFECTED BY THE PILING OPERATIONS.
P2. PILES AND PILING ARE IN ACCORDANCE WITH AS 2150 AND ANY OTHER RELEVANT CODES.
P3. DURING THE DRILLING THROUGH EXISTING FILL, OBSTRUCTION MAY BE EXPECTED.
P4. IF GROUNDWATER IS ENCOUNTED DURING THE PILING, THE ENGINEER TO BE CONTACTED IMMEDIATELY.
P4. ENSURE THAT MINIMUM PIER DEPTH IS 600mm BELOW THE FOOTING TRENCH, U.N.O.
P5. ALL PIERS TO BE POURED SEPARATE FROM FOOTINGS AND SLABS.
P6. ALL PIERS ARE TO BE INSPECTED TO ENSURE THEY ARE CLEANED AND FREE OF LOOSE MATERIAL AND WATER PRIOR TO POURING CONCRETE, WHICH SHOULD BE WITH MINIMAL DELAY AND ON THE SAME DAY AS BORING.
P7. PIER TOPS ARE TO BE FINISHED SMOOTH. ITS IS RESPONSIBILITY OF CONTRACTOR TO ENSURE PIER TOPES ARE CLEAN AND FREE OF FOREIGN MATERIAL PRIOR TO THE LAYING OF THE MEMBRANE AND CONCRETE SLAB. ENGINEER SPOT CHECK DOES NOT RELEASE THE CONTRACTOR FROM THIS RESPONSIBILITY.
P8. ALL EDGE BEAMS & INTERNAL BEAMS TO BE FOUNDED ON CONSISTENT NATURAL STRATA OF (MIN.) BEARING CAPACITY 700KPa (CLASS IV ROCK) OTHERWISE, ø450 PIERS AT 2000 CTS (MAX.) ARE REQUIRED TO BY-PASS FILL & ACHIEVE UNIFORM BEARING CAPACITY (MIN. 700kPa) ON NATURAL STRATA AS SAME BEARING CAPACITY OF BASEMENT FOOTING. REFER STRUCTURAL PLANS PIER DIAMETER AND LOCATIONS.
P9. WHERE DEPTH OF PIERS EXCEED 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
P10. AT THE COMPLETION OF PIERS, THE CONTRACTOR SHALL PROVIDE AN AS-BUILT SURVEY AND CERTIFICATION THAT ALL PIERS ARE WITHIN SPECIFIED TOLERANCES.



SERVICE TRENCH DETAILS

REINFORCED CONCRETE

C1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 3600 CURRENT EDITION, EXCEPT WHERE VARIED BY THE CONTACT DOCUMENTS.
C2. CONCRETE QUALITY (ENGINEER TO APPROVE ANY ADMIXTURE USED IN CONCRETE MIX)

ELEMENT	SLUMP	MAX SIZE AGGREGATE	Fc (28 DAYS)
SLAB	80mm	20mm	32/40 MPa
FOOTINGS	80mm	20mm	32 MPa
PIERS	100mm	20mm	32 MPa

C3. ALL CONCRETE PLACED SHALL BE WELL VIBRATED AND COMPACTED USING AN APPROVED VIBRATOR AND TECHNIQUE.
C4. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE ENGINEER.
C5. SIZES OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF ANY APPLIED FINISHES.
C6. NO HOLES OR CHASES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS. WITHOUT THE PRIOR APPROVAL OF THE ENGINEER.
C7. WATER IS NOT TO BE ADDED TO THE CONCRETE ON SITE SO AS TO INCREASE THE SLUMP ABOVE THAT SPECIFIED.
C8. THE CONTRACTOR SHALL NOTIFY THE ENGINEER 24 HOURS BEFORE REINFORCEMENT IS COMPLETED. THE CONTRACTOR SHALL ALLOW AFTER THE COMPLETION OF THE REINFORCEMENT TWO HOURS FOR THE ENGINEERS INSPECTIONS, UPON APPROVAL CONCRETE MAY BE ORDERED.
C9. CONCRETE TO BE CURED A MINIMUM OF 7 DAYS AFTER POURING.
C10. CURING OF ALL CONCRETE SHALL BE ACHIEVED BY KEEPING SURFACES COMPLETELY WET FOR A PERIOD OF 3 DAYS, AND PREVENTION OF LOSS OF MOISTURE FOR A TOTAL OF 7 DAYS FOLLOWED BY A GRADUAL DRYING OUT. APPROVED SPRAYED ON CURING COMPOUNDS THAT COMPLY WITH AS 3799 MAY BE USED WHERE FLOOR FINISHES WILL NOT BE AFFECTED (REFER MANUFACTURERS SPECIFICATION). POLYTHENE SHEETING OR WET HESSIAN OR SIMILAR PRODUCT MAY BE USED TO RETAIN CONCRETE MOISTURE WHERE PROTECTED FROM WIND AND TRAFFIC.
C11. FORMWORK TO BEAMS AND SLABS SPANNING GREATER THAN 5m SHALL BE PRE-CAMBERED UPWARDS BY 1/500 OF CLEAR SPAN IN EACH SPAN UNLESS NOTED DIFFERENTLY ON THE DRAWINGS.
C12. SHRINKAGE CRACKING IS ALMOST INEVITABLE & DOES NOT REPRESENT FAILURE. HOWEVER, IS OF CONCERN UNDER BRITTLE FLOOR COVERINGS DAMAGE MAY BE REDUCED BY USING FLEXIBLE MORTARS & GLUES FOR FIXING TILES & FIXING OPERATION MUST BE DELAYED AS LATE AS POSSIBLE.
C13. CONSTRUCTION SUPPORT PROPPING IS TO BE LEFT IN PLACE WHERE NEEDED O AVOID OVERSTRESSING THE STRUCTURE DUE TO CONSTRUCTION LOADING. NO BRICKWORK OR PARTITION WALLS ARE TO BE CONSTRUCTED ON SUSPENDED LEVELS UNTIL ALL PROPPING IS REMOVED AND THE SLAB HAS ABSORBED ITS DEAD LOAD DEFLECTION.

REINFORCEMENT

R1. MINIMUM CLEAR CONCRETE COVER REINFORCEMENT INCLUDING TIES AND STIRRUPS (OTHER THAN RESIDENTIAL SLABS ON GROUND OR FOOTINGS) SHALL BE FOLLOWS AND.

EXPOSURE CLASSIFICATION	MINIMUM COVER (mm)				
	CONCRETE STRENGTH (Fc)				
	20 MPa	25 MPa	32 MPa	40 MPa	> 50 MPa
A1	20	20	20	20	20
A2	(50)	30	25	20	20
B1	–	(60)	40	30	25
B2	–	–	(65)	45	35
C	–	–	–	(70)	50

FOR BRACKETED FIGURES REFER TO AS3600 CURRENT EDITION TABLE 4.10.3.2

R2. RESIDENTIAL SLAB ON GROUND AND FOOTINGS COVER REQUIREMENTS: (MINIMUM CONCRETE GRADE N20)
- UNPROTECTED GROUND: 40m
- EXTERNAL EXPOSURE: 40mm
- MEMBRANE IN CONTACT WITH GROUND: 30mm
- INTERNAL SURFACE: 20mm
- STRIP AND PAD FOOTING: 40mm

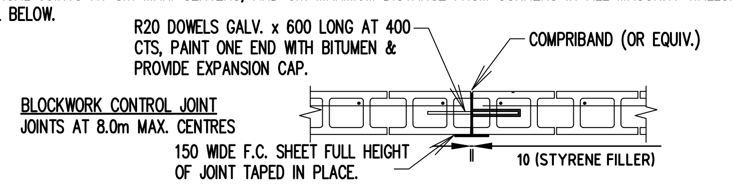
R3. REINFORCEMENT SYMBOLS:
N – DENOTES DEFORMED GRADE 500 NORMAL DUCTILITY REINFORCING BARS TO AS/NZS4671.
R – DENOTES PLAIN ROUND GRADE 250 NORMAL DUCTILITY REINFORCING BARS TO AS/NZS4671.
SL – DENOTES DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH TO AS/NZS4671.
RL – DENOTES DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH TO AS/NZS4671.
L-TM – DENOTES DEFORMED GRADE 500 LOW DUCTILITY TRENCH MESH TO AS/NZS4671.
R4. REINFORCEMENTS IS REPRESENTED DIAGRAMMATICALLY, IT IS NOT NECESSARILY SHOWN IN TRUE PROJECTION.
R5. SPLICES IN REINFORCEMENTS SHALL BE MADE ONLY IN POSITIONS SHOWN OTHERWISE APPROVED BY ENGINEER.
R6. FABRIC REINFORCEMENT SHALL HAVE SPLICES MADE SO THAT THE OVERLAP, MEASURED BETWEEN THE OUTERMOST TRANSVERSE WIRES OF EACH SHEET OR FABRIC IS NOT LESS THAN THE SPACING OF THOSE WIRES PLUS 25mm.
R7. AROUND THE ENTIRE EDGE OF THE SLAB, THERE MUST BE AN INTACT EDGE WIRE OF THE FABRIC PARALLEL TO AND 20mm FROM THE FORM BOARD.- PRESENT, PROVIDE AN ADDITIONAL 1-N12 TO EDGE.
R8. WELDING OF REINFORCEMENT SHALL NOT BE PERMITTED UNLESS SHOWN ON STRUCTURAL DRAWINGS OR APPROVED BY ENGINEER.
R9. TOP AND BOTTOM REINFORCEMENT IN SLABS SHALL BE SUPPORTED ON APPROVED PLASTIC TIPPED CHAIRS, IN BOTH DIRECTIONS AT MAXIMUM CENTRES OF 600mm FOR 10mm DIA BARS, 900mm FOR 12mm AND 16mm DIA BARS, 1200mm FOR 20mm DIA BARS AND 750mm CENTRES FOR FABRIC.
R10. TRUSSED ROOF NOTE: SLAB HAS BEEN DESIGNED FOR ROOF LOADING TO BE SUPPORTED BY PROPRIETARY TRUSSES ONTO EXTERNAL WALLS ONLY.
R11. ARTICULATION NOTE: THIS SLAB IS DESIGNED FOR ARTICULATED MASONRY VENEER TYPE CONSTRUCTION AND ARTICULATION JOINTS ARE TO BE PROVIDED AS PER THE B.C.A. VOL.2 CURRENT EDITION
R12. DEEPEND BEAM NOTE: THE FULL EXTENT OF DEEPENED EDGE BEAMS AND ANY SET-DOWNS ARE TO BE CARRIED OUT IN ACCORDANCE WITH THE ARCHITECTURAL DETAILS
R13. HWS AND ACU NOTE: PADS ARE TO BE LOCATED AS PER ARCHITECTURAL DRAWINGS. WHEN IN FILL, PLACE MASS CONCRETE REINFORCEMENT COVER TABLE PIER UNDER CONCRETE PADS.
R14. CONDUITS, PIPES, ECT. SHALL ONLY BE PLACED IN THE MIDDLE ONE THIRD OF SLAB DEPTH AND SPREAD AT NOT LESS THAN 3 DIAMETERS.
R15. ALL REENTRANT CORNERS AT PENETRATIONS FOR SLUMP, PITS, COLUMN BLOCKOUTS AND THE LIKE, TO HAVE 2N12 TRIMER BARS PLACED AT 45 DEGREE TO CORNER OR IN EACH DIRECTION AT CORNERS U.N.O. TRIMERS BARS TO BE TIED UNDER SIDE OF TOP MESH.

REINFORCEMENT LAP TABLE

BAR SIZE	N12	N16	N20	N24	N28	N32
LAP LENGTH	450	600	800	1200	1350	1650

BRICKWORK AND BLOCKWORK

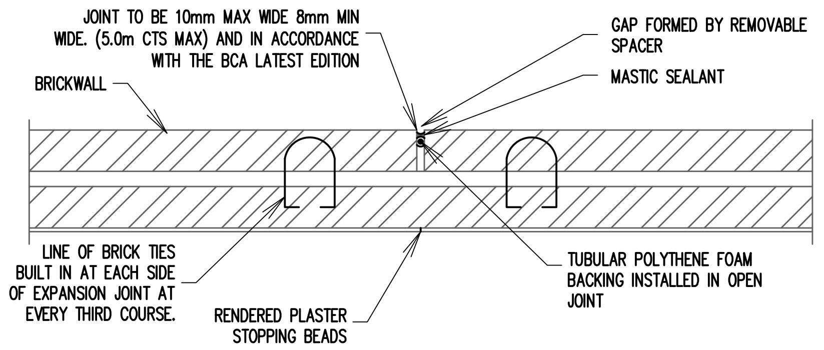
B1. ALL WORKMANSHIP AND MATERIALS TO BE IN ACCORDANCE WITH THE CURRENT AS 3700 AND SAA MASONARY CODES.
B2. INTERNAL BRICKWORK BUILT OFF THE SLAB SHALL BE LAID ON TWO LAYERS OF 'ALCOR' OR '3 PLY MALTHOID' OR SIMILAR SLIP JOINT MATERIAL.
B3. LOAD BEARING MASONRY SHALL COMPLY WITH AS 3700
B4. MASONRY AND MORTAR DURABILITY SHALL COMPLY WITH THE B.C.A CLASS 1 AND 10 BUILDINGS VOLUME 2.
B5. MASONRY SHALL BE ARTICULATED IN ACCORDANCE WITH THE B.C.A CLASS 1 AND 10 BUILDINGS VOLUME 2.
B6. MASONRY WALLS MUST NOT BE BUILT ON CONCRETE SLABS OR BEAMS UNTIL ALL FORMWORK/PROPS SUPPORTING THESE SLABS AND BEAMS HAVE BEEN REMOVED.
B7. ALL BLOCKS SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 25 MPa.
B8. CHASES AND HOLES SHALL NOT BE MADE IN BLOCKWORK WITHOUT THE PRIOR APPROVAL OF SUPERINTENDENT.
B9. PROVIDE CLEAN OUT HOLES AT THE BOTTOM OF ALL CORES FILLED. ALL MORTAR SHALL BE REMOVED FROM BOTTOM OF WALL BEFORE CLEAN OUT HOLES ARE FILLED FOR GROUTING.
B10. ALL CORES SHALL BE CLEANED AND VERTICAL BARS TIED TO STARTER BARS.
B11. CONCRETE TO WALLS TO BE THOROUGHLY VIBRATED AND POURED IN LIFTS OF 1800 MAXIMUM. SECOND LIFT TO BE VIBRATED INTO THE FIRST LIFT.
B12. THE TOP COURSE IN HOLLOW BLOCK WALLS SHALL BE LAID IN SOLID BLOCK.
B13. PROVIDE CLEAN-OUT HOLES AT BASE OF ALL WALLS AND ROD CORE HOLES TO REMOVE PROTRUDING MORTAR FINS.
B14. CORE FILLING GROUT SHALL HAVE MINIMUM CHARACTERISTIC STRENGTH OF 25 MPa, 10mm AGGREGATE, 230mm SLUMP +/- 30mm. THOROUGHLY COMPACT CORE FILL GROUT WITH A MECHANICAL VIBRATOR.
B15. PROVIDE 55mm COVER TO REINFORCING BARS FROM THE OUTSIDE FACE OF THE BLOCKWORK.
B16. BACKFILL TO RETAINING WALLS IS TO BE FREE DRAINING GRANULAR MATERIAL UNLESS NOTED OTHERWISE. PROVIDE A SUBSOIL DRAIN TO WEEP HOLES.
B17. ALL MASONRY WALLS AND PIERS SUPPORTING SLABS AND BEAMS SHALL HAVE A PRE-GREASED GALVANISED STEEL SLIP JOINT BETWEEN THE TOP OF THE MASONRY AND THE CONCRETE SLAB SOFFIT UNLESS NOTED OTHERWISE.
B18. PROVIDE VERTICAL JOINTS AT 8m MAX. CENTERS, AND 5m MAXIMUM DISTANCE FROM CORNERS IN ALL MASONRY WALLS. REFER DETAIL BELOW.



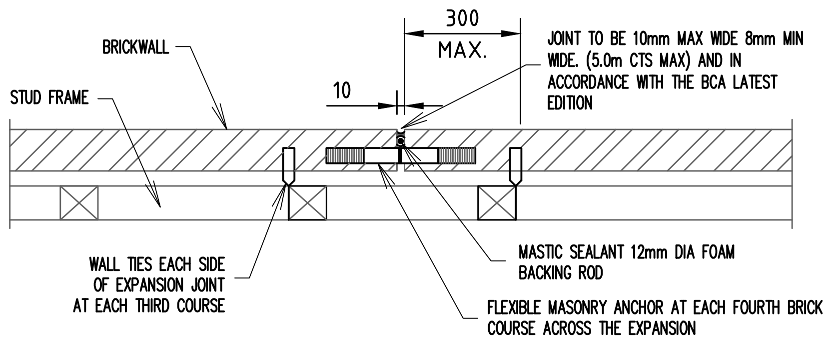
DURABILITY CLASSIFICATION			
CATEGORY (DURABILITY)	WALL TIES AS3700	SALT ATTACK RESISTANCE OF BRICK	MORTAR AS3700
LESS THAN 1km FROM BREAKING SURF OR LESS THAN 100m FROM SALT WATER NOT SUBJECT TO BREAKING SURF	R4 (STAINLESS OR POLYMER)	EXPOSURE - AS3700	M4 (1:4)
1km OR LESS THAN 10km FROM BREAKING SURF OR 100m OR LESS THAN 1km FROM SALT WATER NOT SUBJECT TO BREAKING SURF	R3	EXPOSURE (GENERAL PURPOSE - AS3700) ✳	M3 (1:5)
ALL OTHER AREAS	R2	GENERAL PURPOSE - AS3700	M2 (1:2:8)

THE TABLE ABOVE IS A PART SUMMARY OF THE REQUIREMENTS OF CURRENT EDITION OF AS 3700 & AS 4100. IT IS RECOMMENDED TO REFER THESE AUSTRALIAN STANDARDS FOR MORE OPTIONS/INFORMATIONS. ✳ REFER NOTES: MANDATORY REQUIREMENTS FOR SALINITY AFFECTS.

ARTICULATION JOINT DETAIL FOR DOUBLE BRICK



ARTICULATION JOINT DETAIL FOR BRICK VENEER



FOR CAVITY BRICK WALL AND SINGLE LEAF OR DOUBLE MASONRY ARTICULATION JOINTS DETAILS IN ACCORDANCE WITH THE BCA LATEST EDITION.

DRAINAGE REQUIREMENTS

D1. ALL WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH CURRENT EDITIONS OF AS2870, AS/NZS 2032 INTALL OF PVC PIPES AND AS/NZS 3500 PLUMBING & DRAINAGE.
D2. PLUMBING TRENCHES SHALL BE SLOPED AWAY FROM THE HOUSE AND SHALL BE BACKFILLED WITH CLAY IN THE OP 300mm WITHIN 1.5m OF THE HOUSE. THE CLAY USED FOR BACKFILLING SHALL BE COMPACTED. WHERE PIPES PASS UNDER THE FOOTING SYSTEM, THE TRENCH SHALL BE BACKFILLED WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM.
D3. DRAINAGE SHALL BE CONSTRUCTED TO AVOID WATER PONDING AGAINST OR NEAR THE FOOTING.
D4. ECAVATION NEAR THE EDGE OF THE FOOTING SYSTEM SHALL BE BACKFILLED IN SUCH A WAY AS TO PREVENT ACESS OF WATER TO THE FOUNDATION.
D5. WATER RUN-OFF SHALL BE COLLECTED AND CHANNELLED AWAY FROM THE HOUSE DURING CONSTRUCTION.
D6. PENETRATIONS OF THE EDGE BEMS AND FOOTING BEAMS ARE TO BE AVOIDED, BUT WHERE NECESSARY SHALL BE SLEEVED TO ALLOW FOR MOVEMENT.
D7. CONNECTION OF STORMWATER DRAINS AND WASTE DRAINS SHALL BE INCLUDED FLEXIBLE CONNECTIONS.

STRUCTURAL STEELWORK

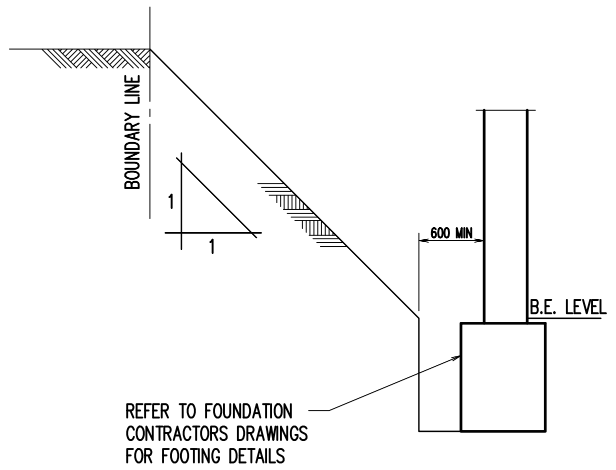
S1. ALL MATERIAL, WORKMANSHIP, FABRICATION AND ERECTION SHALL BE IN ACCORDANCE WITH AS 4100, AS 1538, AS 1554 AND RELEVANT ASSOCIATED STANDARDS.
S2. UNLESS NOTED OTHERWISE, ALL STEEL SHALL BE IN ACCORDANCE WITH AS1204 GRADE 300. ALL STEEL HOLLOW SECTIONS SHALL BE IN ACCORDANCE WITH AS1163 GRADE 350. ALL PRESSED METAL PURLINS AND GIRTS SHALL BE IN ACCORDANCE WITH AS1538 GRADE 450.
S3. THREE SETS OF STEELWORK SHOP DETAIL DRAWINGS SHALL BE SUBMITTED TO THE ENGINEER FOR PPROVAL PRIOR TO COMMENCEMENT OF ANY FABRICATION. THE CHECK SHALL NOT COVER LAYOUT AND MEMBER DIMENSION.
S4. UNLESS NOTED OTHERWISE, USE:
A) 10mm GUSSET, BASE AND END PLATES.
B) M20 8.8/S BOLTS. HIGH STRENGTH STRUCTURAL (AS1252)
C) 6mm CONTINUOUS FILLET WELDS MADE WITH E41XX MILD ELECTRODES.
D) ALL BUT WELDS SHALL BE COMPLETE PENETRATION BUTT WELD TO AS1554.1
S5. SITE WELDING OF HOT DIP GALVANIZED STEEL IS PERMISSIBLE IF UPON COMPLETION THE WELDS ARE TREATED WITH THE APPROPRIATE COATING FOR SEVERE AS PER THE B.C.A AND AS/NZS 2312.
S6. BEAMS SUPPORTED ON BRICKWORK (BEARING NOTED ON PLAN) TO HAVE INCOMPRESSIBLE PACKING AS REQUIRED UNDER THE ENDS OF THE BEAM TO ENSURE EVEN BEARING ON THE BRICKWORK.
S7. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER AND OTHER ELEMENTS TO STEEL WHETHER OR NOT DETAILED IN THE DRAWINGS.
S8. WELDS TO BE PROPERLY CHIPPED TO REMOVE SLAG.
S9. STEELWORK NOT ENCASED SHALL HAVE THE FOLLOWING SURFACE TREATMENTS

EXPOSURE CLASSIFICATION	STEELWORK PROTECTION REQUIRED
A1/A2	POWDER TOOL CLEAN TO AS1627 CLASS 1, 1 COT ALKYD PRIMER (ZINC PHOSPHATE)
B1	ABRASIVE BLAST TO AS1627 CLASS 2.5, 1 COT ZINC PHOSPHATE SILICATE
B2	HOT DIPPED GALVANISED TO AS4680, 600gm/M2

S11. THE BUILDER SHALL PROVIDE ALL CLEATS AND DRILL ALL HOLES NECESSARY FOR FIXING STEEL TO STEEL AND TIMBER AND OTHER ELEMENTS TO STEEL WHETHER OR NOT DETAILED IN THE DRAWINGS.
S12. ALL STEELWORK SHALL HAVE ADEQUATE TEMPORARY BRACING PROVIDED TO STABILIZE THE STRUCTURE DURING CONSTRUCTION.
S13. THE FABRICATION AND ERECTION OF THE STRUCTURAL STEEL WORK SHALL BE SUPERVISED BY QUALIFIED ENGINEER, EXPERIENCE IN SUCH SUPERVISION, TO ENSURE THAT ALL REQUIREMENTS OF THE OF THE DESIGN ARE MET.
S14. SURFACE FINISHES FOR ALL STRUCTURAL STEELWORK TO BE IN ACCORDANCE WITH THE ARCHITECTURAL SPECIFICATION.
S15. THE INSTALLATION OF GALINTELS AND 'T' BARS TO BE STRICTLY IN ACCORDANCE WITH MANUFACTURERS SPECIFICATIONS. PLACE GALINTEL 'T' BAR OVER OPENING ALLOWING A MINIMUM OF 200mm END BEARING EACH END.GALINTEL 'T' BAR MUST BE PROPPED BEFORE BRICKLAYING. FROM 2.4m TO 3.3m SPAN 2 PROPS. FROM 3.6m TO 4.5m SPAN 3 PROPS. FROM 4.8m TO 5.7m SPAN 4 PROPS. SPACED EQUALLY ALONG THE LENGTH AND UNDER THE BASE OF THE BAR. WHEN LAYING BRICKS MORTAR MUST BE APPLIED TO ALL BRICK FACES COMING IN CONTACT WITH THE 'T' BAR. PROPS TO REMAIN IN PLACE UNTIL MORTAR ACHIEVES FULL STRENGTH (7 DAYS MIN.). A MINIMUM 1:4 MORTAR MIX IS TO BE USED AND APPLIED TO ALL FACES BETWEEN STEEL AND BRICKS (VERTICAL AND HORIZONTAL LEGS) AND BETWEEN BRICKS ABOVE THE STEEL SECTION.

MANDATORY REQUIREMENTS FOR SALINITY AFFECTS

FOR HOUSE SLAB FOTING AND BRICKWORKS
SN1. FOR SLAB ON GROUND CONSTRUCTION, A LAYER OF SAND AT LEAST 50mm DEEP UNDER THE SLAB MUST BE PROVIDED.
SN2. A HIGH IMPACT DAMP PROOF MEMBRANE (RATHER THAN A VAPOUR PROOF MEMBRANE) MUST BE LAID UNDER SLAB (NSW BCA 3.2.2.6)
SN3. THE DAMP PROOF MEMBRANE MUT EXTEND TO THE OUTSIDE FACE OF THE EXTERNAL EDGE BEAM UP TO THE FINISHED GROUND LEVEL.
SN4. ALL CLASS 32MPA (N32) CONCRETE MUST BE USED OR A SULPHATE RESISTING. TYPE SR CEMENT WITH A WATER CEMENT RATIO OF 0.5 MUST BE USED.
SN5. IT IS IMPERATIVE THAT WATER IS NOT ADDED AT THE CONSTRUCTION SITE FOR 32 MPA CONCRETE AS THESE WILL REDUCE THE CONCRETE STRENGTH.
SN6. SLABS MUST BE VIBRATED AND CURED FOR A MINIMUM OF THREE DAYS. CARE MUST BE TAKEN NOT TO OVER VIBRATE THE CONCRETE DURING PLACEMENT, AS SEGREGATION OF THE CONCRETE AGGREGATES WILL OCCUR.
SN7. THE MINIMUM COVER TO REINFORCEMENT MUST BE 50MM FROM UNPROTECTED GROUND. CHAIRS INCLUDING LATERAL SUPPORTS SHOULD BE IN POSITION PRIOR TO INSPECTION AND SUBSEQUENTLY DURING POURING OF THE CONCRETE.
SN8. THE MINIMUM COVER TO REINFORCEMENT MUST BE 30MM FROM THE MEMBRANE IN CONTACT WITH THE GROUND.
SN9. THE MINIMUM COVER TO REINFORCEMENT MUST BE 50MM FOR STRIP FOOTINGS AND BEAMS IRRESPECTIVE OF WHETHER A DAMP PROOF MEMBRANE IS USED.
SN10. APPROVED ADMIXTURES FOR WATERPROOFING AND/ OR CORROSION PREVENTION MAY BE USED.
SN11. THE FOLLOWING MEASURES MUST BE USED FOR BRICKWORK:
- THE DAMP PROOF COURSE MUST CONSIST OF POLYETHYLENE OR POLYETHYLENE COATED METAL AND BE CORRECTLY PLACED. (NSW BCA 3.3.4.4).
- EXPOSURE CLASS MASONRY UNITS MUST BE USED BELOW THE DAMP PROOF COURSE LEVEL. (CLAUSE 3.3.1.5(B) AND TABLE 3.3.1.1 OF THE BCA). APPROPRIATE MORTAR AND MIXING RATIO MUST BE USED WITH EXPOSURE CLASS MASONRY UNITS. (CLAUSE 3.3.1.6 OF THE BCA).
- APPROVED ADMIXTURES FOR WATERPROOFING AND/ OR CORROSION PREVENTION MAY BE USED. WORKMANSHIP AND MATERIAL SHALL BE IN ACCORDANCE WITH CURRENT EDITIONS OF AS2870, AS/NZS 2032 INTALL OF PVC PIPES AND AS/NZS 3500 PLUMBING & DRAINAGE.



TYPICAL TEMPORARY WORKS BATTER

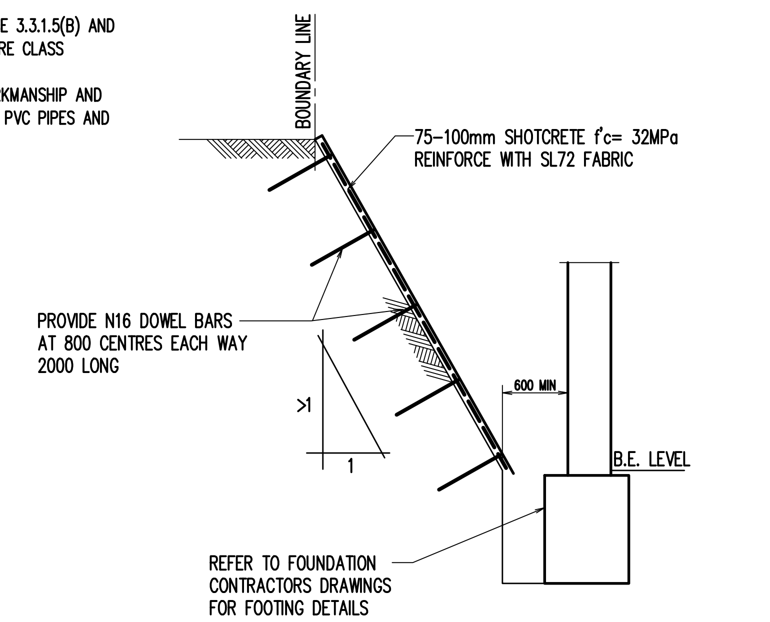
BATTER SLOPE TO BE CONFIRMED BY GEOTECHNICAL ENGINEERS (SHOWN 1:1)
NOTES:
1. SEEK APPROVAL FROM GEOTECHNICAL ENGINEER BEFORE PROCEEDING THE WORKS
2. GEOTECHNICAL ENGINEER TO APPROVE THE STABILITY BATTER
3. WRITTEN APPROVAL FROM GEOTECHNICAL ENGINEER TO BE SUBMITTED TO THE STRUCTURAL ENGINEER

FORMWORK NOTES

F1. THE DESIGN CERTIFICATION, CONSTRUCTION AND PERFORMANCE OF THE FORMWORK AND FALSE WORK SHALL BE THE RESPONSIBILITY OF THE BUILDER/SUBCONTRACTOR, UNLESS SPECIFIC DESIGN REQUIREMENTS ARE SHOWN ON THE ENGINEERING DRAWING.
F2. FORMWORK DESIGN, CONSTRUCTION TOLERANCES AND STRIPPING TIMES SHALL COMPLY WITH AS 3610 AND AS 3600 UNLESS OTHERWISE APPROVED BY THE PROJECT STRUCTURAL ENGINEER.
F3. DURING CONSTRUCTION, SUPPORT PROPPING WILL BE REQUIRED WHERE LOADS FROM STACKED MATERIALS, FORMWORK AND OTHER SUPPORTED SLABS INDUCE LOADS IN A SLAB OR BEAM WHICH EXCEED THE DESIGN LOAD FOR STRENGTH OR SERVICEABILITY AT THAT AGE. ONCE THE NOMINATED 28 DAY STRENGTH HAS BEEN ATTAINED, THESE LOADS SHALL NOT EXCEED THE DESIGN SUPERIMPOSED LOADS SET OUT IN THE GENERAL NOTES.
F4. STRIPPING OF FORMWORK AND BACKPROPPING DETAILS SHALL BE IN ACCORDANCE WITH AS 3600 CLAUSE 17.6 , AND SHALL BE CARRIED OUT BY A SUITABLY QUALIFIED AND EXPERIENCED PERSON. REFER TO THE PROJECT ENGINEER FOR ANY SPECIFIC REQUIREMENTS WHICH MAY BE SPECIFIED. REFER TO NOTE G6.
F5. THE FORMWORK SHALL NOT BE DESIGNED TO RELY ON RESTRAINT OR SUPPORT FROM THE PERMANENT STRUCTURE WITHOUT PRIOR APPROVAL FROM THE PROJECT STRUCTURAL ENGINEER.
F6. FORMWORK SHALL BE DESIGNED TO ACCOMMODATE MOVEMENTS AND LOAD REDISTRIBUTION DUE TO POST TENSIONING.
F7. CONCRETE FORMED SURFACES SHALL HAVE FINISHES IN ACCORDANCE WITH AS 3610, AS SPECIFIED BY THE PROJECT ARCHITECT.
F8. DO NOT PLACE PERMANENT LOADS, INCLUDING MASONRY WALLS AND THE LIKE, ON THE CONCRETE STRUCTURE UNTIL AFTER THE FORMWORK AND BACKPROPPING HAS BEEN REMOVED. REFER TO NOTE RC15.
F9. DESIGN INFORMATION CONCERNING THE FOUNDATION FORMWORK SHALL BE DETERMINED FROM THE CONDITIONS EXISTING ON SITE AT THE TIME OF CONSTRUCTION. ALSO REFER GEOTECHNICAL REPORT WHERE AVAILABLE FOR FURTHER DETAILS.
F10. REFER TO ARCHITECTS DRAWINGS FOR TEST PANEL DETAILS. REINFORCEMENT FOR TEST PANELS SHALL BE SIMILAR TO THAT IN THE PERMANENT STRUCTURE BEING REPRESENTED.
F11. FORMWORK AND SCAFFOLDING SHALL BE EXTENDED BEYOND TENDON ANCHORAGE POINT TO PROVIDE SPACE FOR STRESSING OPERATIONS WHERE REQUIRED. SAFETY HOARDING FOR STRESSING SHALL BE PROVIDED AS NECESSARY.
F12. PROVIDE INDEPENDENT THIRD PARTY CERTIFICATION OF ADEQUACY OF ALL FORMWORK PRIOR TO POURING ANY CONCRETE. CERTIFICATION SHALL BE BY A SUITABLY QUALIFIED STRUCTURAL ENGINEER REGISTERED ON THE NATIONAL PROFESSIONAL ENGINEERS REGISTER.
F13. THE CONTRACTOR SHALL DEVELOP A WRITTEN CONSTRUCTION PROCEDURE FOR THE PLACING OF CONCRETE IN TALL EMBERS. THIS MAY REQUIRE THE PROVISION OF INSPECTION AND POURING OPENINGS IN FORMWORK, OR CONSTRUCTION JOINTS AT LOCATIONS APPROVED BY THE ARCHITECT. THE CONSTRUCTION PROCEDURE SHALL BE SUBMITTED TO THE ENGINEER FOR ACCEPTANCE WITH RESPECT TO COMPLIANCE WITH THE DESIGN INTENT.
F14. ALL SLABS, BEAMS, COLUMNS, WALLS ETC HAVE BEEN DESIGNED FOR THEIR FINAL DESIGN LOADS WHEN ACTING AS A PART OF A TOTAL STRUCTURE. PROPPING AND TEMPORARY SUPPORTS THAT MAY BE REQUIRED TO MAINTAIN THE TEMPORARY STRUCTURAL ADEQUACY DURING CONSTRUCTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.

NOTES:

1. 50% BACKPROPPING DENOTES PROPS AT NO GREATER THAN 3m CENTRES FOR 200mm SLABS.
2. ANY UNSTRESSED DECK NEEDS TO BE SUPPORTED BY TWO FULLY STRESSED CONCRETE DECKS.
3. ALL FORMWORK TO BE CERTIFIED BY STRUCTURAL ENGINEER BEFORE EACH CONCRETE POUR.



TEMPORARY BATTER TREATMENT WHERE REQ'D

NOTES:
1. BATTER SLOPE TO GEOTECHNICAL ENGINEERS REQUIREMENTS
2. WHERE EXCAVATION HEIGHT IS GREATER THAN 4500mm, SHORING TYPE A IS TO BE USED IN PLACE OF BATTER STABLISATION DETAIL.
3. SEEK APPROVAL FROM GEOTECHNICAL ENGINEER BEFORE PROCEEDING THE WORKS
4. GEOTECHNICAL ENGINEER TO APPROVE THE STABILITY BATTER
5. WRITTEN APPROVAL FROM GEOTECHNICAL ENGINEER TO BE SUBMITTED TO THE STRUCTURAL ENGINEER

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ISSUE	DESCRIPTION	DATE	ISSUE	DESCRIPTION	DATE
B	ISSUED FOR CC	22.09.25			
A	ISSUED FOR CC	20.08.25			

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PROPOSED RESIDENCE
LOT 515 (NO. 18) THORPE WAY
BOX HILL, NSW
FOR -

BUILDER
DEZIRE HOMES

CONSTRUCTION NOTES
SCALE: 1:100, 1:20 U.N.O.
DESIGN BY: R.H. | DRAWN BY: C.V. | CHECK BY: R.H. | PAGE: 1 OF 16
JOB NUMBER: 230608SD | ISSUE: B

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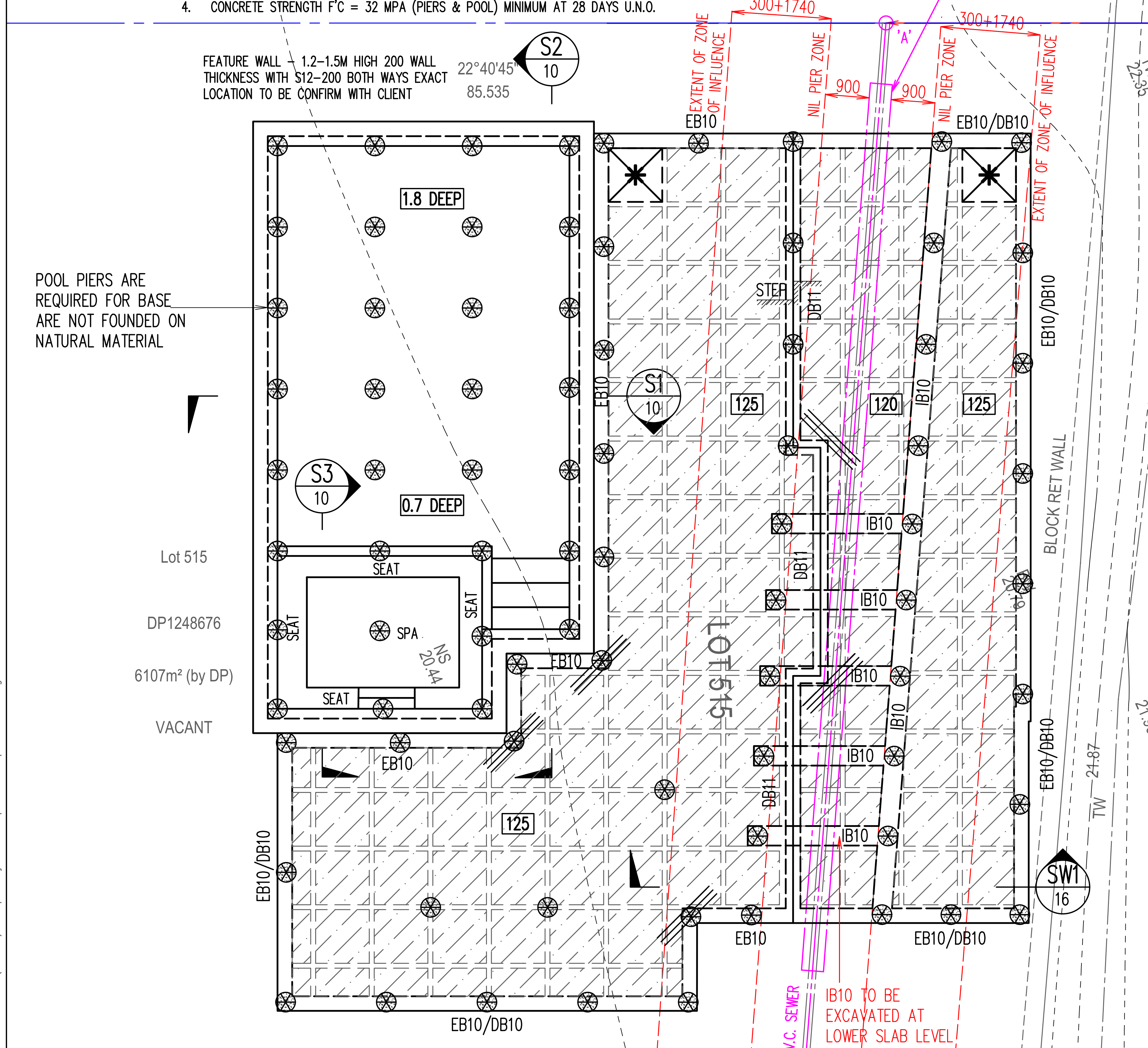
IN-GROUND POOL PLAN
SCALE 1:100

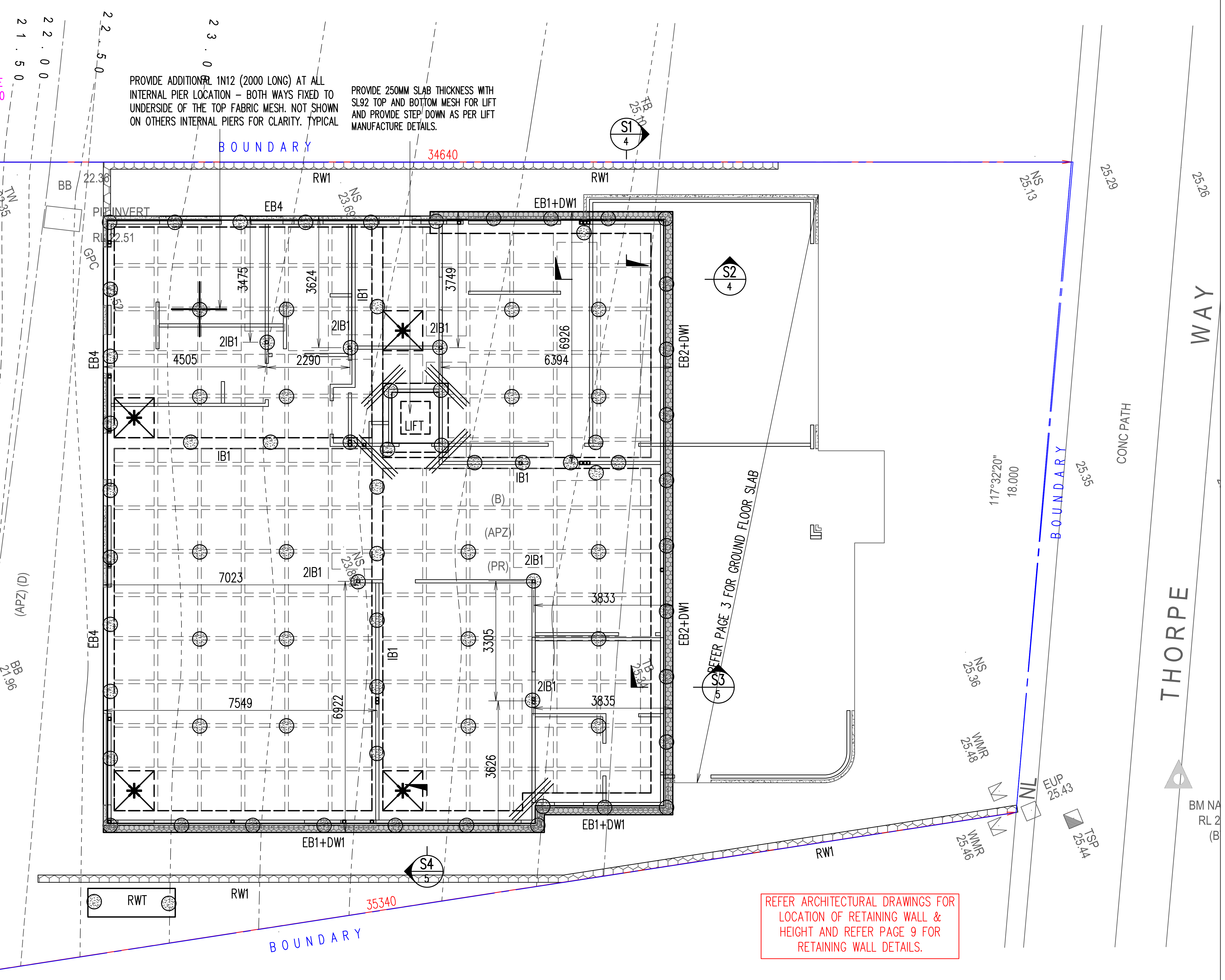
1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS.
3. POOL BASE NEED TO FOUND NATURAL MATERAIL, ALTERNATIVELY 400 DIA MASS CONCRETE REQUIRED TO SHALE AT 1.8M CTS MAX.
4. CONCRETE STRENGTH F'C = 32 MPA (PIERS & POOL) MINIMUM AT 28 DAYS U.N.O.

NOTE:
THE SEWER LOCATION & DEPTH IS IN ACCORDANCE WITH SERVICE PROTECTION REPORT OF RMA INFRASTRUCTURE PTY LTD REPORT NO. B-21002. DATED 01/07/2025.
SEWER MAIN AT 'A' – 1.74m TO INVERT
SEWER MAIN AT 'B' – 1.61m TO INVERT



CABANA WAFFLE SLAB PLAN & FOOTING LAYOUT PLAN
SCALE 1:100

1. 150mm DEEP PODS U.N.O. + SLAB THICKNESS = 125mm
2. PROVIDE SL92 TOP MESH (20 COVER) FOR 100mm SLAB THICKNESS
3. CONCRETE STRENGTH f'c = 32 MPa (PIERS) AND f'c = 32 MPa (SLAB) MINIMUM AT 28 DAYS U.N.O.
4. LAID 0.2 MICRON HIGH IMPACT DAMP-PROOF MEMBRANE UNDER WAFFLE POD SLAB.
5. IF GROUND CONDITIONS CHANGE DURING EXCAVATION, PLEASE NOTIFY ENGINEER AND SEEK FURTHER INSTRUCTIONS.
6. IF PIER DEPTH EXCEEDS 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
7. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
8. THIS DRAWINGS IS SIGNED SUBJECT TO CERTIFICATE OF INSPECTION ISSUED BY AN ENGINEER FROM THIS OFFICE. ALL PIERS AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.



BASEMENT WAFFLE SLAB PLAN & FOOTING LAYOUT PLAN
SCALE 1:100

1. 150/225/300mm DEEP PODS U.N.O. + SLAB THICKNESS = 100mm
2. PROVIDE SL92 TOP MESH (20 COVER) FOR 100mm SLAB THICKNESS
3. CONCRETE STRENGTH f'c = 32 MPa (PIERS) AND f'c = 32 MPa (SLAB) MINIMUM AT 28 DAYS U.N.O.
4. LAID 0.2 MICRON HIGH IMPACT DAMP-PROOF MEMBRANE UNDER WAFFLE POD SLAB.
5. IF GROUND CONDITIONS CHANGE DURING EXCAVATION, PLEASE NOTIFY ENGINEER AND SEEK FURTHER INSTRUCTIONS.
6. IF PIER DEPTH EXCEEDS 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
7. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
8. THIS DRAWINGS IS SIGNED SUBJECT TO CERTIFICATE OF INSPECTION ISSUED BY AN ENGINEER FROM THIS OFFICE. ALL PIERS AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.

WALL SCHEDULE

MARK	FLOOR LEVEL	THICKNESS (mm)	VERT. REINF.	HORZ. REINF.	f'c
DINCEL DW1	AS MARKED ON PLAN	200	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS1	AS MARKED ON PLAN	200	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW2	AS MARKED ON PLAN	150	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS2	AS MARKED ON PLAN	150	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW3	AS MARKED ON PLAN	110	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS3	AS MARKED ON PLAN	110	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW4	AS MARKED ON PLAN	300	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS4	AS MARKED ON PLAN	300	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa

LEGEND

- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL OVER
- DENOTES 200 DINCEL WALL OVER
- [XXX] DENOTES SLAB THICKNESS
- DENOTES 3N12 OR EQUIV. (OR 3L11TM) CRACK CONTROL BARS, 2000mm LONG TIED UNDER TOP MESH.
- DENOTES 300 DEEP PODS (D1). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES 225 DEEP PODS (D2). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES 150 DEEP PODS (D3). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES POD STARTING POINT
- DENOTES 450 DIA. MASS CONCRETE PIERS (450 KPA MIN. BEARING CAPACITY) TO UNIFORM BEARING ON NATURAL STRATA ROCK THROUGHOUT. ALL PIERS: 5.5M MIN. DEEP OR ROCK – WHICHEVER IS LESSER. ALL PIERS NEED TO HAVE SAME BEARING CAPACITY OF LOWER GROUND FLOOR PIERS.(ALT. USE 100KN SCREW PIER AS PER MANUFACTURERS DETAILS. EXACT PIER DEPTH TO BE DETERMINE BY SCREW PIER CONTRACTOR TO ACHIVE 100KN BEARING CAPACITY.)
- DENOTES 400 DIA. MASS CONCRETE PIERS (250 KPA MIN. BEARING CAPACITY) TO UNIFORM BEARING ON NATURAL STRATA ROCK THROUGHOUT. ALL PIERS: 3.0M MIN. DEEP OR ROCK – WHICHEVER IS LESSER. (ALT. USE 85KN SCREW PIER AS PER MANUFACTURERS DETAILS. EXACT PIER DEPTH TO BE DETERMINE BY SCREW PIER CONTRACTOR TO ACHIVE 85KN BEARING CAPACITY.)

JOB SPECIFICATION	
SITE CLASSIFICATION	**P (H1 SLAB)**
TYPE OF CONSTRUCTION	**ARTICULATED HEBEL VENEER**
SALINITY AFFECTED	**YES – REFER PAGE 2**
WIND CLASSIFICATION	**N2**
EXPOSURE CLASSIFICATION CONCRETE SURFACE	**INTERNAL (A1) EXTERNAL (B1)**
THE SITE CLASSIFICATION REPORT PREPARED BY IDEAL GEOTECH PTY LTD REPORT NO. 7549-IDF DATED 15/04/25. WE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS.	
PROVIDE ADDITIONAL 1N12 (2000 LONG) AT PIER LOCATION – BOTH WAYS FIXED TO UNDERSIDE OF THE TOP FABRIC MESH.	

REINFORCEMENT COVER SCHEDULE

ELEMENT	COVER (mm) TOP	BOTTOM	SIDES	EXPOSURE CLASSIFICATION
FOOTINGS	50mm	50mm	50mm	A2
SLAB ON GROUND	30mm	30mm	30mm	A1

CONCRETE QUALITY

ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f'c
FOOTINGS	80mm	20mm	A	32MPa
SLAB ON GROUNDDD	80mm	20mm	A	32MPa

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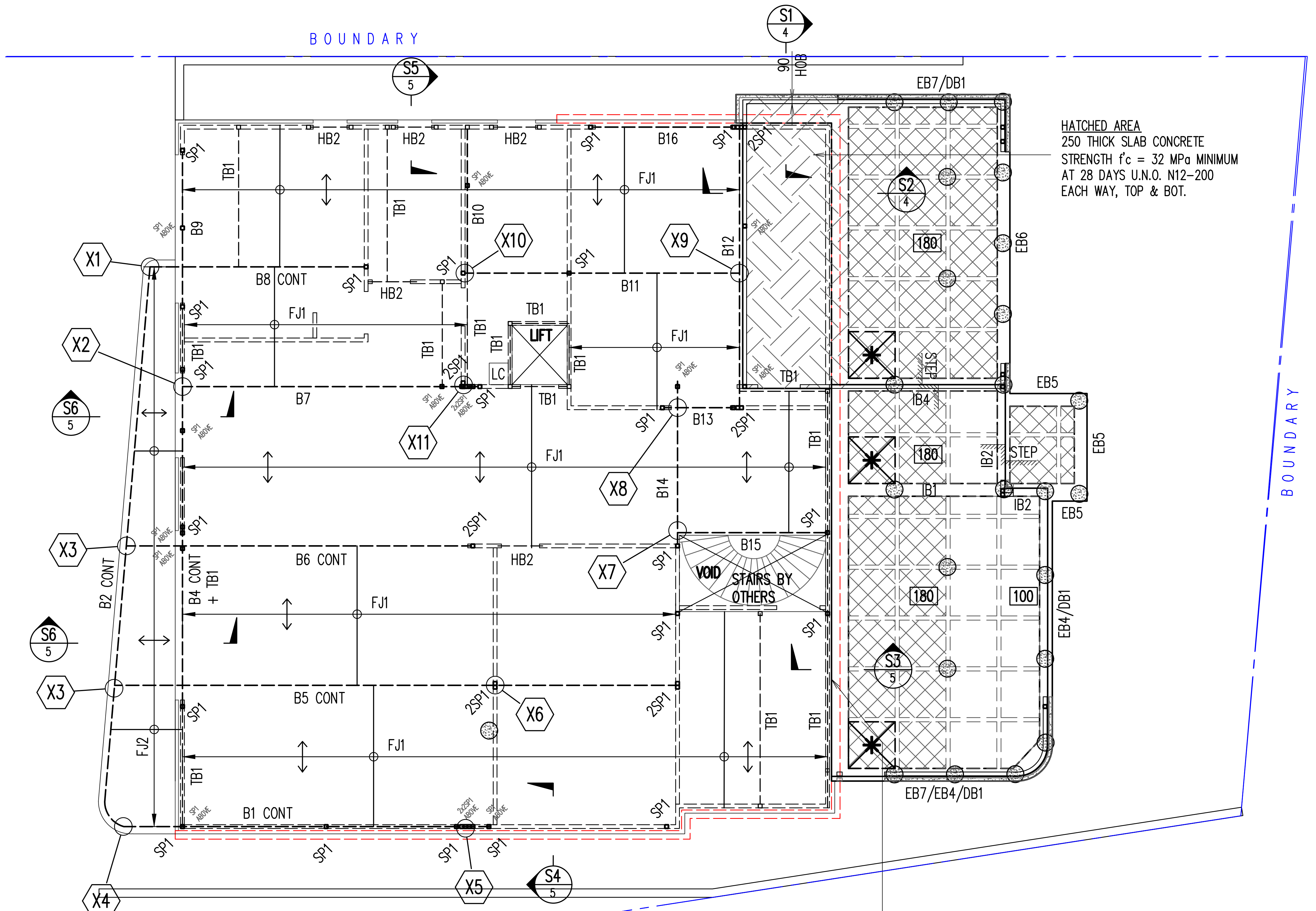
PROPOSED RESIDENCE
LOT 515 (NO. 18) THORPE WAY
BOX HILL, NSW
FOR –

BUILDER
DEZIRE HOMES

GROUND FLOOR SLAB PLAN

SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 2 OF 16
JOB NUMBER: 230608SD			ISSUE: B



IMPORTANT NOTE:
ADEQUATE EXPANSION MUST BE USED FOR ANY FLOOR COVERING BRIDGING THE TIMBER FLOOR AND CONCRETE SLAB.

FIRST FLOOR – STEEL BEAM AND FLOOR JOIST LAYOUT PLAN

SCALE 1:100

1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
3. PROVIDE DOUBLE FLOOR JOISTS UNDER ALL WALLS RUNNING PARALLEL TO JOISTS
4. TB1 TO BE PROVIDED ON OVER ALL OPENINGS
5. PROVIDE DOUBLE STUD UNDER ALL TIMBER AND HEADER BEAM.

GROUND FLOOR WAFFLE SLAB PLAN

SCALE 1:100

1. 225/300mm DEEP PODS U.N.O. + SLAB THICKNESS = 180 & 100mm
2. PROVIDE SL92 TOP MESH (20 COVER) FOR 100mm SLAB THICKNESS
3. PROVIDE SL92 TOP & BOTTOM MESH (20 COVER) FOR 180mm SLAB THICKNESS
4. CONCRETE STRENGTH f'c = 32 MPa (PIERS) AND f'c = 32 MPa (SLAB) MINIMUM AT 28 DAYS U.N.O.
5. LAID 0.2 MICRON HIGH IMPACT DAMP-PROOF MEMBRANE UNDER WAFFLE POD SLAB.
6. IF GROUND CONDITIONS CHANGE DURING EXCAVATION, PLEASE NOTIFY ENGINEER AND SEEK FURTHER INSTRUCTIONS.
7. IF PIER DEPTH EXCEEDS 2.5m, ENGINEER TO BE NOTIFIED FOR DECISION.
8. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
9. THIS DRAWINGS IS SIGNED SUBJECT TO CERTIFICATE OF INSPECTION ISSUED BY AN ENGINEER FROM THIS OFFICE. ALL PIERS AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.

LAID 1ST & 4TH N12-200 CTS
LAID 2ND & 3RD N12-200 CTS

BAR LAYING SEQUENCE

U.N.O.

REINFORCEMENT COVER SCHEDULE				
ELEMENT	COVER (mm) TOP	BOTTOM	SIDES	EXPOSURE CLASSIFICATION
INTERNAL SLABS	25mm	25mm	25mm	A1
EXTERNAL SLABS	40mm	40mm	40mm	B1

CONCRETE QUALITY				
ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f'c
INTERNAL SLABS	80mm	20mm	A	32MPa U.N.O.
EXTERNAL SLABS	80mm	20mm	A	32MPa U.N.O.

REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.

REFER ARCHITECTURAL DRAWINGS FOR LOCATION OF RETAINING WALL & HEIGHT AND REFER PAGE 9 FOR RETAINING WALL DETAILS.

1. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
2. BUILDER TO CONFIRM CONCRETE EDGE REBATE WITH HEBEL (OR SIMILAR PRODUCT) MANUFACTURERS DETAILS AND SPECIFICATIONS.
3. PROVIDE ARTICULATION JOINTS – LOCATION & DETAIL AS PER HEBEL MANUFACTURE DETAILS.

JOB SPECIFICATION	
SITE CLASSIFICATION	**P (H1 SLAB)**
TYPE OF CONSTRUCTION	**ARTICULATED HEBEL VENEER**
SALINITY AFFECTED	**YES – REFER PAGE 2**
WIND CLASSIFICATION	**N2**
EXPOSURE CLASSIFICATION CONCRETE SURFACE	**INTERNAL (A1) EXTERNAL (B1)**
THE SITE CLASSIFICATION REPORT PREPARED BY CORE GEOTECH PTY LTD REPORT NO. CG23-0798/A DATED 10/09/23. WE TAKE NO RESPONSIBILITY FOR VARIATIONS THAT MAY OCCUR DUE TO VARIATIONS IN SITE CONDITIONS.	
PROVIDE ADDITIONAL 1N12 (2000 LONG) AT PIER LOCATION – BOTH WAYS FIXED TO UNDERSIDE OF THE TOP FABRIC MESH.	

LEGEND

- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL OVER
- DENOTES 200 DINCEL OR BLOCK WALL UNDER
- DENOTES 200 DINCEL WALL OVER
- XXX DENOTES SLAB THICKNESS
- DENOTES 3N12 OR EQUIV. (OR 3L11TM) CRACK CONTROL BARS, 2000mm LONG TIED UNDER TOP MESH.
- DENOTES 300 DEEP PODS (D1). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES 225 DEEP PODS (D2). POD SIZE: 1090x1090 (CUT PODS AS REQUIRED)
- DENOTES POD STARTING POINT
- DENOTES 450 DIA. MASS CONCRETE PIERS (450 KPA MIN. BEARING CAPACITY) TO UNIFORM BEARING ON NATURAL STRATA ROCK THROUGHOUT. ALL PIERS: 5.5.0M MIN. DEEP OR ROCK – WHICHEVER IS LESSER. ALL PIERS NEED TO HAVE SAME BEARING CAPACITY OF LOWER GROUND FLOOR PIER.(ALT. USE 100KN SCREW PIER AS PER MANUFACTURERS DETAILS. EXACT PIER DEPTH TO BE DETERMINE BY SCREW PIER CONTRACTOR TO ACHIVE 100KN BEARING CAPACITY.)
- ⟷ DENOTES FLOOR JOIST DIRECTION

MEMBER SCHEDULE		
MARK	SIZE	COMMENTS
B1,B2,B4,B16	300 PFC	300 GRADE + 10 DIA HOLES T&B
B5,B6,B7,B12	310UC96.8	@ 1.0M CTS BOTH SIDES
B8,B10,B11	310UB46.2	300 GRADE + 10 DIA HOLES T&B
B9,B13	300 PFC	300 GRADE + 10 DIA HOLES T&B
B14,B15	310UB46.2	300 GRADE + 10 DIA HOLES T&B
SP1	89 x 89 x 5.0 SHS	350 GRADE
FJ1	HJ 300 63 HYJOIST	AT 450 CTS OR MANUFACTURES DETAILS
FJ2	240 X 45 MGP10 OR F7	AT 350 CTS
TB1,HB1	300 x 63 HYSPAN	TIMBER BEAM
TB2	2/300 X 63 HYSPAN	M10 4.6/S @ 450 CTS TOP & BOTTOM
HB2	2/190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
HB3	240 X 63 HYSPAN	TIMBER BEAM
HB4	190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
TB,HB	STRUCTURAL TIMBER BEAM	TO MANUFACTURES DETAILS
TP1	STRUCTURAL TIMBER POST	TO MANUFACTURES DETAILS

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BOX HILL, NSW
FOR –

BUILDER
DEZIRE HOMES

GROUND/FIRST FLOOR SLAB

SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 3 OF 16
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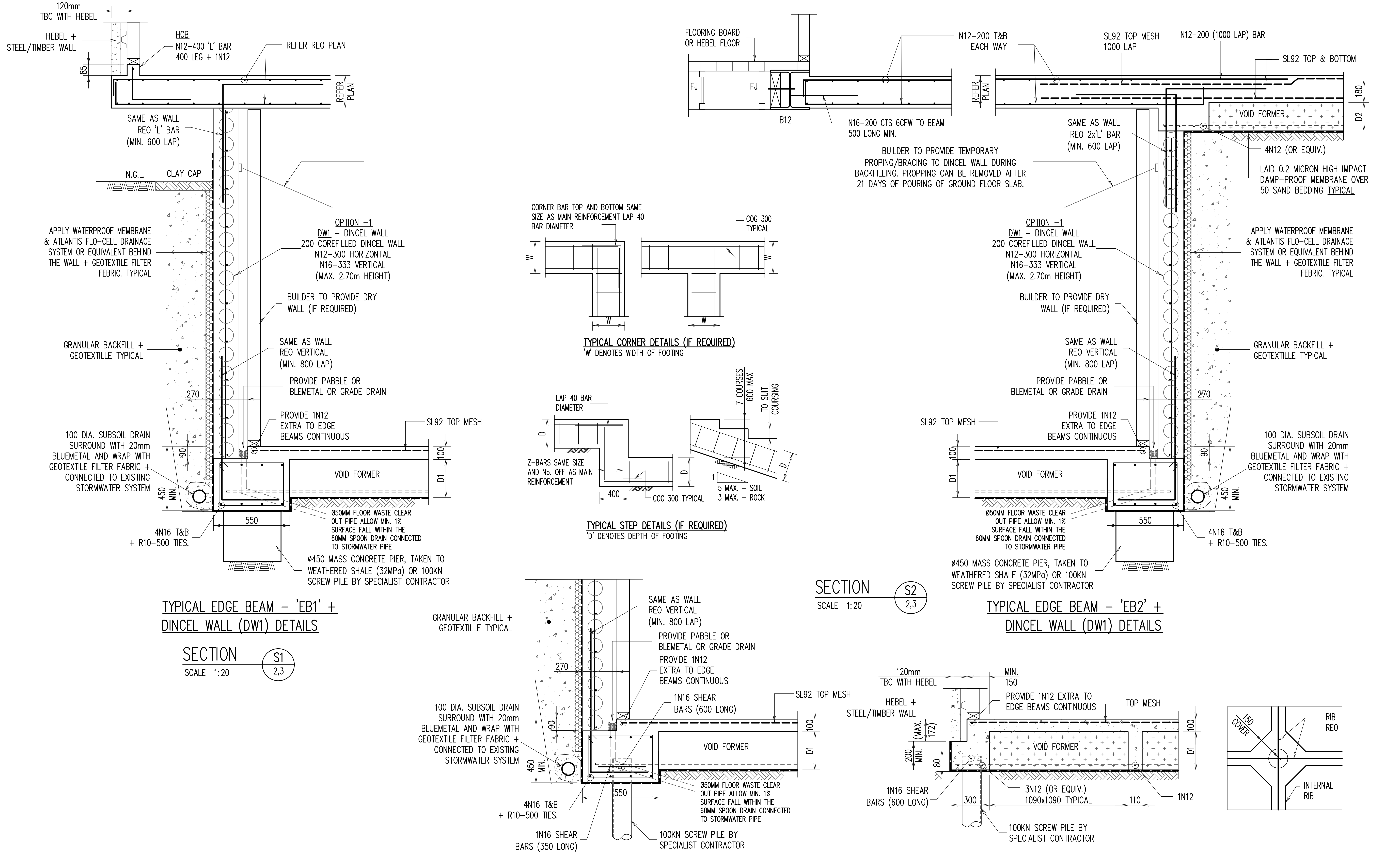

TYPICAL SLAB TO SCREW PILE CONNECTION DETAILS

A2 0 1 2 3 4 5 6 7 8 9 10

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DEZIRE HOMES

SLAB DETAILS - 1

SCALE: 1:100, 1:20 U.N.O.

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R.H.	C.V.	R.H.	4 OF 16

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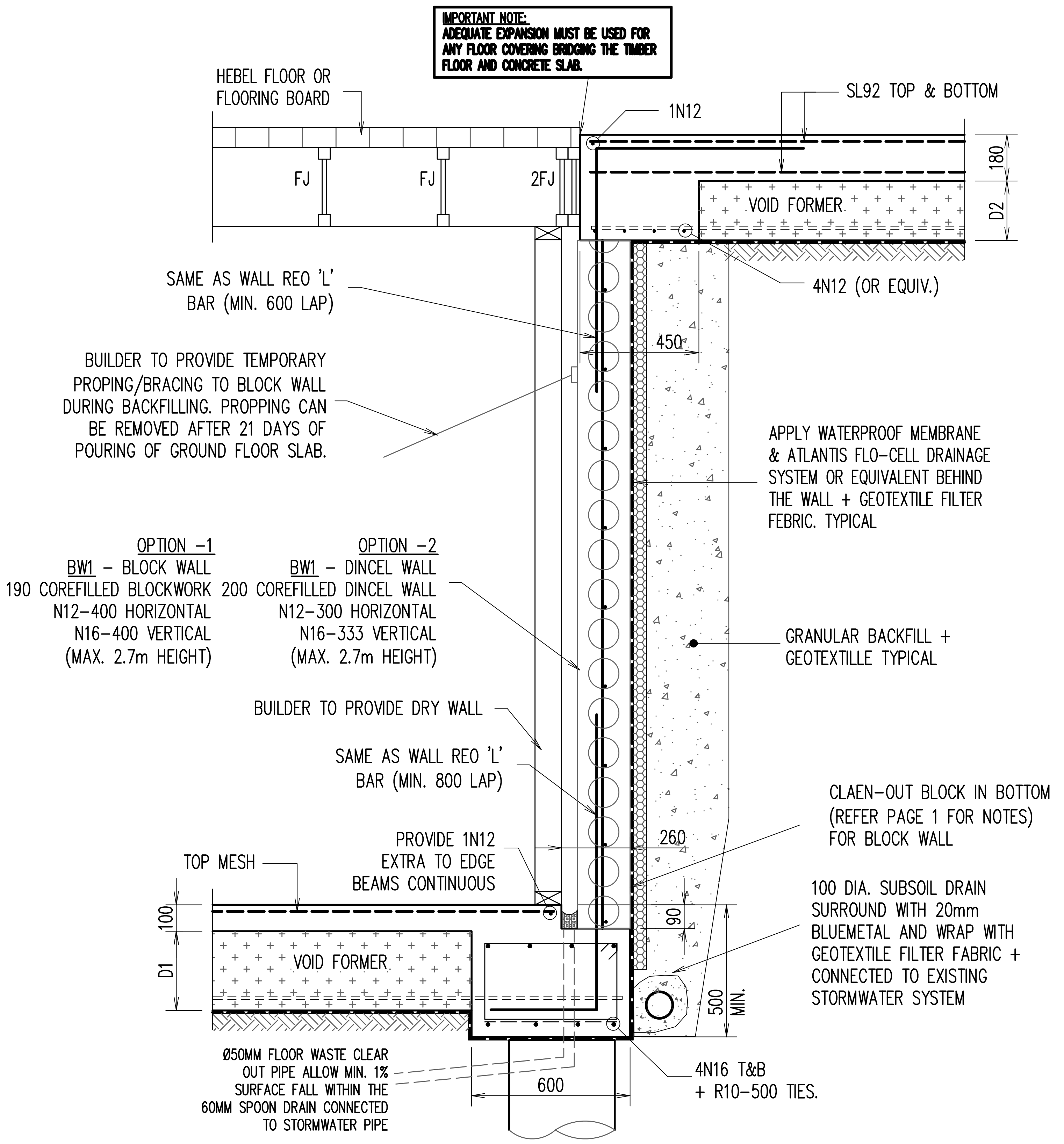
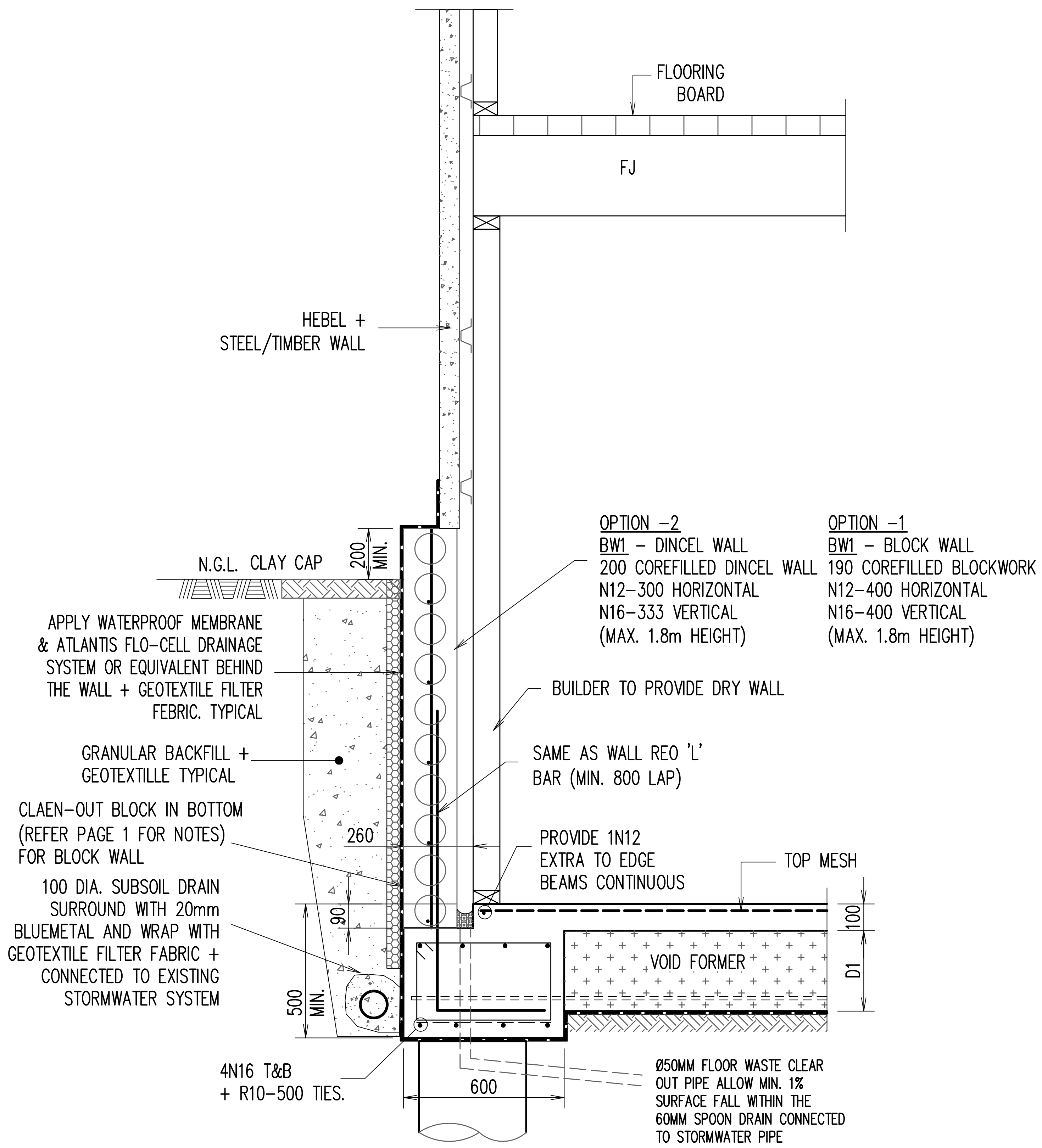


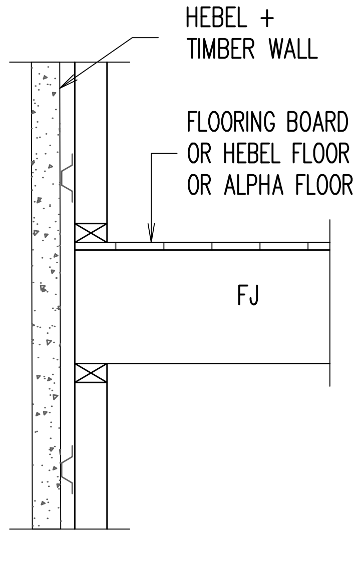
TYPICAL EDGE BEAM - 'EB1' + DINCEL WALL (DW1) DETAILS

SECTION S3 (2,3)
SCALE 1:20

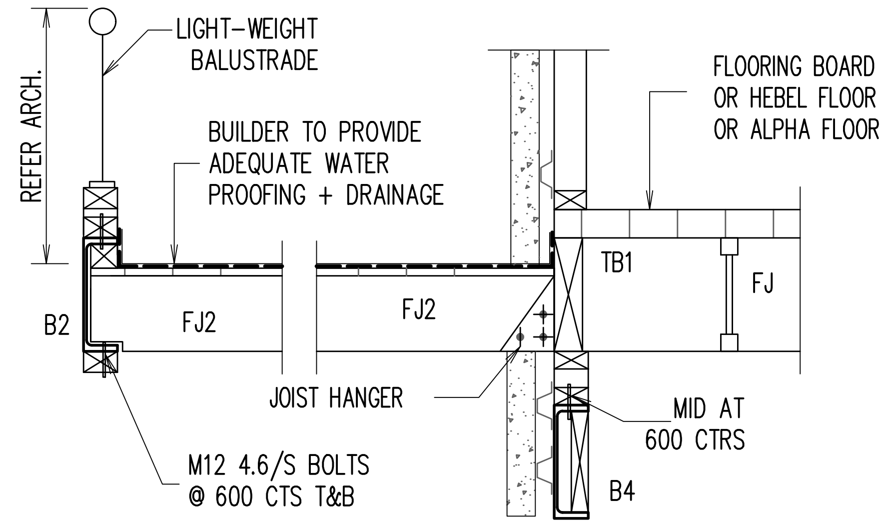


TYPICAL EDGE BEAM - 'EB1' + DINCEL WALL (DW1) DETAILS

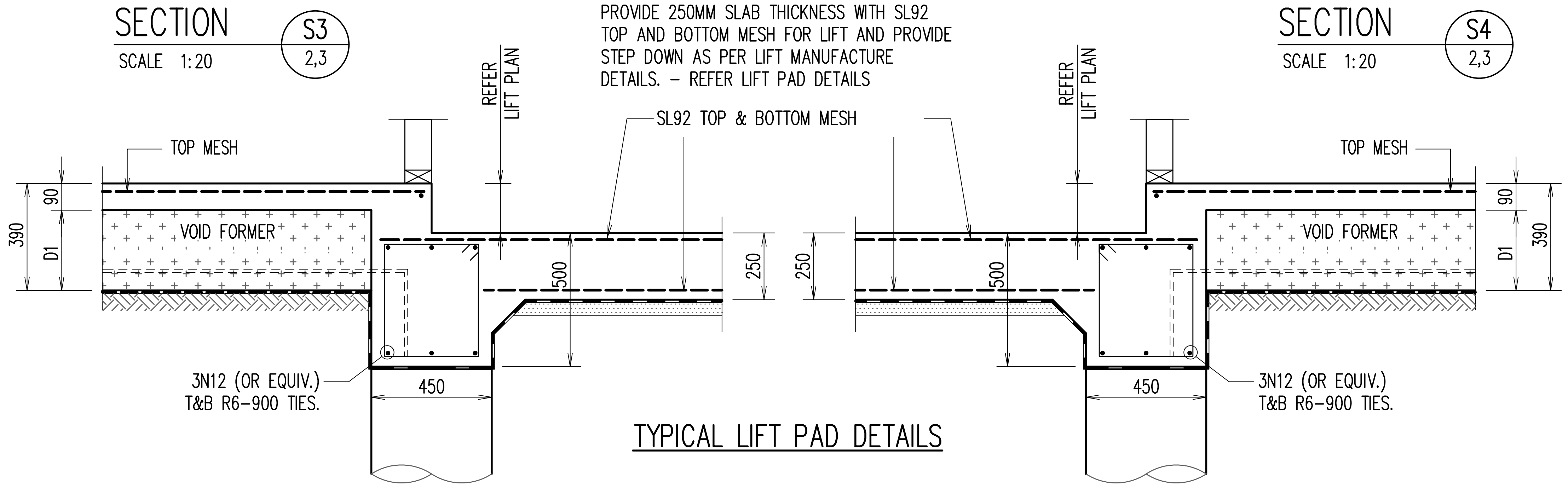
SECTION S4 (2,3)
SCALE 1:20



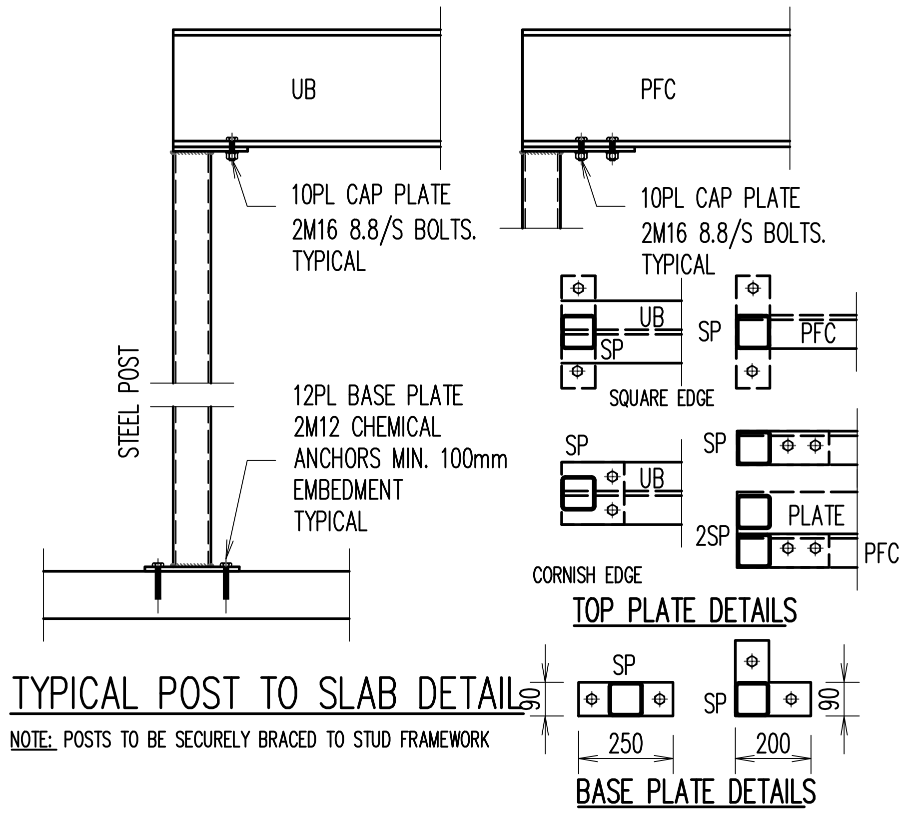
SECTION S5 (3)
SCALE 1:20



SECTION S6 (3)
SCALE 1:20



TYPICAL LIFT PAD DETAILS



TYPICAL POST TO SLAB DETAIL

NOTE: POSTS TO BE SECURELY BRACED TO STUD FRAMEWORK

ISSUE	DESCRIPTION	DATE	ISSUE	DESCRIPTION	DATE
B	ISSUED FOR CC	22.09.25			
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PROPOSED RESIDENCE
LOT 515 (NO. 18) THORPE WAY
BOX HILL, NSW
FOR -

BUILDER
DEZIRE HOMES

SLAB DETAILS - 2			
SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 5 OF 16
JOB NUMBER: 230608SD			ISSUE: B

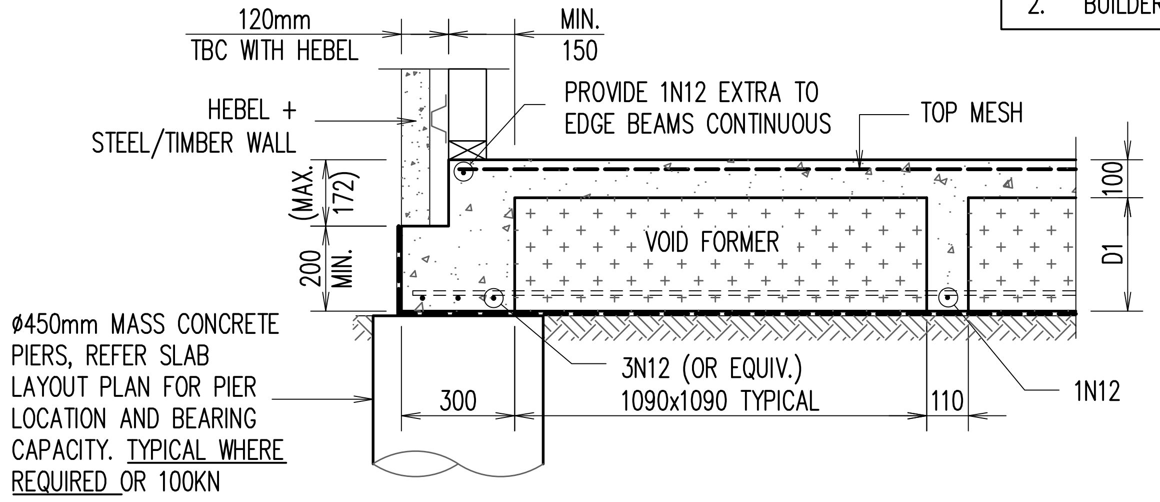
A2

1. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
2. BUILDER TO CONFIRM CONCRETE EDGE REBATE WITH HEBEL (OR SIMILAR PRODUCT) MANUFACTURERS DETAILS AND SPECIFICATIONS.

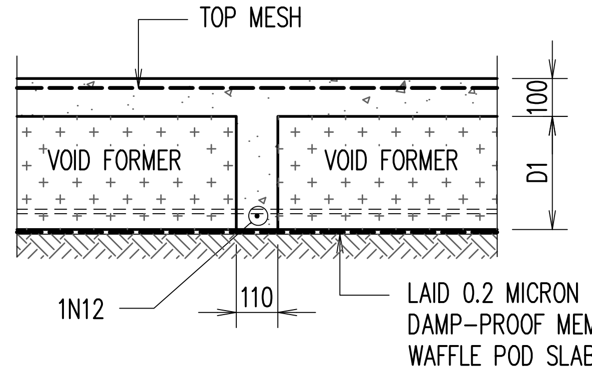
JOB SPECIFICATION	
POD DEPTH (D1)	300mm
POD DEPTH (D2)	225mm
SLAB THICKNESS	100mm
TOP MESH	SL92
BEAM REINFORCEMENT: 3N12 (ALTERNATIVELY 3-L11TM)	

REINFORCEMENT FOR EXTERNAL BEAMS WHERE WIDTH EXCEEDS 300mm

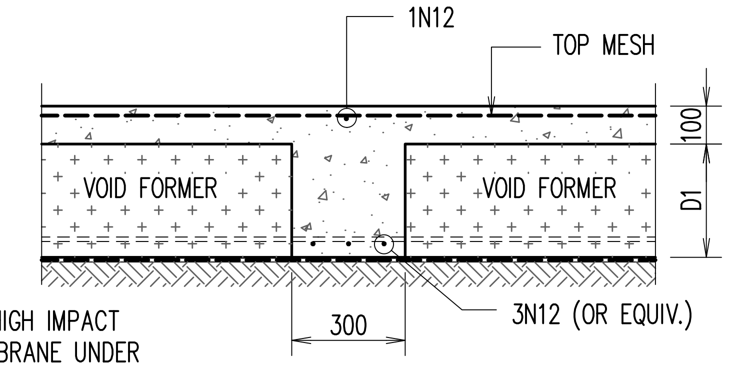
WIDTH	TOP STEEL	BTM STEEL EXTRA
301-330	1N12	1N12
331-440	1N12	2N12
441-550	2N12	3N12
551-660	3N12	4N12



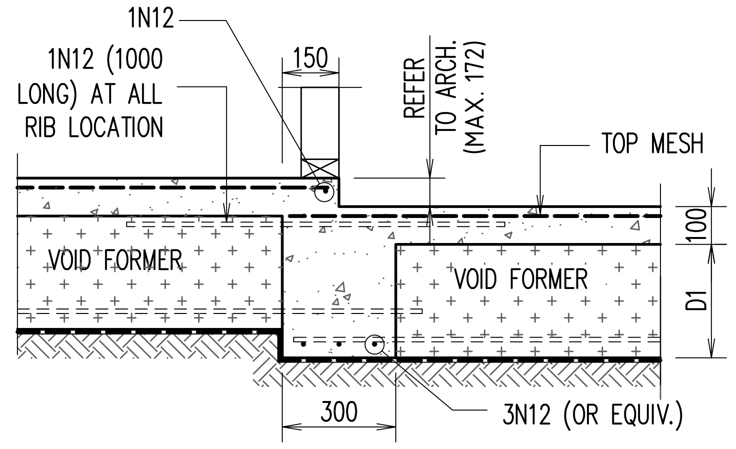
TYPICAL EDGE BEAM – 'EB4'
ALTERNATIVELY REFER TO 'DB1' FOR DETAILS



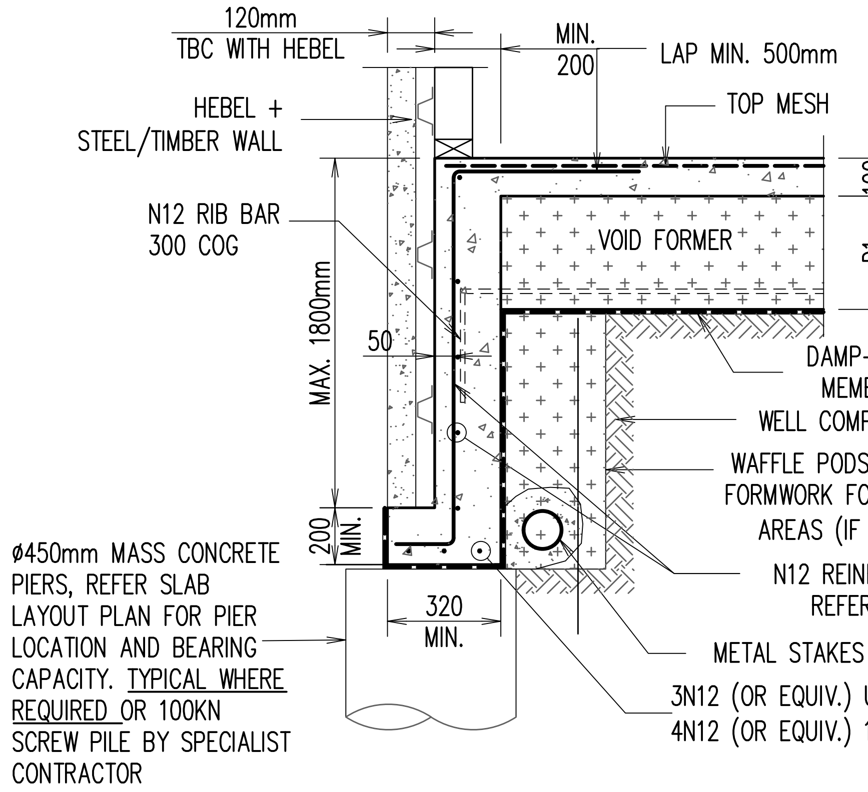
INTERNAL RIB



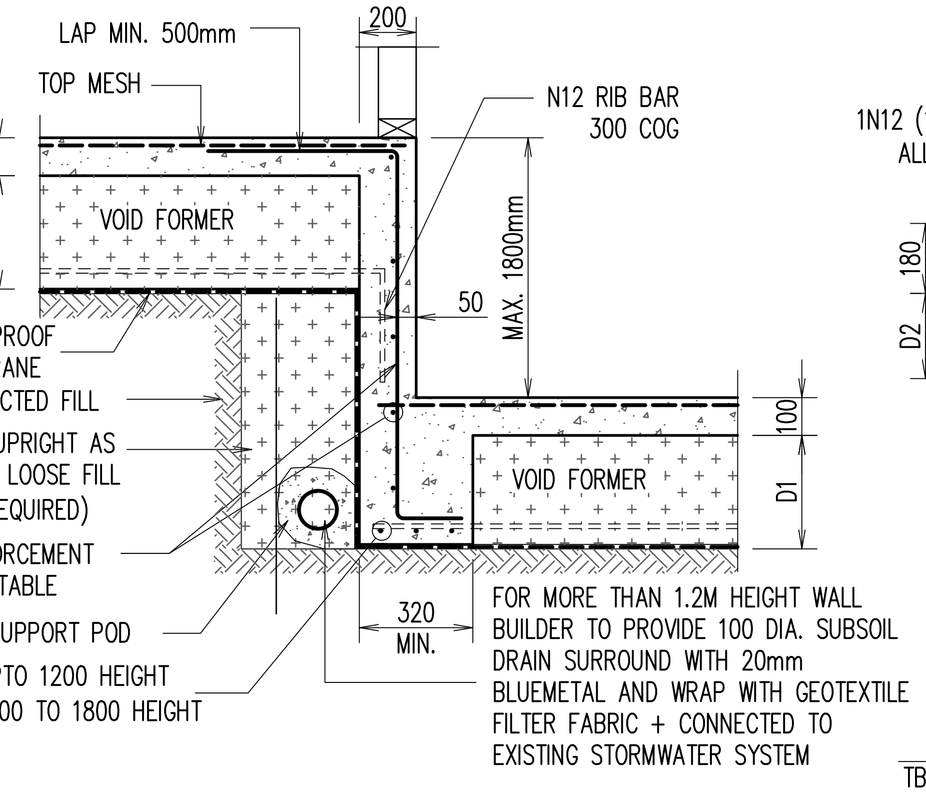
INTERNAL BEAM – 'IB2'



INTERNAL STEPDOWN BEAM – 'IB3'
ALTERNATIVELY REFER TO 'DB2' FOR DETAILS



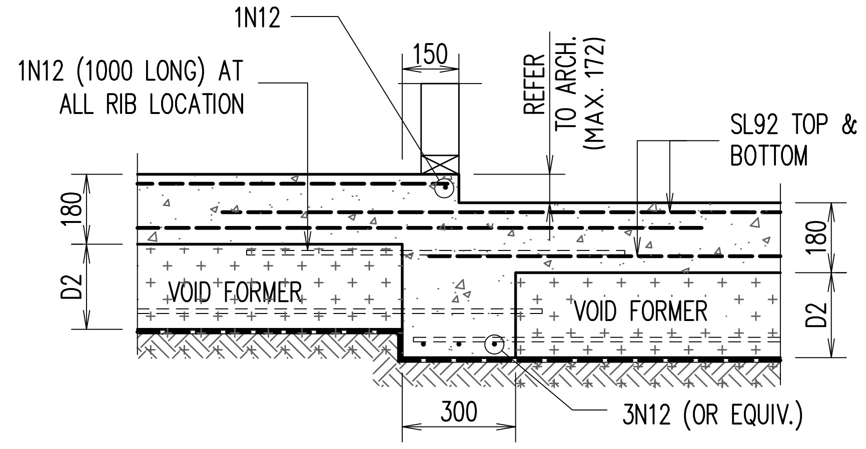
'EXTERNAL' DROP BEAM – 'DB1'



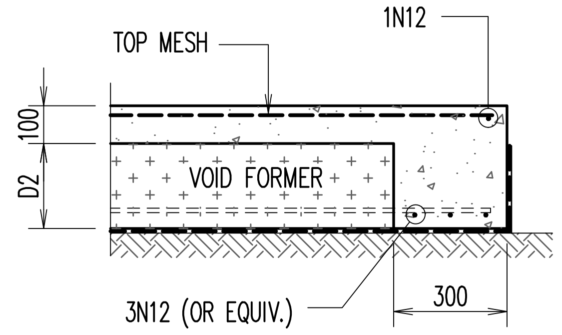
'INTERNAL' DROP BEAM – 'DB2'

DROP EDGE BEAM REINFORCEMENT TABLE

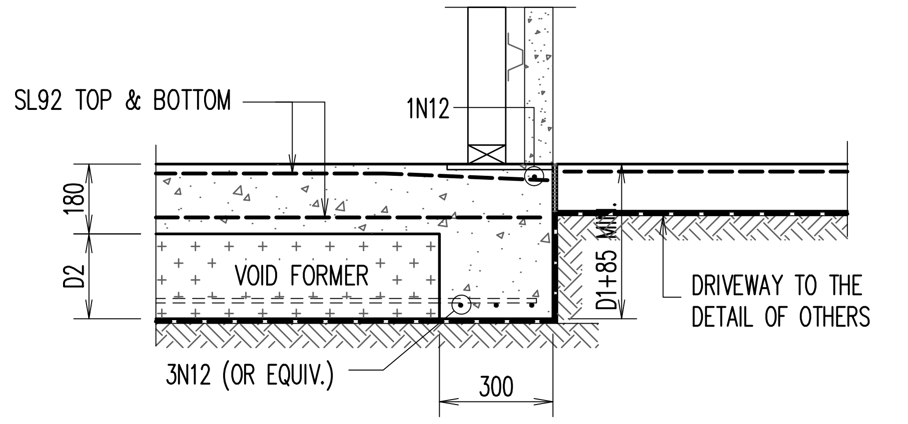
DROP EDGE BEAM DEPTH	HORIZONTAL REO	VERTICAL REO
UP TO 400mm	1N12	N12-900 CTS
410 – 900mm	N12-400 CTS	N12-400 CTS
910 – 1500mm	N12-300 CTS	N12-300 CTS
1510 – 1800	N12-200 CTS	N12-200 CTS



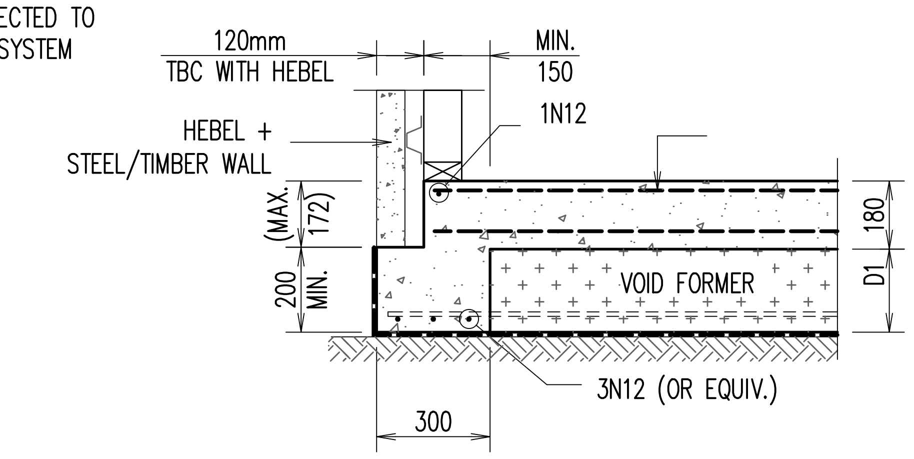
INTERNAL STEPDOWN BEAM – 'IB4'
ALTERNATIVELY REFER TO 'DB2' FOR DETAILS



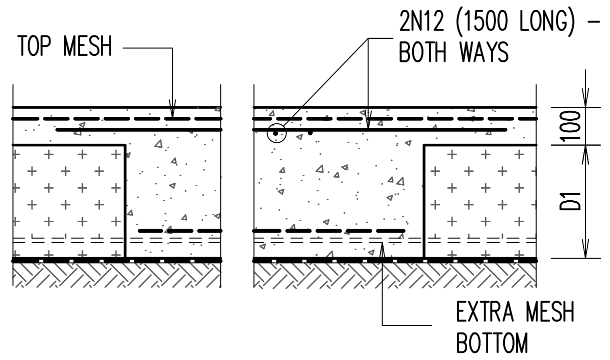
PORCH/ALFRESCO EDGE BEAM – 'EB5'
ALTERNATIVELY REFER TO 'DB3' FOR DETAILS



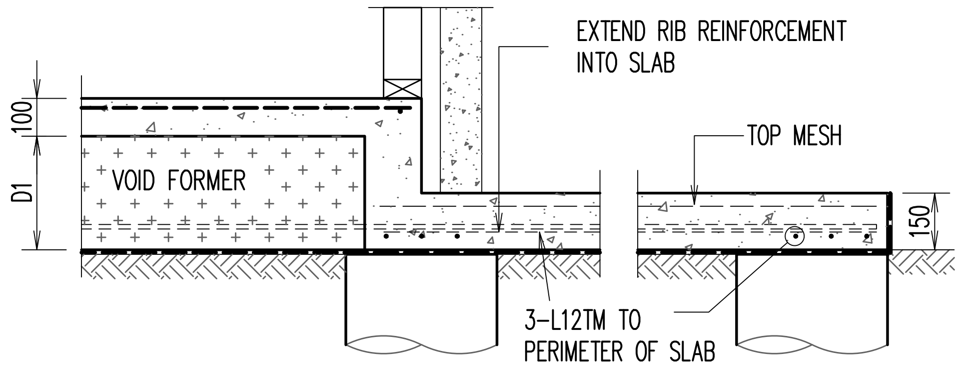
GARAGE EDGE BEAM – 'EB6'
ALTERNATIVELY REFER TO 'DB3' FOR DETAILS



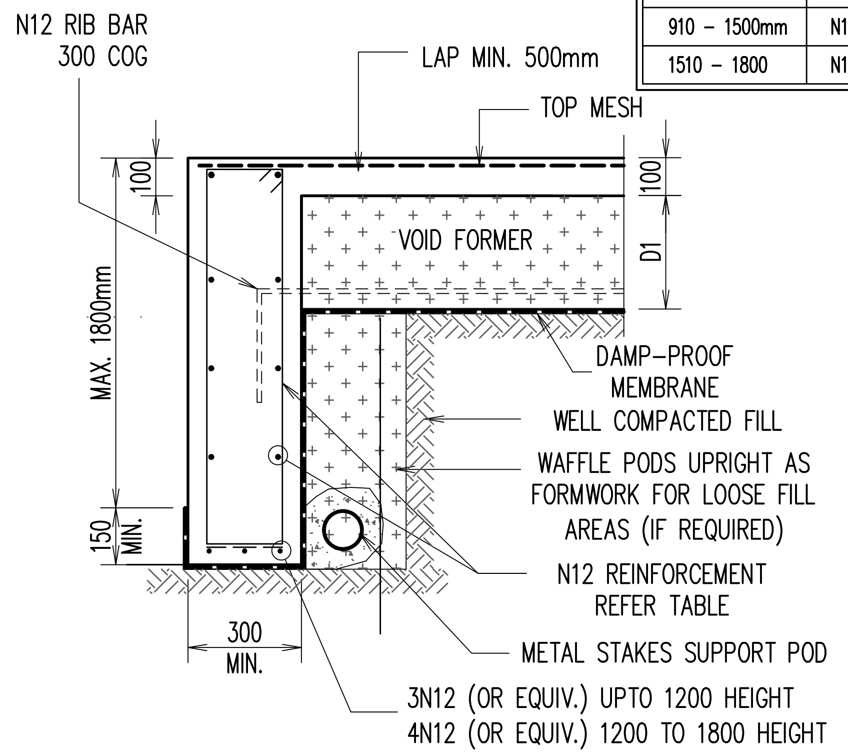
TYPICAL EDGE BEAM – 'EB7'
ALTERNATIVELY REFER TO 'DB1' FOR DETAILS



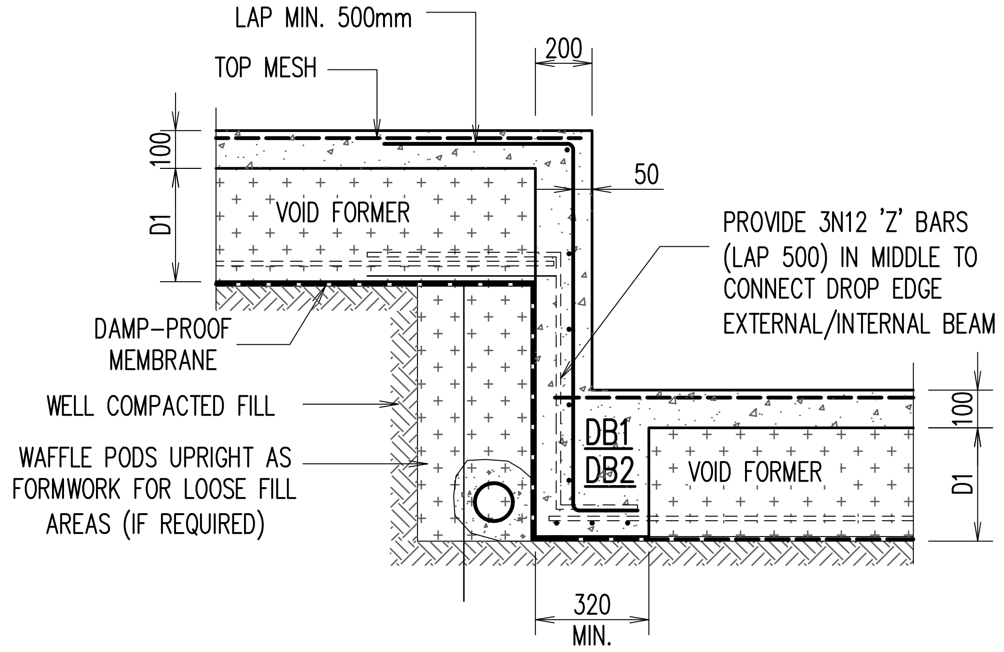
LOCAL THICKENING DETAIL



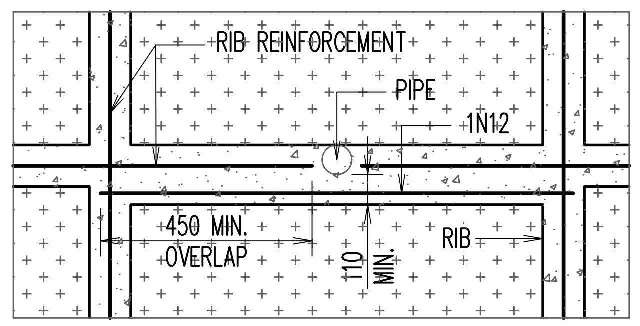
TYPICAL ATTACHED TANK/AC SLAB DETAIL



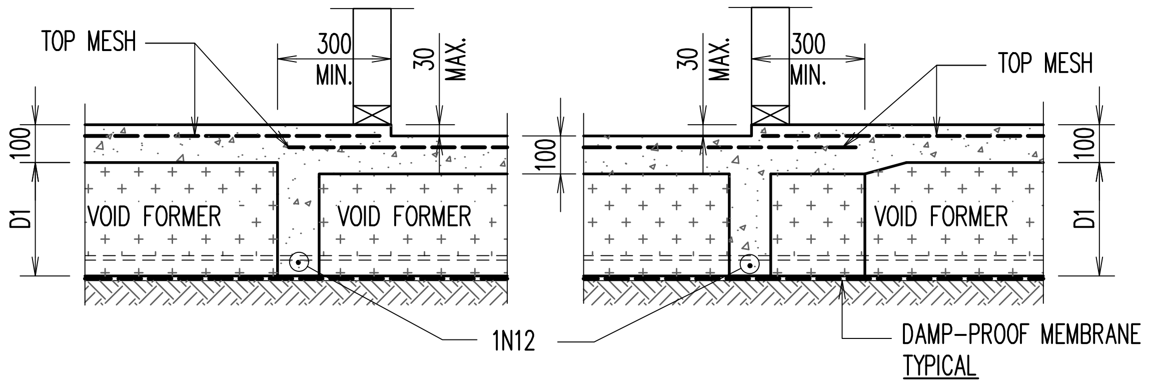
'EXTERNAL' DROP BEAM – 'DB3'



DROP EDGE BEAM TRANSITION DETAIL



PIPE PENETRATION THROUGH RIB



TYPICAL WET AREA STEP DETAIL

A2 0 1 2 3 4 5 6 7 8 9 10

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SLAB DETAILS – 3

SCALE: 1:100, 1:20 U.N.O.

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JOB NUMBER: 230608SD
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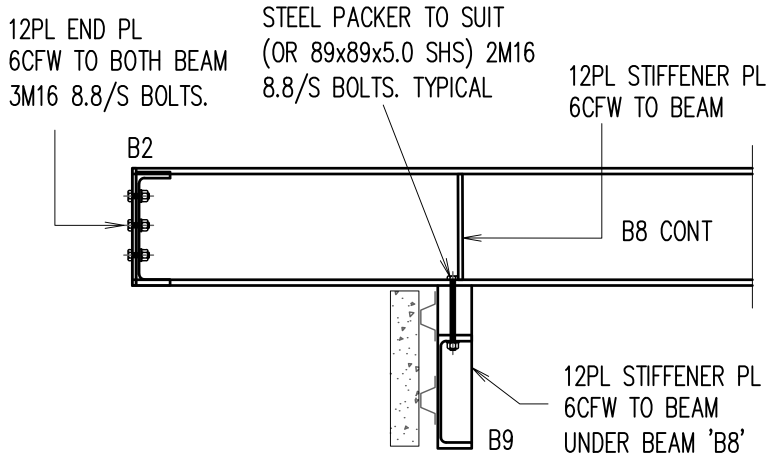
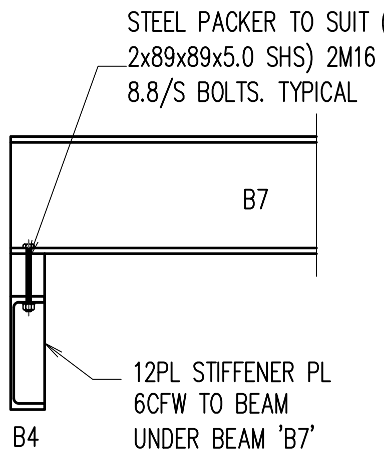


DETAIL – X1
SCALE 1:20



DETAIL – X2
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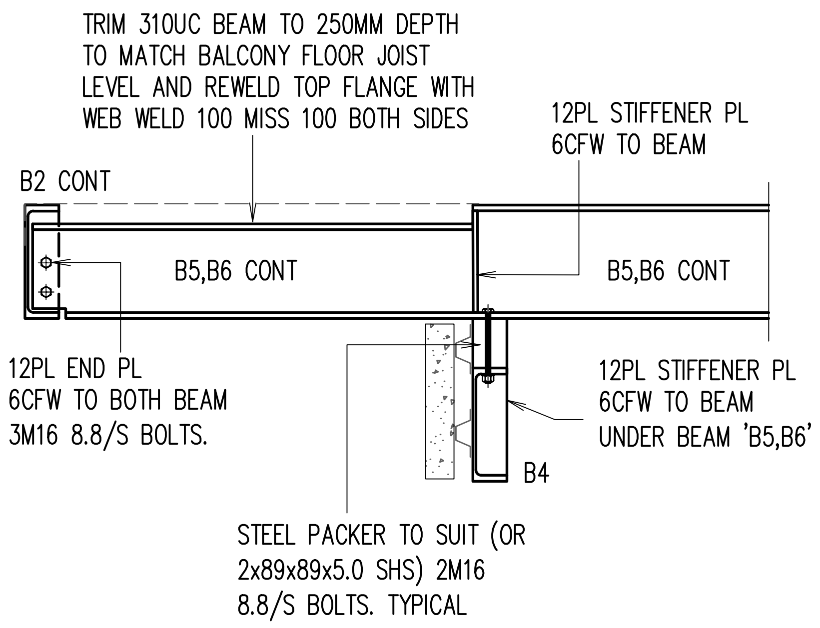
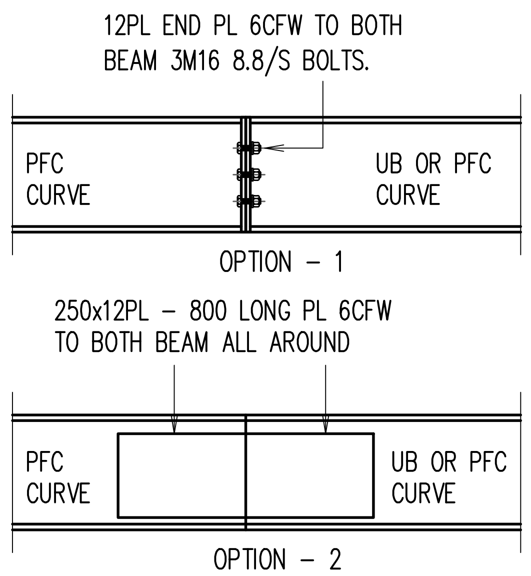
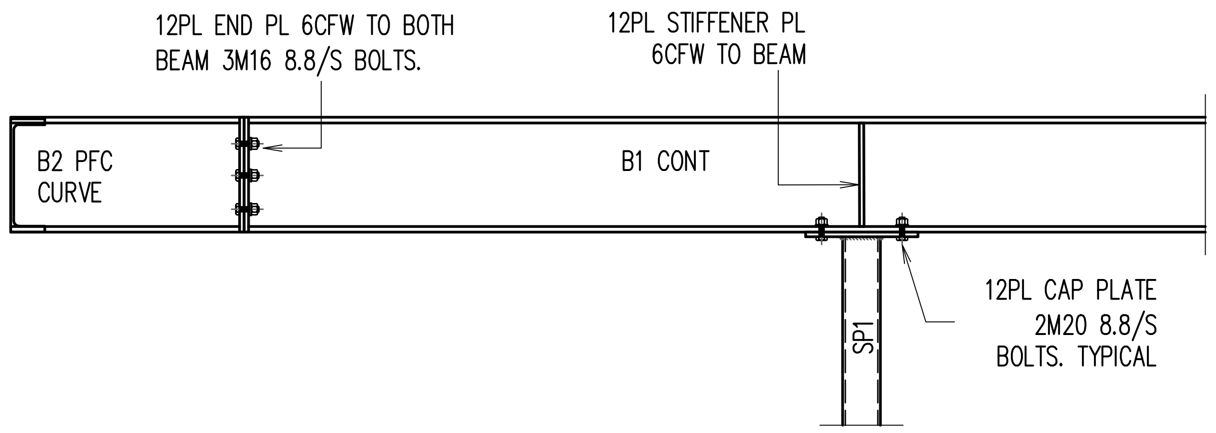


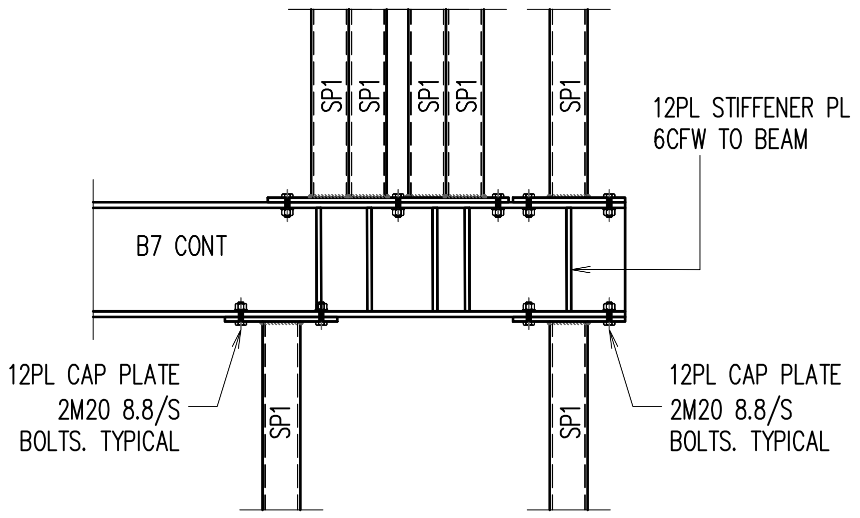
DETAIL – X3
SCALE 1:20



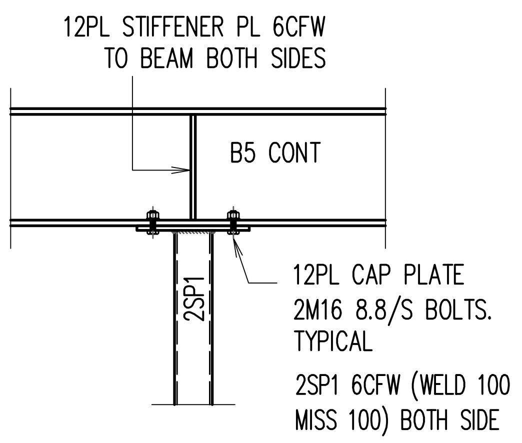
TYPICAL CURVE STEEL BEAM DETAIL



DETAIL – X4
SCALE 1:20



DETAIL – X5
SCALE 1:20



DETAIL – X6
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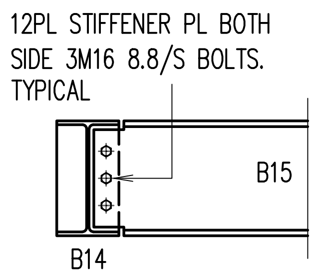
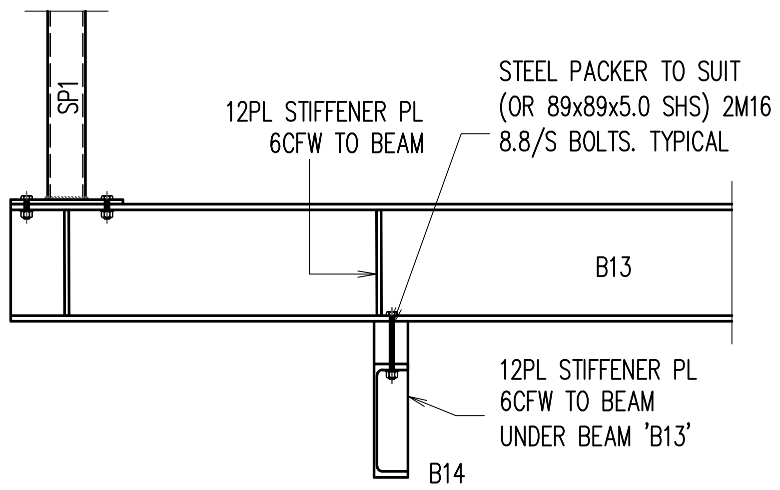
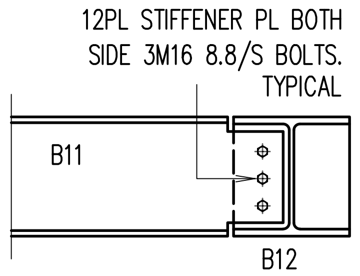


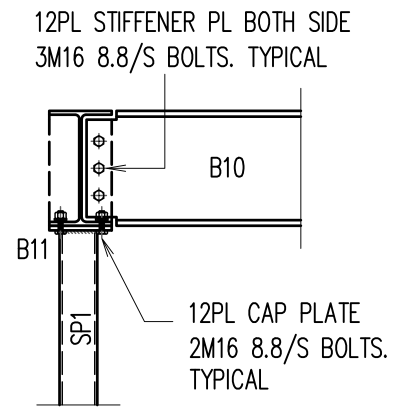
DETAIL – X7
SCALE 1:20



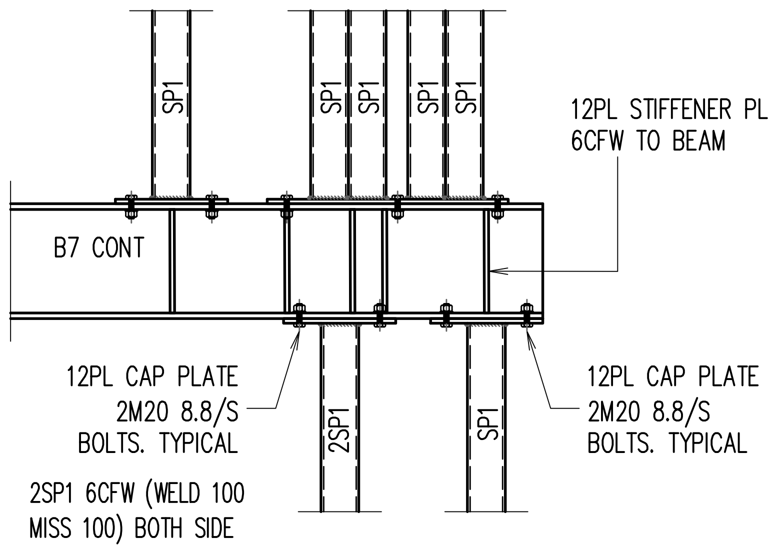
DETAIL – X8
SCALE 1:20



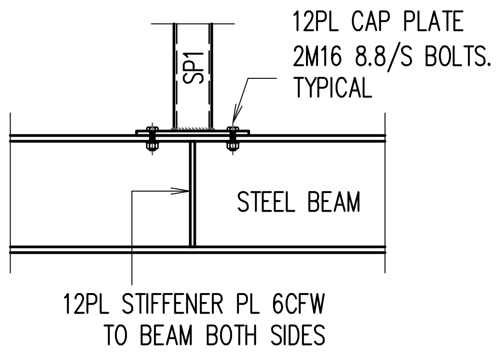
DETAIL – X9
SCALE 1:20



DETAIL – X10
SCALE 1:20



DETAIL – X11
SCALE 1:20



TYPICAL STEEL POST
OVER STEEL BEAM DETAIL

RAMESHH Sep 22, 2025 – 5:41PM G:\My Drive\ASE\ASE PROJECTS\2023\230608SD – Lot 515 (No. 18) Thorpe Way Box Hill\Structural\230608SD.dwg

ISSUE	DESCRIPTION	DATE	ISSUE	DESCRIPTION	DATE
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ASE
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LOT 515 (NO. 18) THORPE WAY
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STEEL WORK DETAILS

SCALE: 1:100, 1:20 U.N.O.

DESIGN BY:	DRAWN BY:	CHECK BY:	PAGE:
R.H.	C.V.	R.H.	7 OF 16

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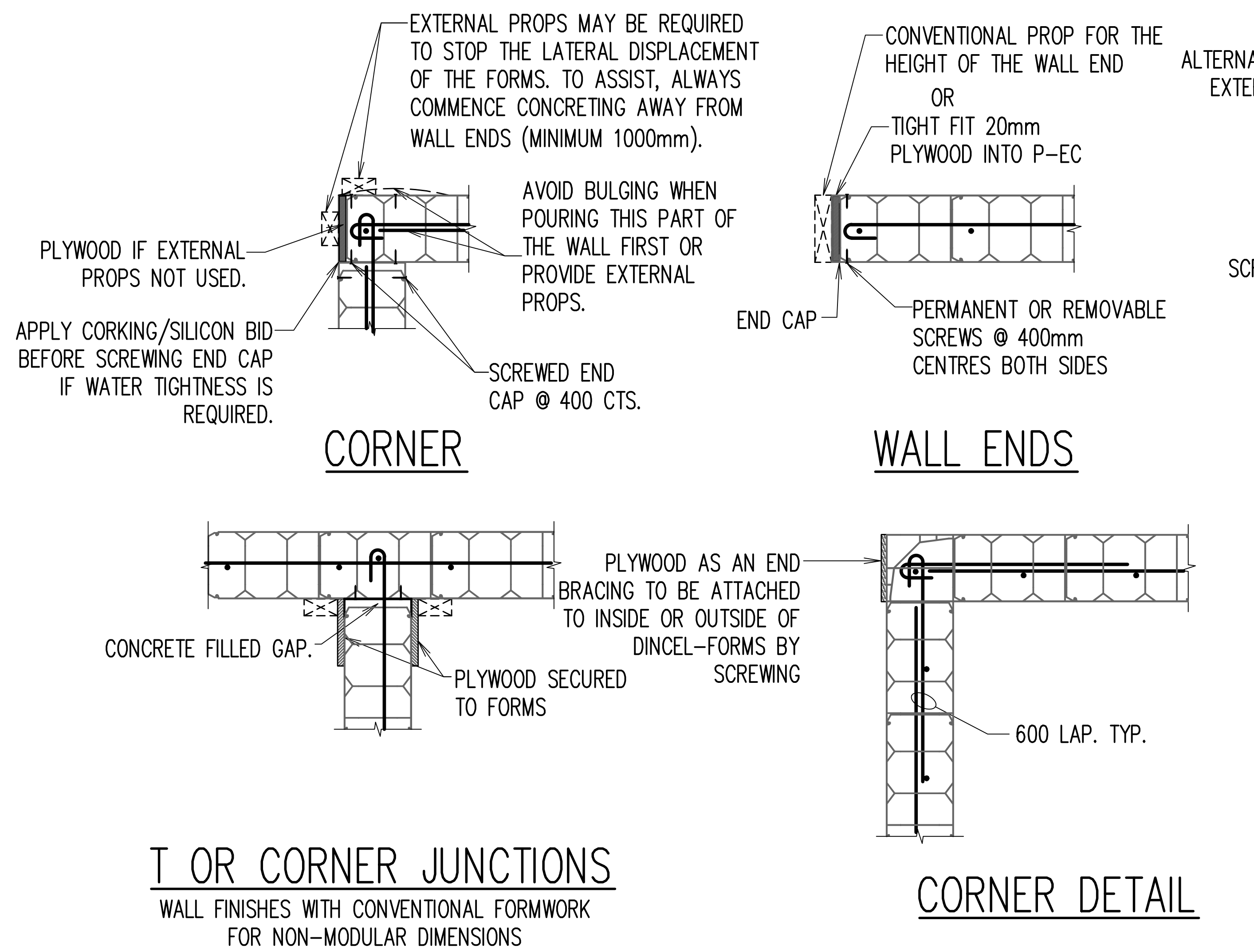
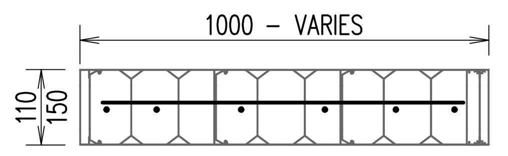


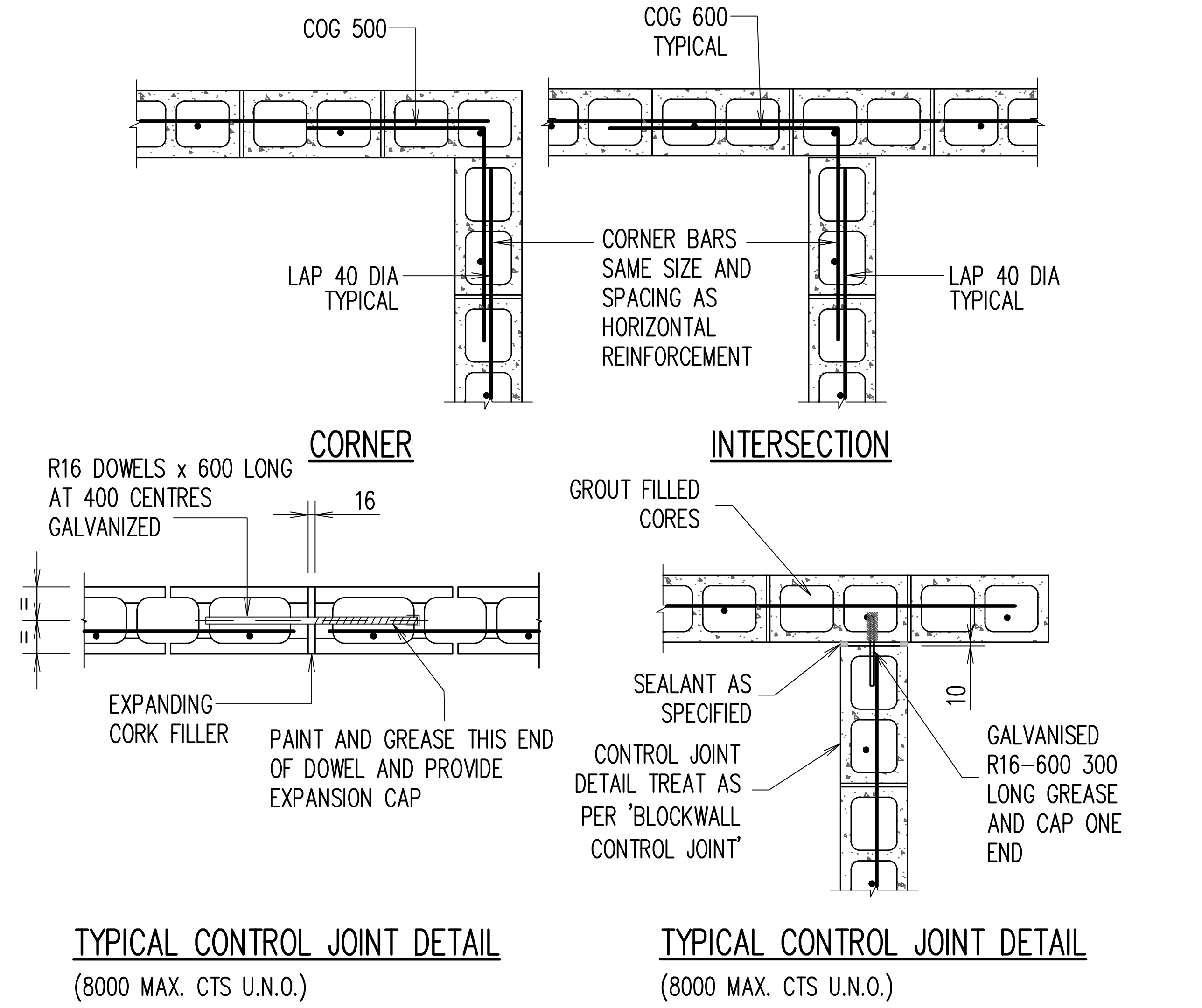
TYPICAL DINCEL WALL PLANS



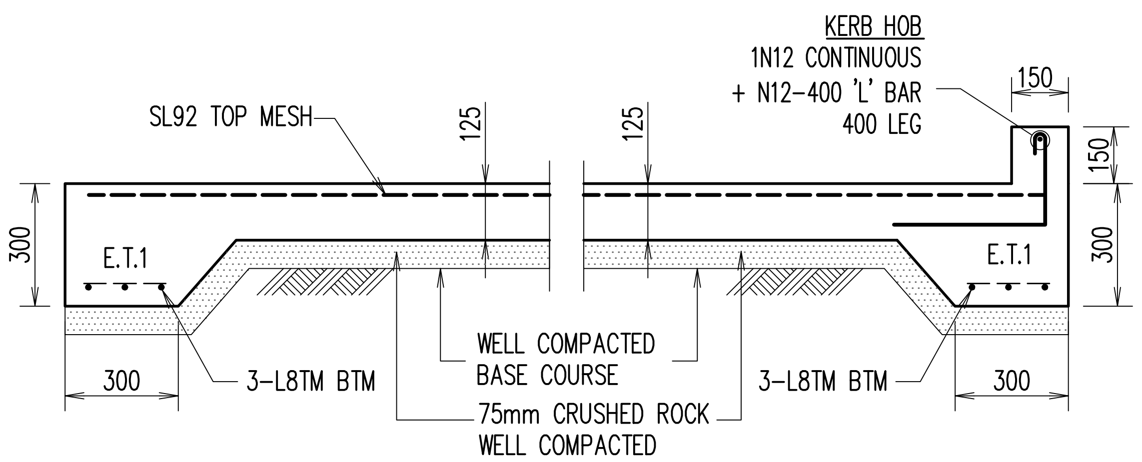
TYPICAL DINCEL WALL COLUMN DETAIL

WALL SCHEDULE

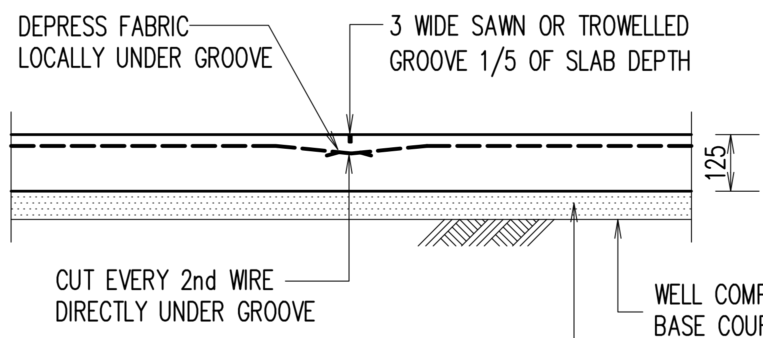
MARK	FLOOR LEVEL	THICKNESS (mm)	VERT. REINF.	HORZ. REINF.	f'c
DINCEL DW1	AS MARKED ON PLAN	200	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS1	AS MARKED ON PLAN	200	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW2	AS MARKED ON PLAN	150	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS2	AS MARKED ON PLAN	150	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW3	AS MARKED ON PLAN	110	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS3	AS MARKED ON PLAN	110	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa
DINCEL DW4	AS MARKED ON PLAN	300	N16-333 CENTRAL	N12-300 CENTRAL	32 MPa
OR AFS4	AS MARKED ON PLAN	300	N16-250 CENTRAL	N12-300 CENTRAL	32 MPa



TYPICAL BLOCKWALL PLANS

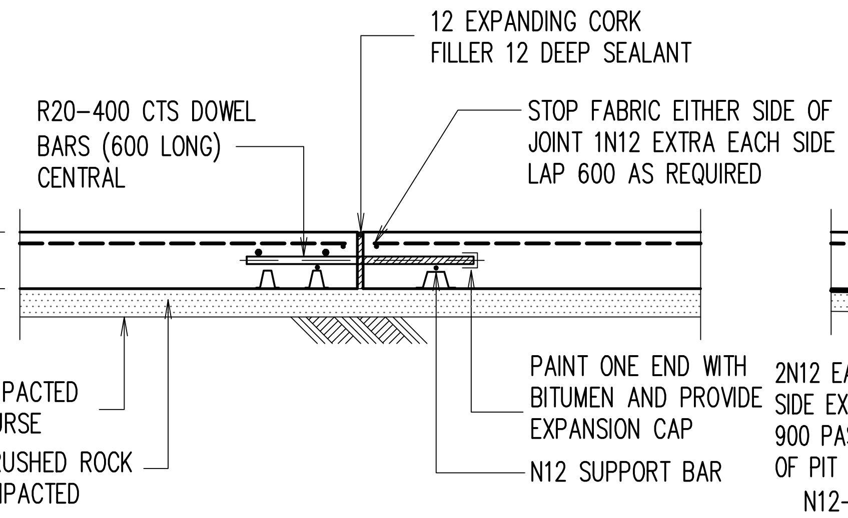


TYPICAL RAFT SLAB TO BE USED FOR DRIVEWAY & GARAGE SLAB

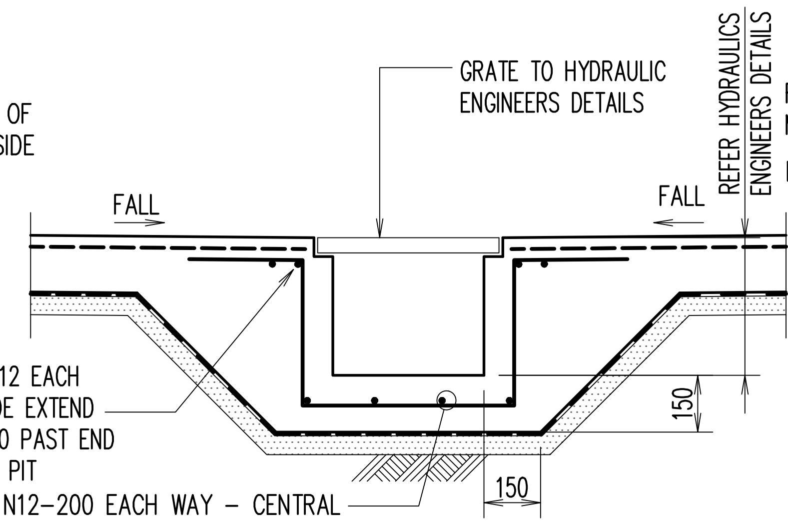


S.J. - SAWN CONTROL JOINT

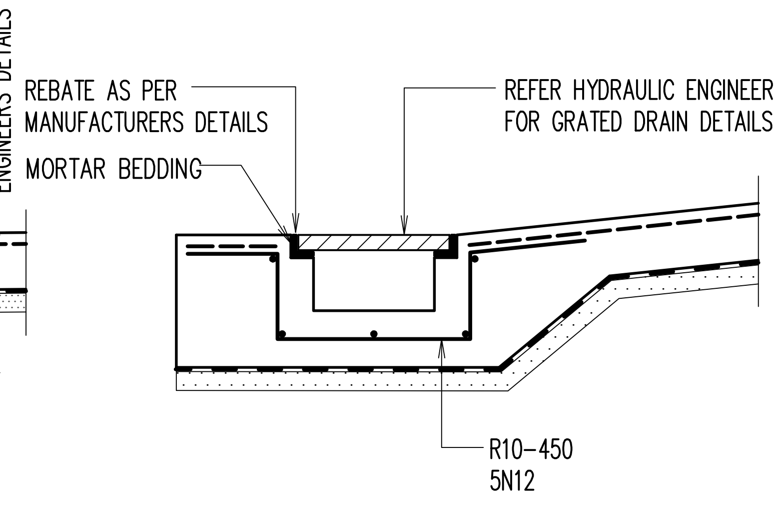
NOTE: SAWCUT TO BE MADE WITHIN 24 HOURS OF CASTING SLAB



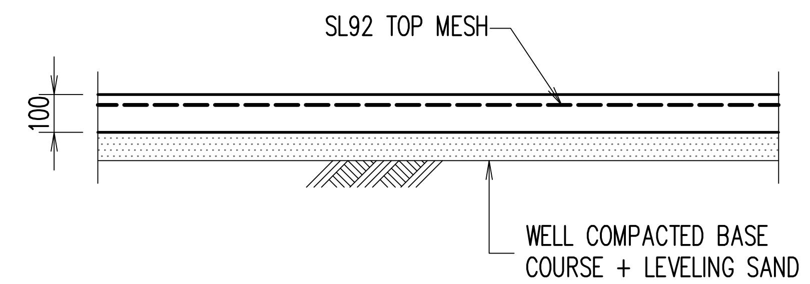
E.J. - EXPANSION JOINT



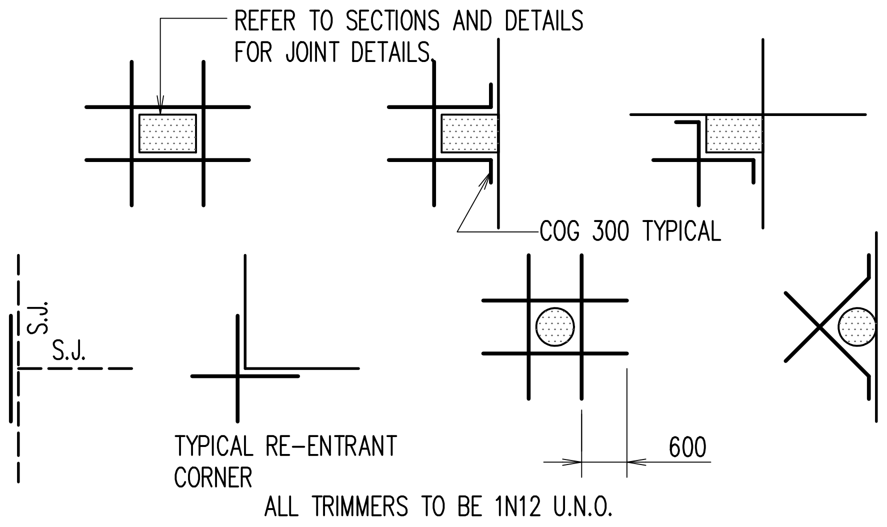
TYPICAL DRAIN PIT DETAIL
(TO BE USED AS REQUIRED)



TYPICAL GRATED DRAIN DETAIL
(TO BE USED AS REQUIRED)

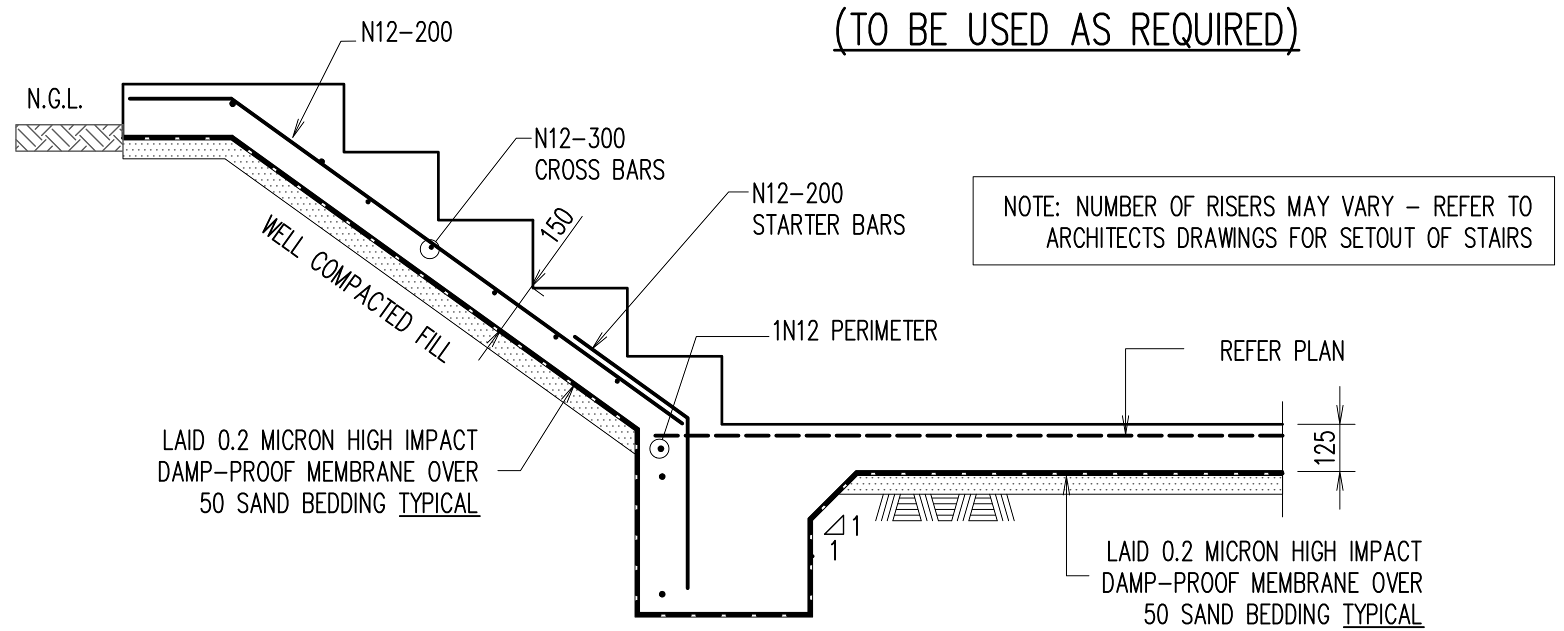


TYPICAL FOOTPATH SLAB



TYPICAL SLAB ON GROUND TRIMMER DETAILS

AT ALL COLUMNS, WALLS, PITS, FLOOR WASTES, ETC THAT CAUSE A PENETRATION THROUGH THE SLAB.



NOTE: NUMBER OF RISERS MAY VARY - REFER TO ARCHITECTS DRAWINGS FOR SETOUT OF STAIRS

TYPICAL STAIR DETAILS - SLAB ON GROUND STAIRS

A2 0 1 2 3 4 5 6 7 8 9 10

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DINCEL WALL DETAILS

SCALE: 1:100, 1:20 U.N.O.

DESIGN BY:	DRAWN BY:	CHECK BY:	PAGE:
R.H.	C.V.	R.H.	8 OF 16

JOB NUMBER: 230608SD
ISSUE: B

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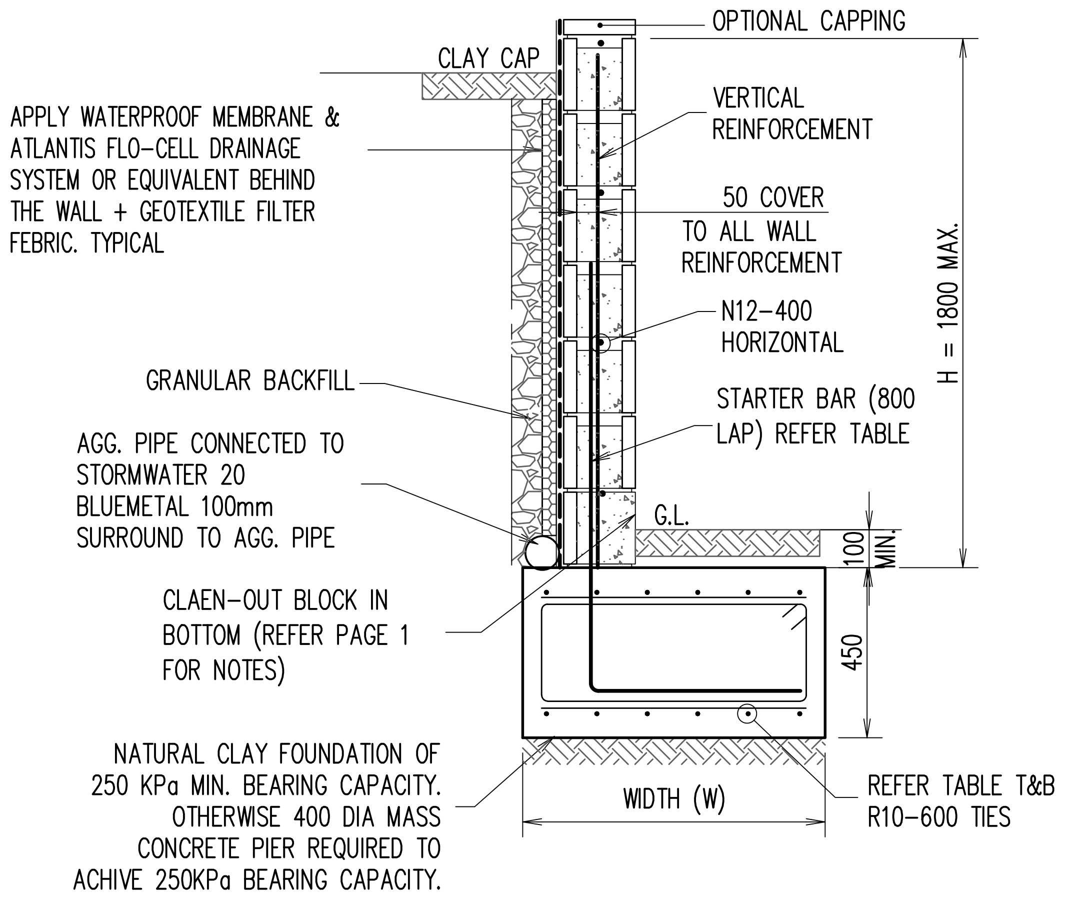
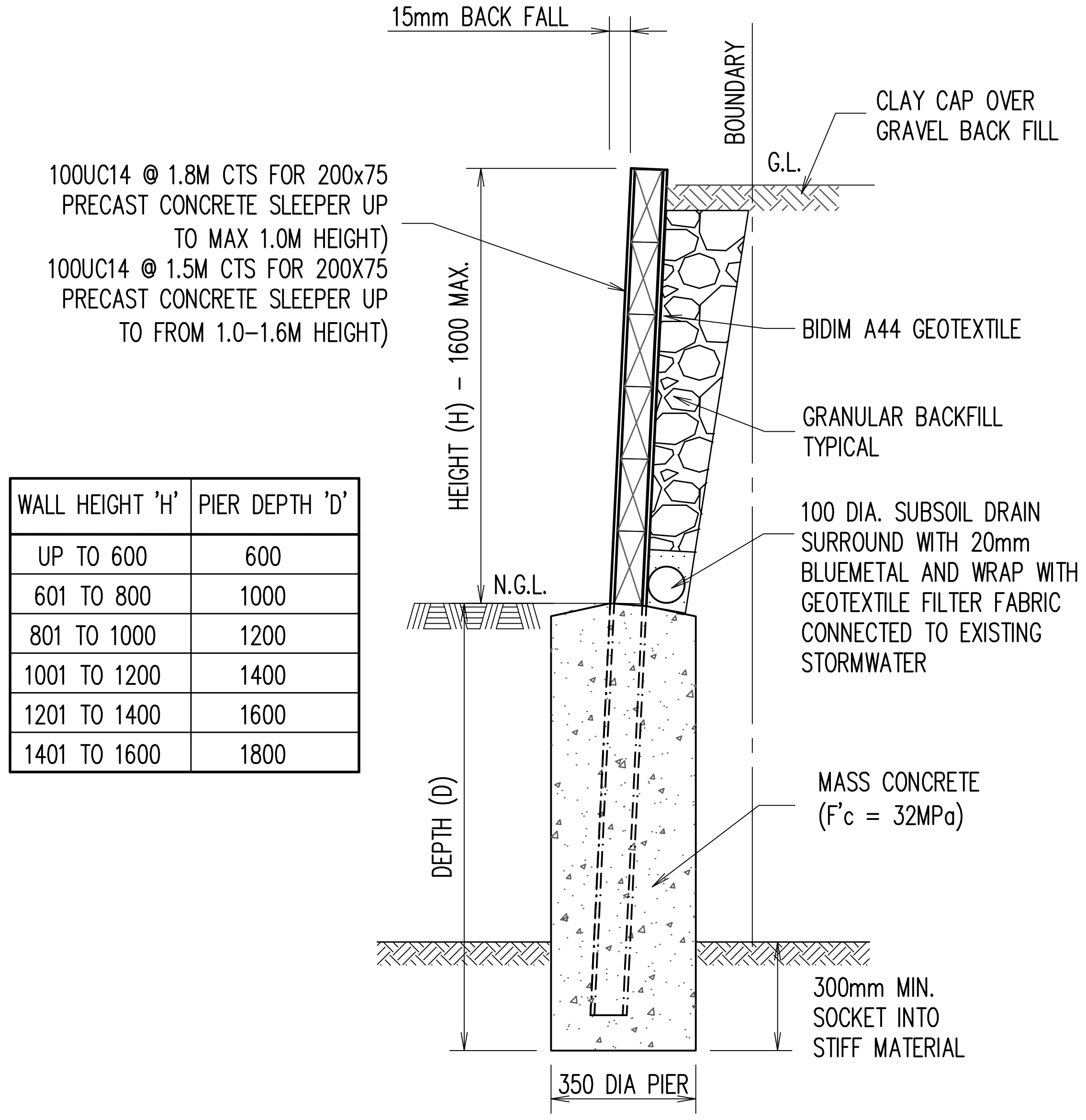
OPTION 2 – CONCRETE SLEEPER RETAINING WALL MAX 1.6M HIGH



TYPICAL BLOCK RETAINING WALL RW1

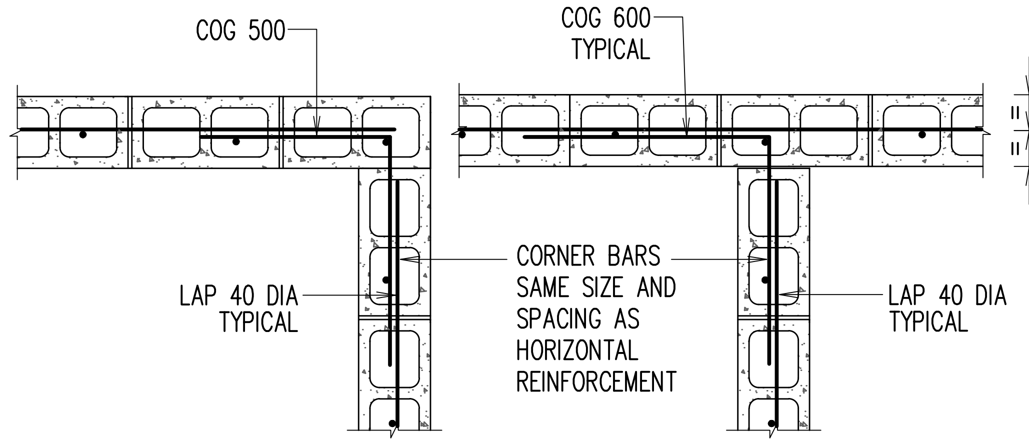
(TO BE USED AS REQUIRED)

RETAINING WALL – RW1 SCHEDULE			
RETAINING WALL HEIGHT	FOOTING WIDTH + REINFORCEMENT	STARTER BAR	VERTICAL BAR
UP TO 600mm	400MM + 3L11TM TOP & BOTTOM	N12-400 CTS	N12-400 CTS
601 – 1000mm	500MM + 4L11TM TOP & BOTTOM	N12-400 CTS	N12-400 CTS
1001 – 1200mm	600MM + 5L11TM TOP & BOTTOM	N12-400 CTS	N12-400 CTS
1201 – 1400mm	800MM + 2/3L11TM TOP & BOTTOM	N16-400 CTS	N16-400 CTS
1401 – 1600mm	1000MM + 2/4L11TM TOP & BOTTOM	N16-400 CTS	N16-400 CTS
1601 – 1800mm	1200MM + 2/5L11TM TOP & BOTTOM	N16-400 CTS	N16-400 CTS

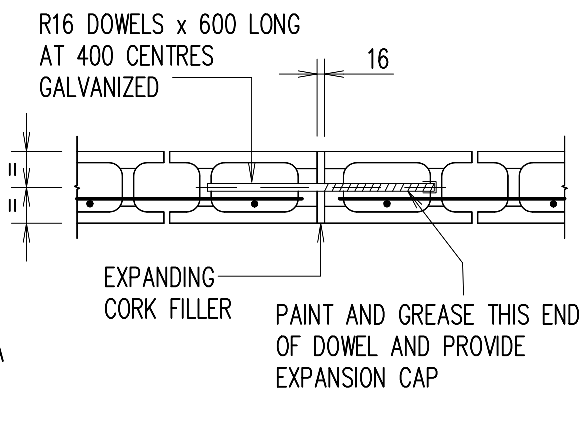


WALL HEIGHT 'H'	PIER DEPTH 'D'
UP TO 600	600
601 TO 800	1000
801 TO 1000	1200
1001 TO 1200	1400
1201 TO 1400	1600
1401 TO 1600	1800

TYPICAL RETAINING WALL 'RW2' DETAIL

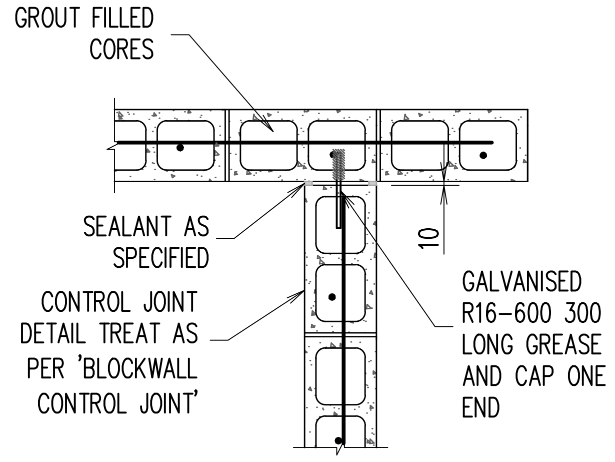


TYPICAL BLOCKWALL PLANS



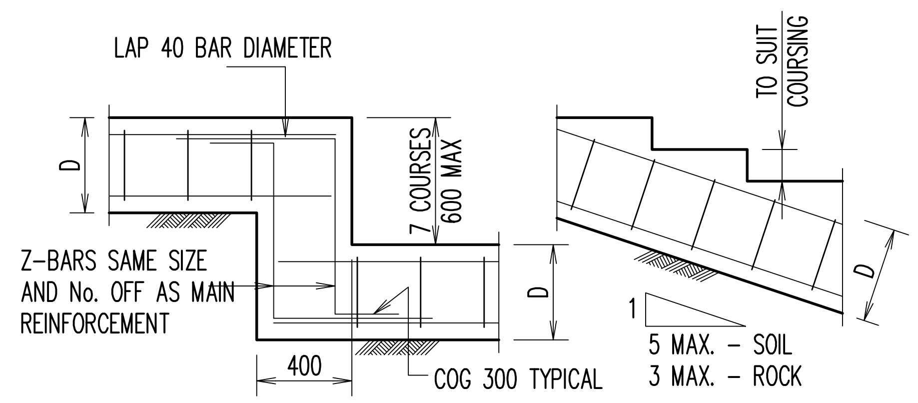
TYPICAL CONTROL JOINT DETAIL

(8000 MAX. CTS U.N.O.)



TYPICAL CONTROL JOINT DETAIL

(8000 MAX. CTS U.N.O.)



TYPICAL STEP DETAILS (IF REQUIRED)

'D' DENOTES DEPTH OF FOOTING

ISSUE	DESCRIPTION	DATE	ISSUE	DESCRIPTION	DATE
B	ISSUED FOR CC	22.09.25			
A	ISSUED FOR CC	20.08.25			

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PROPOSED RESIDENCE
LOT 515 (NO. 18) THORPE WAY
BOX HILL, NSW
FOR –

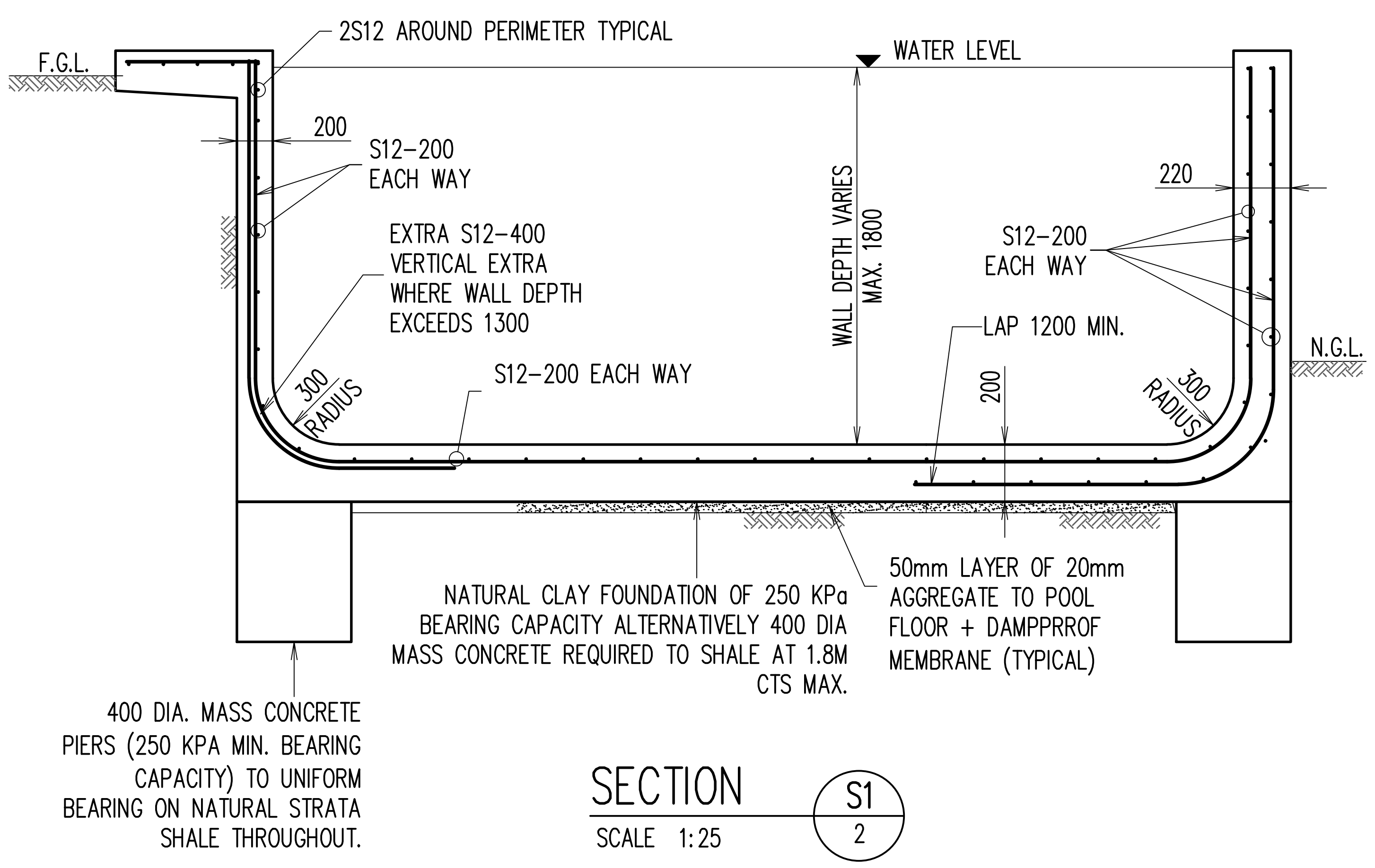
BUILDER
DEZIRE HOMES

RETAINING WALL DETAIL

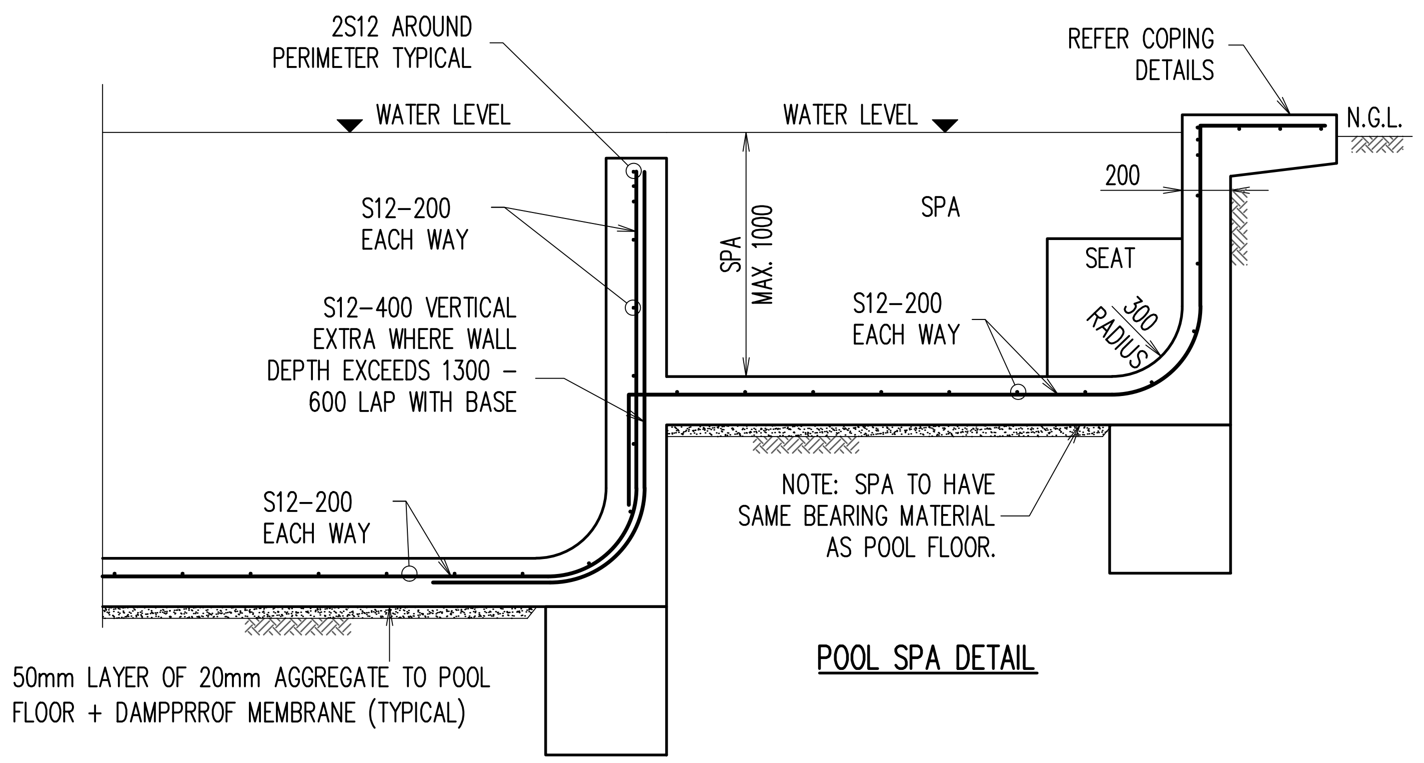
SCALE: 1:100, 1:20 U.N.O.

DESIGN BY:	DRAWN BY:	CHECK BY:	PAGE:
R.H.	C.V.	R.H.	9 OF 16

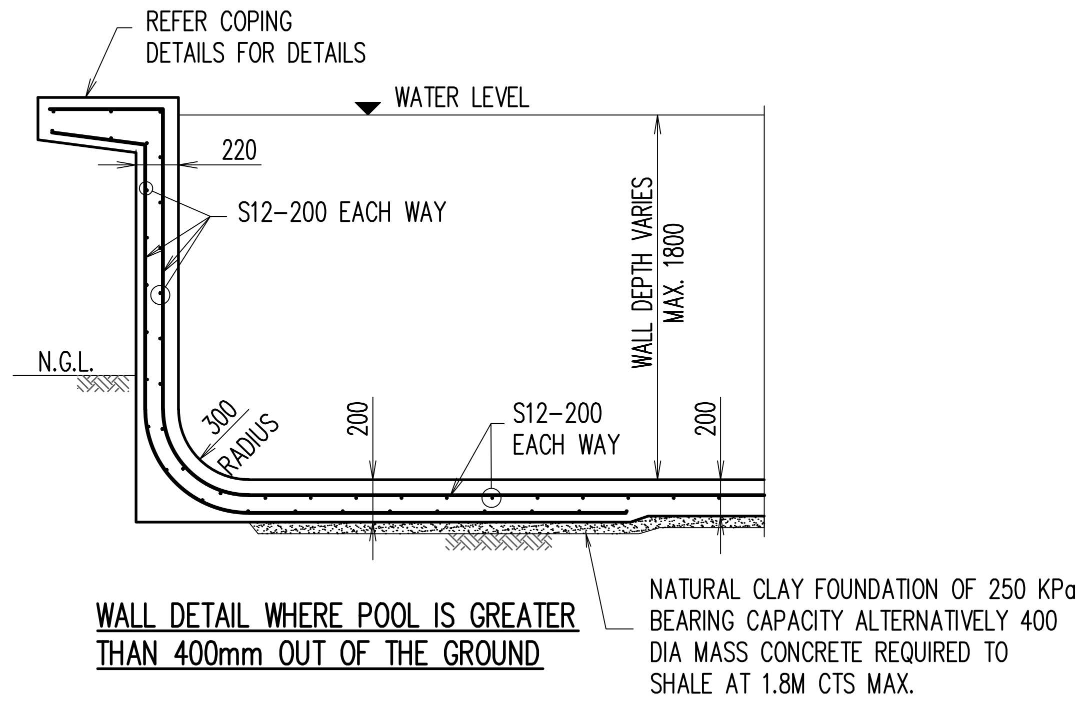
JOB NUMBER: 230608SD
ISSUE: B



SECTION S1 / 2
SCALE 1:25



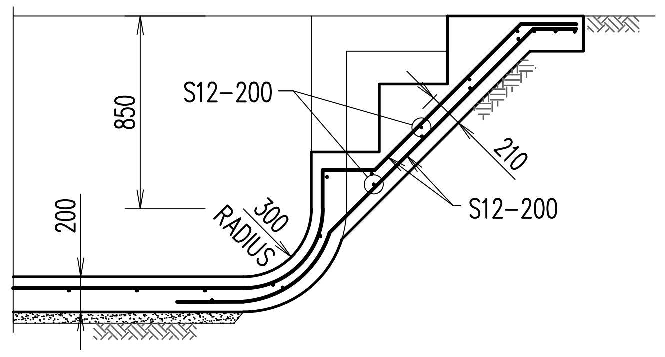
POOL SPA DETAIL



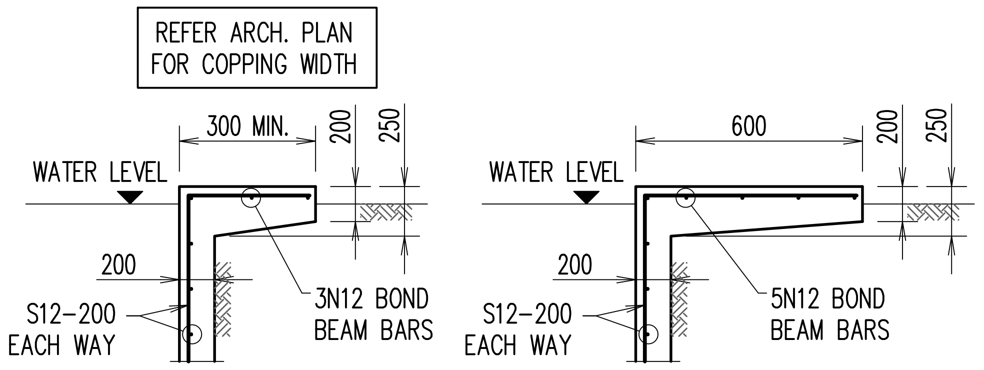
WALL DETAIL WHERE POOL IS GREATER THAN 400mm OUT OF THE GROUND

SECTION S3 / 2
SCALE 1:25

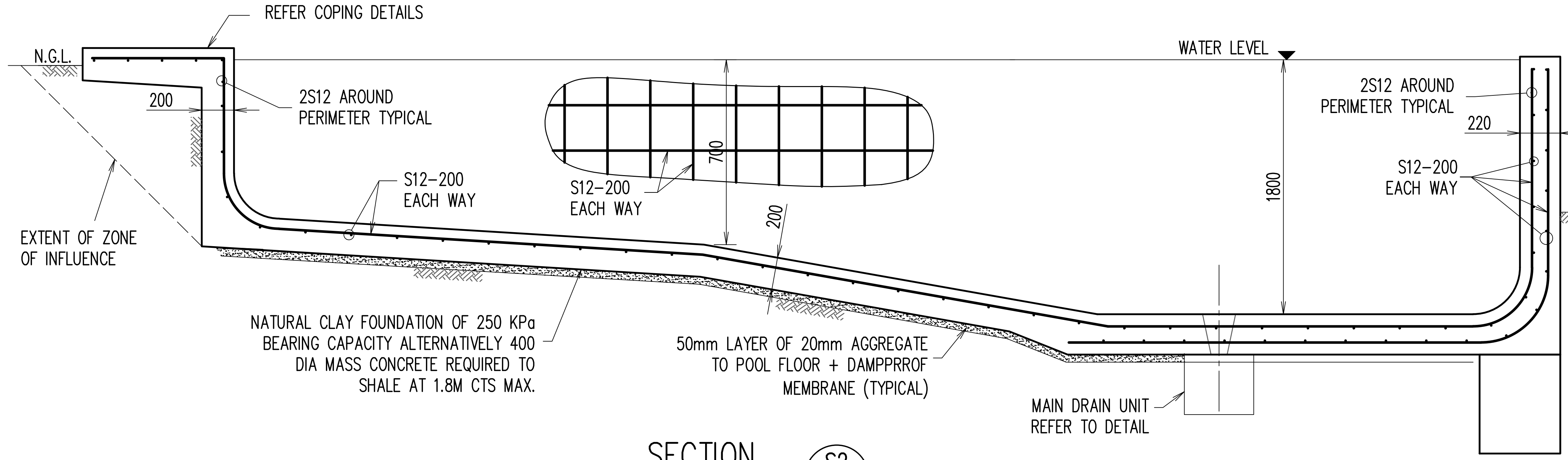
THIS DRAWINGS IS SIGNED SUBJECT TO CERTIFICATE OF INSPECTION ISSUED BY AN ENGINEER FROM THIS OFFICE. ALL FOOTING AND REINFORCEMENT SHALL BE INSPECTED BY THIS OFFICE PRIOR TO PLACING CONCRETE.



SECTION THROUGH STEPS



COPING DETAIL



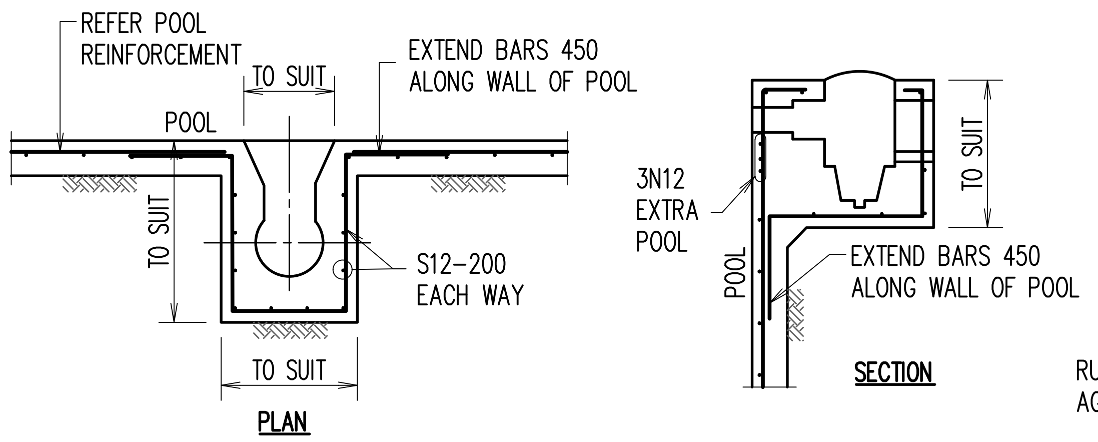
SECTION S2 / 2
SCALE 1:50

POOL SPECIFICATIONS

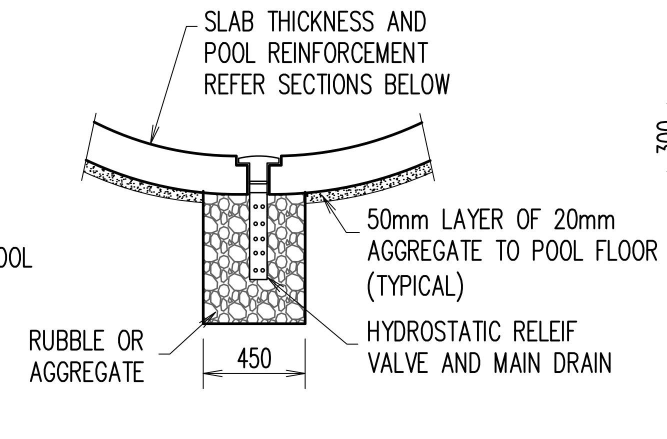
1. ALL WORKMANSHIP AND MATERIALS TO BE IN ACCORDANCE WITH AUSTRALIAN STANDARD AS2783-1992, 'USE OF REINFORCEMENT FOR SMALL SWIMMING POOLS'
2. DIMENSIONS SHALL NOT BE OBTAINED BY SCALING THE DETAILS.
3. PROVIDE FILTER WITH MATCHED PUMP AND PLUMBING TO MANUFACTURES RECOMMENDATIONS.
4. SUPPORTING FOUNDATION MATERIAL TO BE STABLE SAND NATURAL OF UNIFORM MOISTURE CONTENT WITH SAFE BEARING CAPACITY OF 150kPa.
5. ADVISE ENGINEER IF EXCAVATION IN FILL OR GROUND WATER IS ENCOUNTERED. PROVIDE TEMPORARY PENETRATIONS TO FLOOR SLAB IF GROUND WATER LEVEL EXCEEDS 500mm ABOVE DEEP FLOOR LEVEL.
6. THE EXCAVATION BASE IS TO BE PROVIDED WITH AN UNDER SHELL DRAINAGE LAYER AS FOLLOWS:
   A). 75 MIN. BLUE METAL DRAINAGE LAYER, OR 50 MIN. THICK LAYER WITH PLASTIC COVER.
   B). CORRUGATED IRON SHEETING & MEMBRANE IF OVER ROCK.
   C). PLASTIC LAYER ONLY IF BASE IS ENTIRELY IN SAND.
   D). MAIN DRAIN PIT IS TO BE BLUE METAL FILLED IRRESPECTIVE OF DRAINAGE LAYER TYPE.
7. ALL REINFORCEMENT TO BE OF AUSTRALIAN MANUFACTURE IN ACCORDANCE WITH AS/NZS4671 & SAA STANDARDS.
   N - GRADE 500 NORMAL DUCTILITY REINFORCING BARS.
   R - PLAIN ROUND GRADE 250 NORMAL DUCTILITY REINFORCING BAR.
   SL- DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH.
   RL- DEFORMED GRADE 500 LOW DUCTILITY REINFORCING MESH.
   S - GRADE 230 STRUCTURAL GRADE REINFORCING BARS.
   L-TM- DENOTES DEFORMED GRADE 500 LOW DUCTILITY TRENCH MESH.
8. REINFORCING BARS ARE TO BE LAPPED 40 BAR DIA. MIN., FABRIC TO BE LAPPED 400mm MIN. ALL LAPS SHOULD PREFERABLY BE STAGGERED. U.N.O.
9. WATER FACE REINFORCEMENT TO HAVE 65mm CONCRETE COVER, REAR FACE REINFORCEMENT TO HAVE 50mm MINIMUM COVER FROM REAR FACE IF FORMED AND 65mm MINIMUM COVER IF SPRAYED AGAINST GROUND.
10. CONCRETE TO BE PNEUMATICALLY PLACED, HAVE MIN. DESIGN STRENGTH OF F'c 32 MPa AT 28 DAYS.
11. UPON COMPLETION OF CONCRETE THE HYDROSTATIC VALVE IS TO BE CLEANED & CHECKED TO ENSURE CORRECT OPERATION.

POOL NOTES

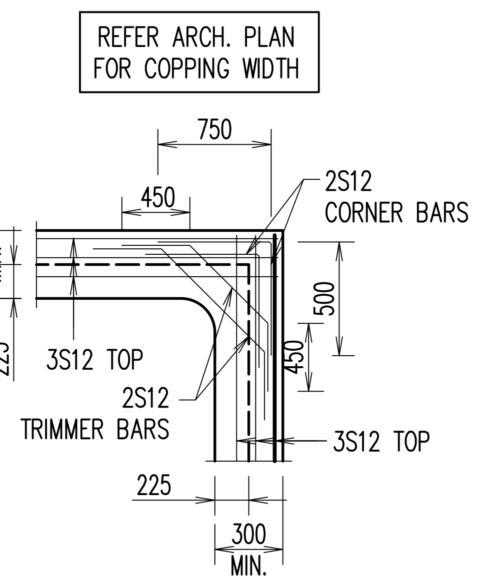
1. POOL SET-OUT SIZE, LOCATION AND HEIGHT LEVEL IS DEEMED TO BE ACCEPTABLE TO THE OWNER UNLESS THE BUILDER IS ADVISED OTHERWISE. SUCH ADVICE MUST BE PRIOR TO PLACEMENT OF REINFORCEMENT. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE ARCHITECTURAL DRAWINGS.
2. AFTER CONCRETING, THE POOL SHELL IS TO BE THOROUGHLY WETTED DOWN TWICE DAILY FOR AT LEAST SEVEN DAYS (TEN DAYS IN SUMMER).
3. SAFETY FENCING IS TO BE COUNCIL APPROVED PRIOR TO THE POOL BEING FILLED.
4. LIGHTS MUST BE FULLY SUBMERGED DURING USE.
5. PLUMBING IS TO BE IN ACCORDANCE WITH WRITTEN RECOMMENDATIONS OF FILTER MANUFACTURER.
6. WALKWAYS AND COPINGS ARE DESIGNED FOR A 2 KPa LIVE LOAD AND ARE NOT DESIGNED TO SUPPORT MASONRY WALLS U.N.O.



SKIMMER BOX DETAIL



MAIN DRAIN DETAIL



BOND BEAM CORNER DETAIL

A2 0 1 2 3 4 5 6 7 8 9 10

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BOX HILL, NSW
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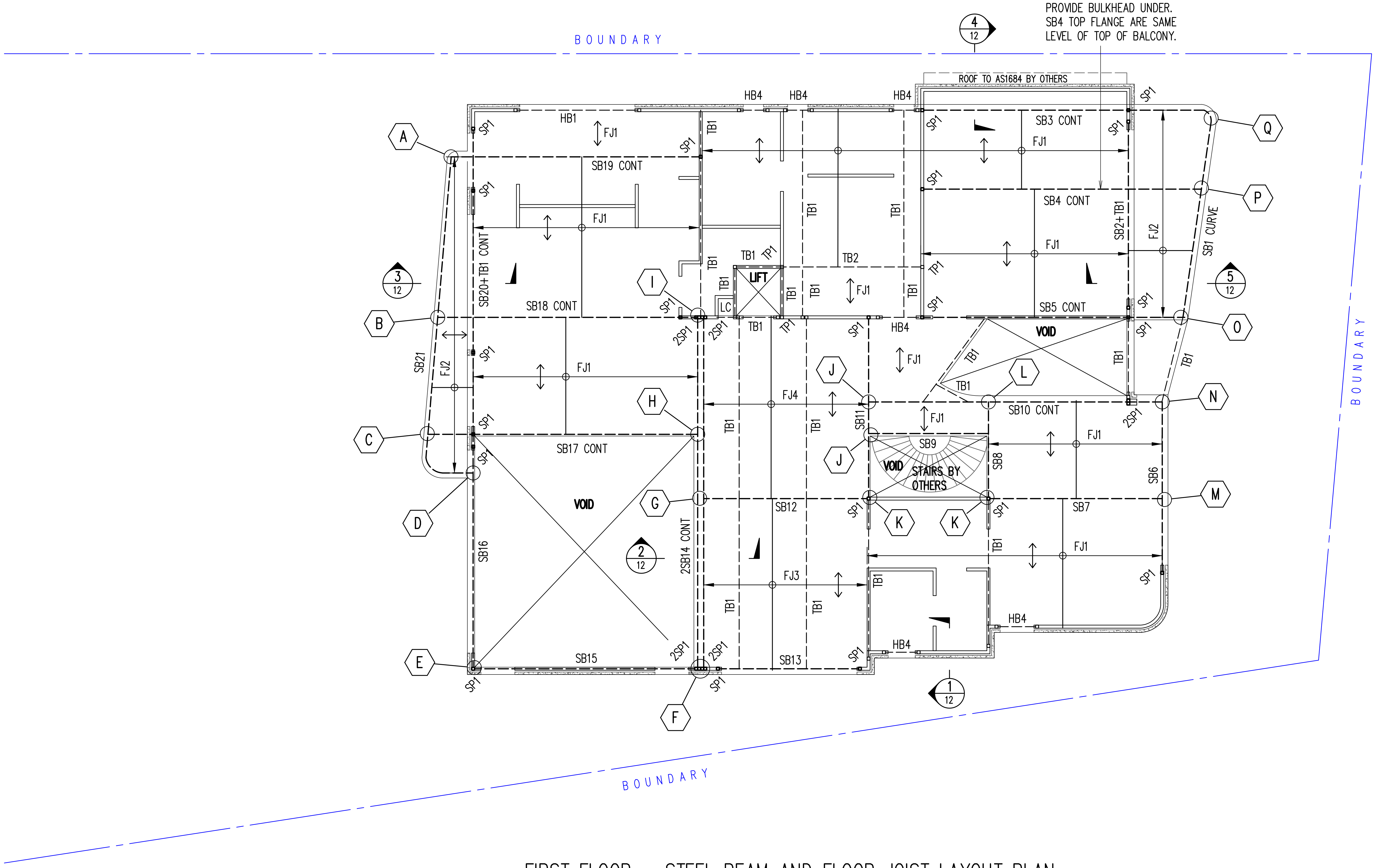
BUILDER
DEZIRE HOMES

IN-GROUND POOL DETAILS

SCALE:	1:100, 1:20 U.N.O.		
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 10 OF 16
JOB NUMBER: 230608SD			ISSUE: B



1. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
2. PROVIDE ARTICULATION JOINTS – LOCATION & DETAIL AS PER HEBEL MANUFACTURE DETAILS.

LEGEND

- DENOTES BRICK WALL UNDER
- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL UNDER
- DENOTES FLOOR JOIST DIRECTION

NOTE:

1. IF IN DOUBT FOR LINTEL SIZES OVER REMAINING WINDOW/DOOR OPENINGS CONTACT STRUCTURAL ENGINEER.
2. FIRST FLOOR PARAPET WALLS AND SUPPORT DETAILS BY OTHERS.
3. FLOOR JOIST ARE NOT DESIGN TO TAKE ROOF LOAD, FOR CONVENTIONAL ROOF, CARPENTER NEED TO PROVIDE TIMBER BEAM AS PER AS1684.
4. TIMBER TRUSS ROOF AND WALLS BRACING AND TIE DOWN DETAILS BY OTHERS & COMPLY WITH RELEVANT SECTIONS OF AS1684.

MEMBER SCHEDULE

MARK	SIZE	COMMENTS
SB1,SB2	300 PFC	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB3,SB4,SB5	310UB46.2	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB6,SB13	300 PFC	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB7,SB8,SB9	310UB46.2	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB10	310UC96.8	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB11,SB12	310UB46.2	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB14	2x310UC158	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB15,SB16	300 PFC	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB17,SB18,SB19	310UB46.2	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SB20,SB21	300 PFC	300 GRADE + Ø 10 DIA HOLES @ 1.0M CTS
SP1	89 x 89 x 5.0 SHS	350 GRADE
FJ1	HJ 300 63 HYJOIST	AT 450 CTS OR MANUFACTURES DETAILS
FJ2	190 X 45 MGP10 OR F7 LVL	AT 450 CTS
FJ3	HJ 300 63 HYJOIST	AT 350 CTS OR MANUFACTURES DETAILS
FJ4	HJ 300 63 HYJOIST	AT 300 CTS OR MANUFACTURES DETAILS
TB1,HB1	300 x 63 HYSPAN	TIMBER BEAM
TB2	2/300 X 63 HYSPAN	M10 4.6/S @ 450 CTS TOP & BOTTOM
HB2	2/190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
HB3	240 X 63 HYSPAN	TIMBER BEAM
HB4	190 X 45 MGP10 OR F7	TIMBER HEADER BEAM
TB,HB	STRUCTURAL TIMBER BEAM	TO MANUFACTURES DETAILS
TP1	STRUCTURAL TIMBER POST	TO MANUFACTURES DETAILS

FIRST FLOOR – STEEL BEAM AND FLOOR JOIST LAYOUT PLAN

SCALE 1:100

1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
3. PROVIDE ADEQUATE TIMBER BEAM UNDER ALL WALLS RUNNING PARALLEL TO JOISTS AND ROOF POINT LOAD FOR CUT ON SITE FRAME.
4. TB TO BE PROVIDED ON OVER ALL OPENINGS
5. PROVIDE DOUBLE STUD UNDER ALL TIMBER AND HEADER BEAM.

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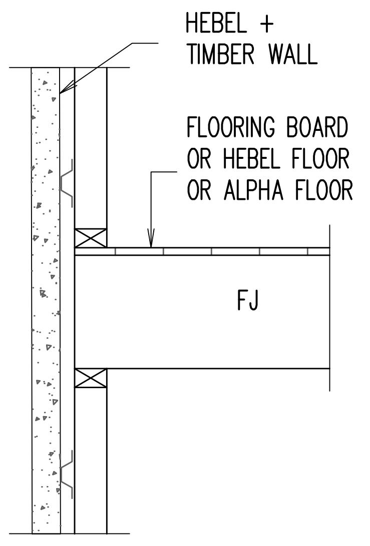
DEZIRE HOMES

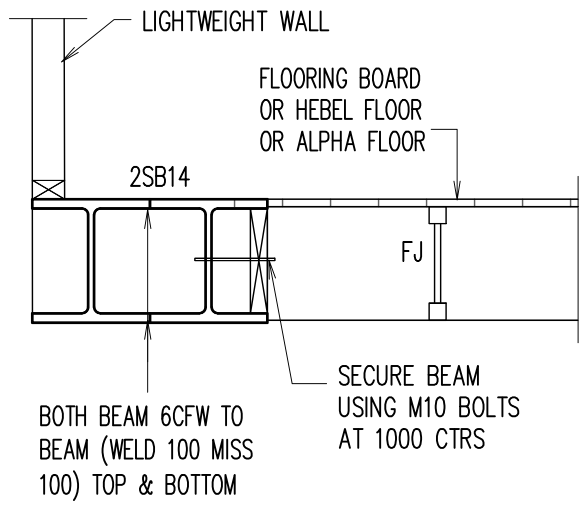
FIRST FLOOR STEEL/JOIST PLAN

SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 11 OF 16
JOB NUMBER: 230608SD			ISSUE: B

A2



SECTION 1 (11)
SCALE 1:20



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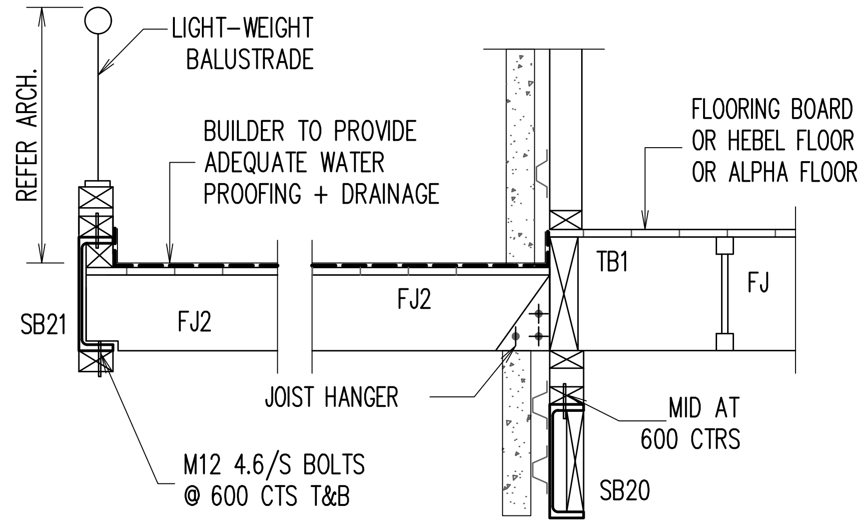

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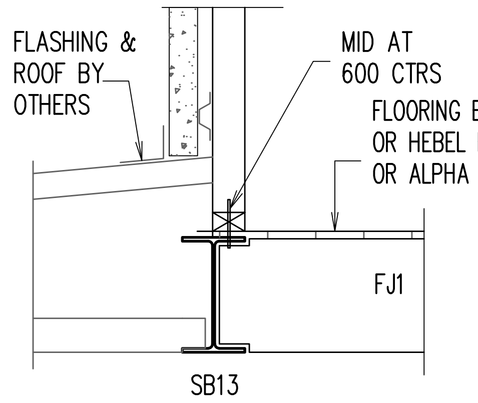

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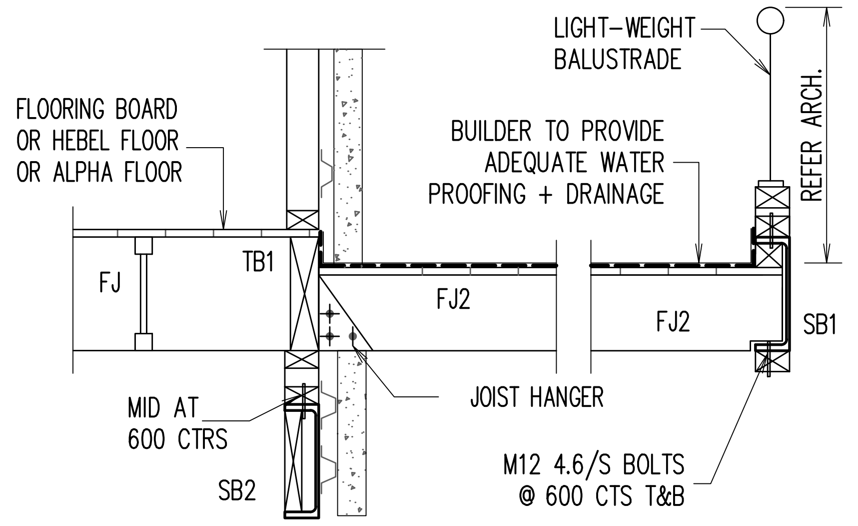

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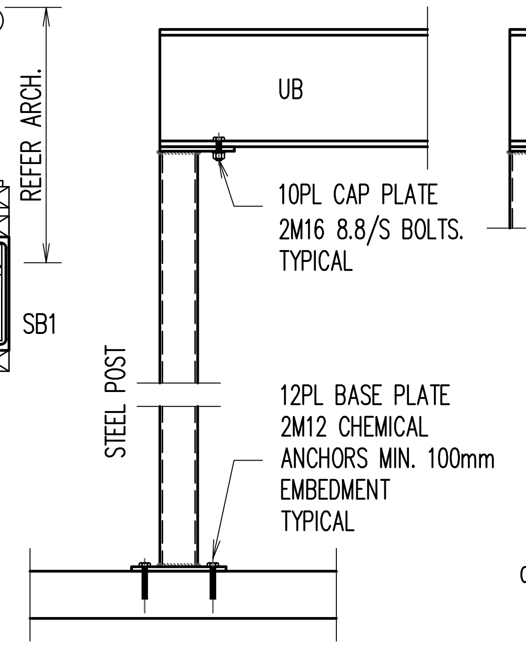
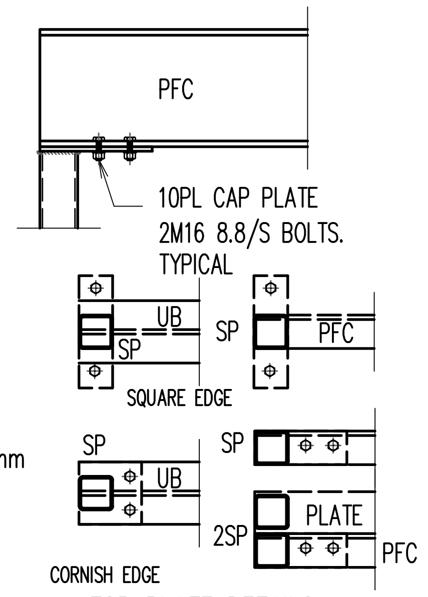
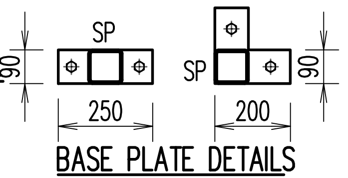


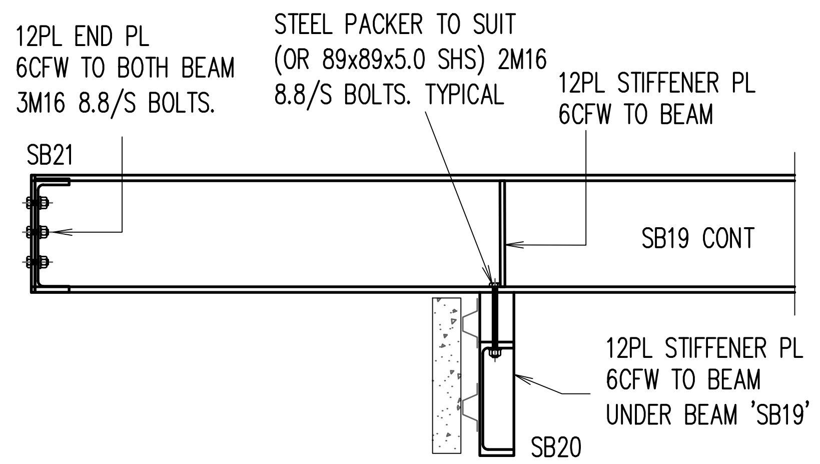
TYPICAL POST TO SLAB DETAIL
NOTE: POSTS TO BE SECURELY BRACED TO STUD FRAMEWORK



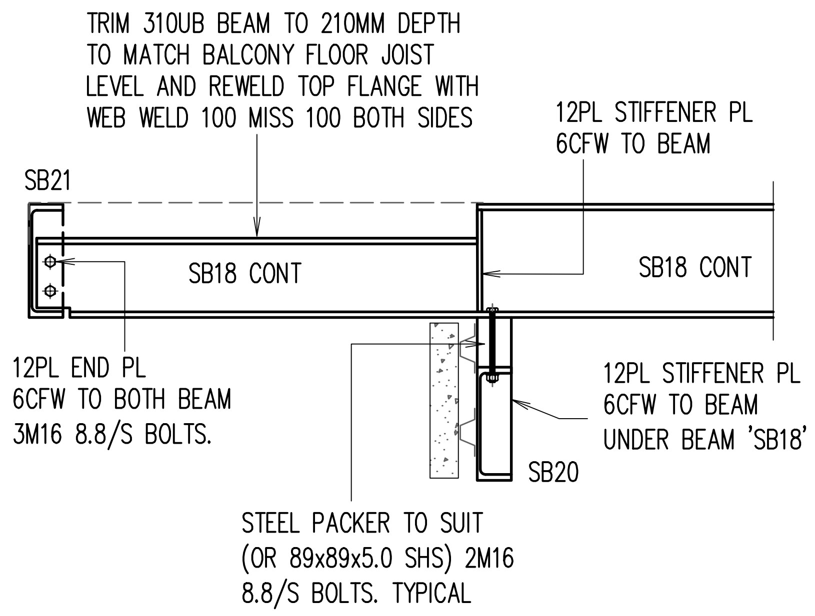
TOP PLATE DETAILS



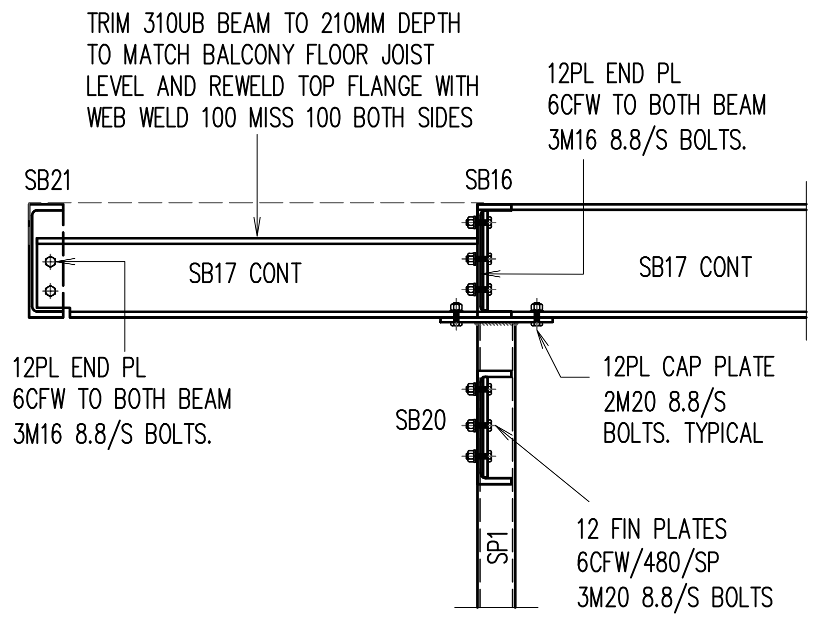
BASE PLATE DETAILS



DETAIL – A
SCALE 1:20



DETAIL – B
SCALE 1:20



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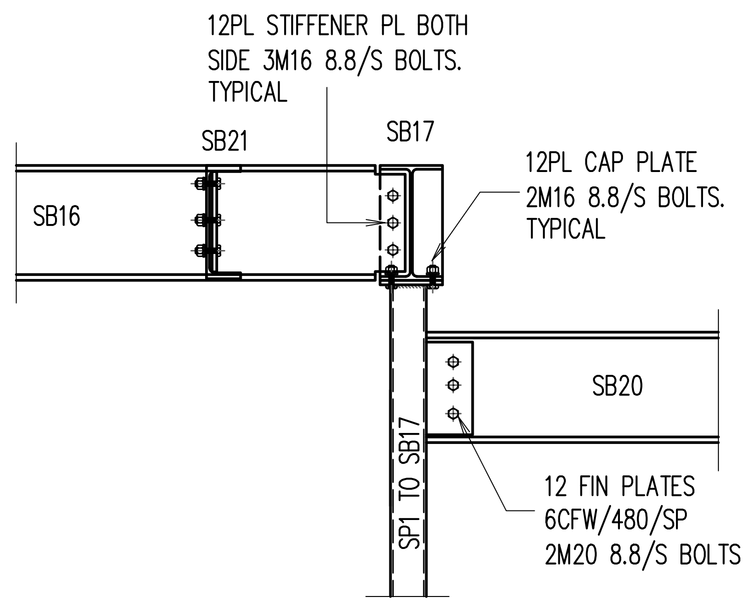

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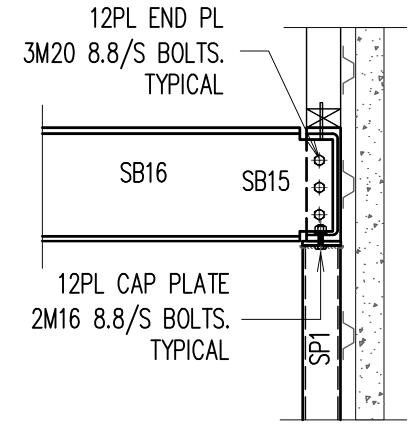

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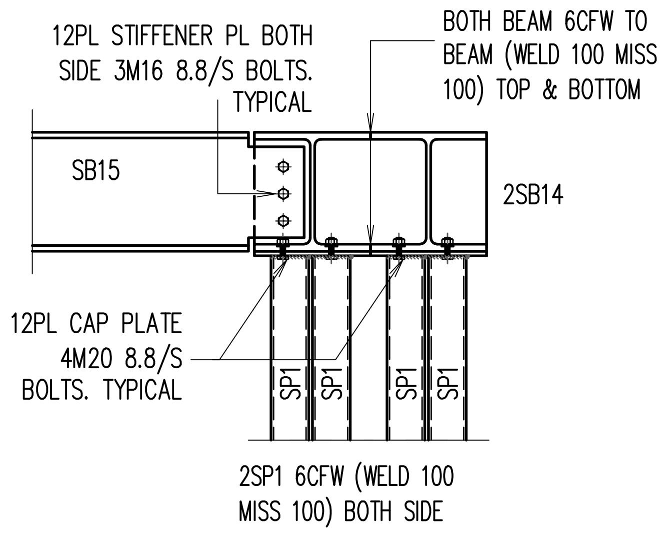

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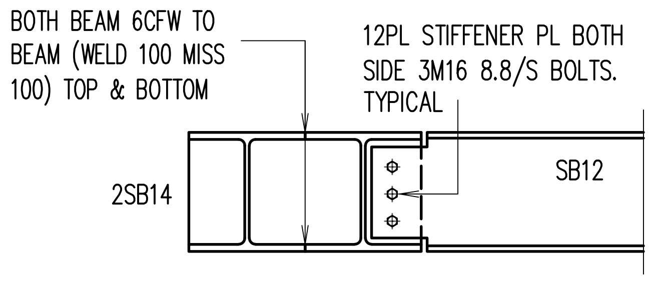
SCALE 1:20



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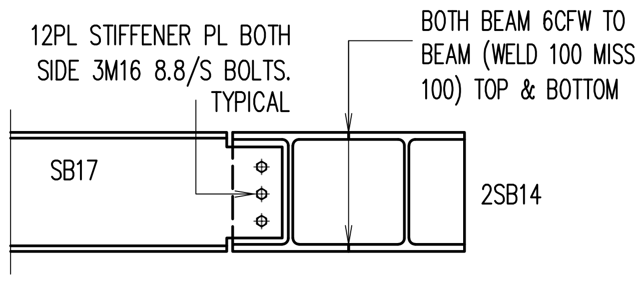
SCALE 1:20



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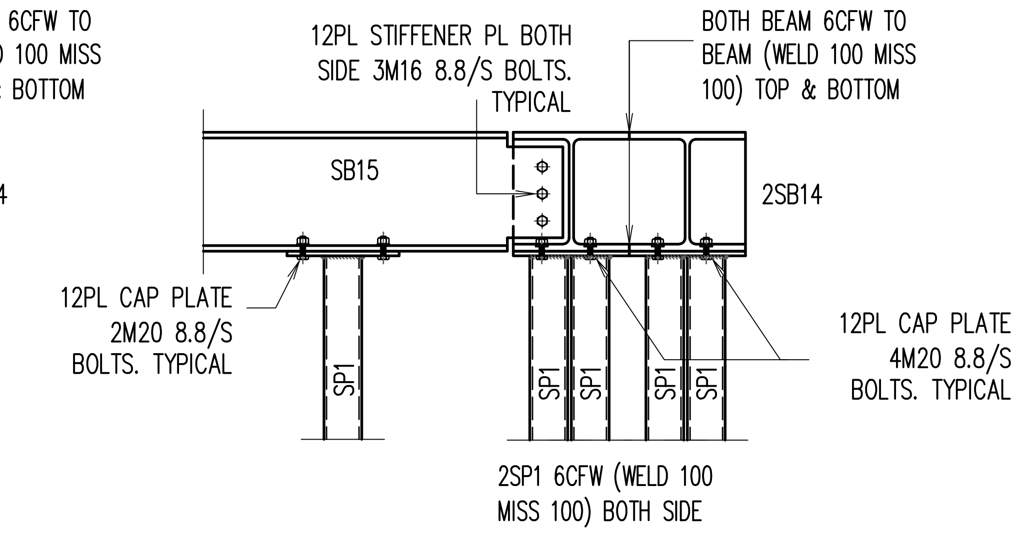

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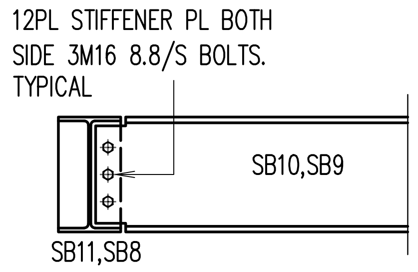
SCALE 1:20



DETAIL – J
SCALE 1:20

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STEEL WORK DETAILS

SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 12 OF 16
JOB NUMBER: 230608SD			ISSUE: B

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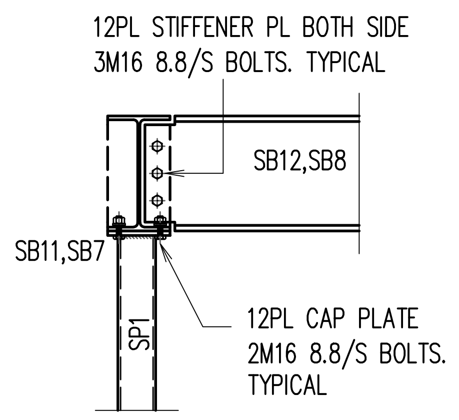

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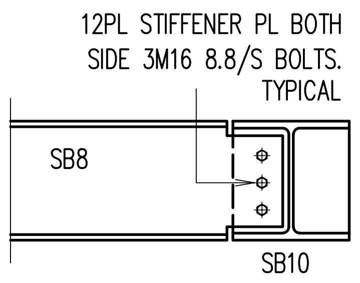

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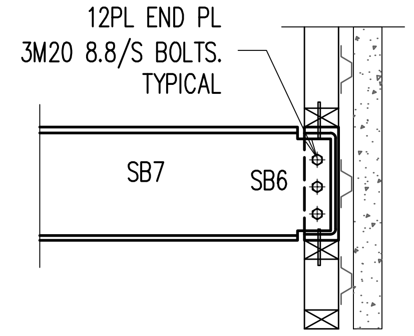

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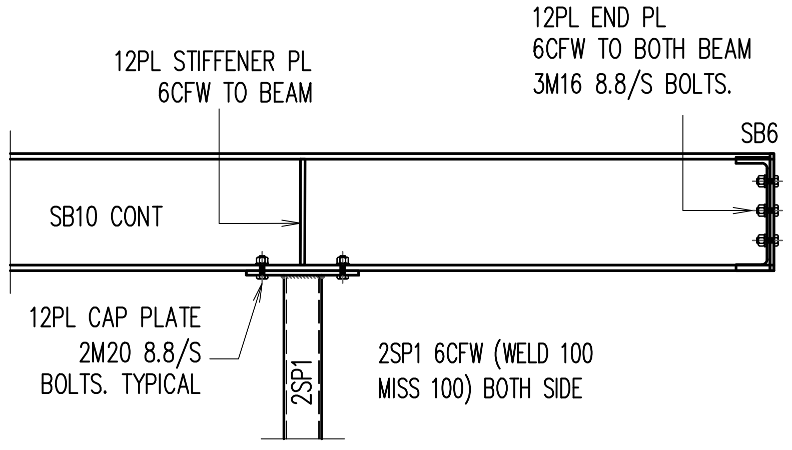

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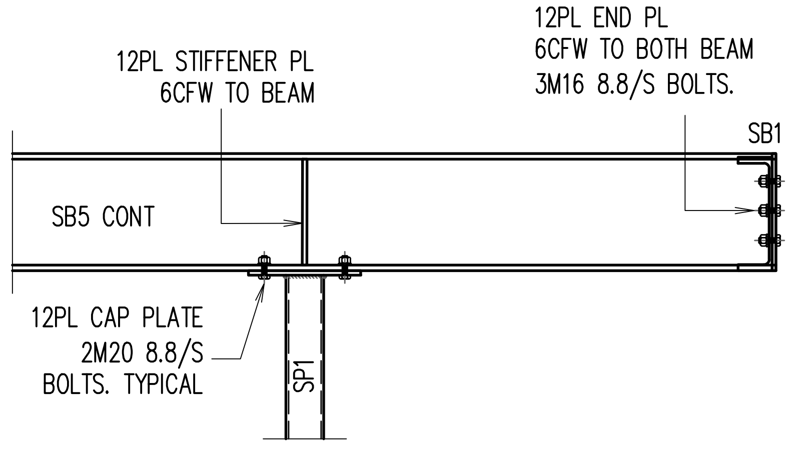
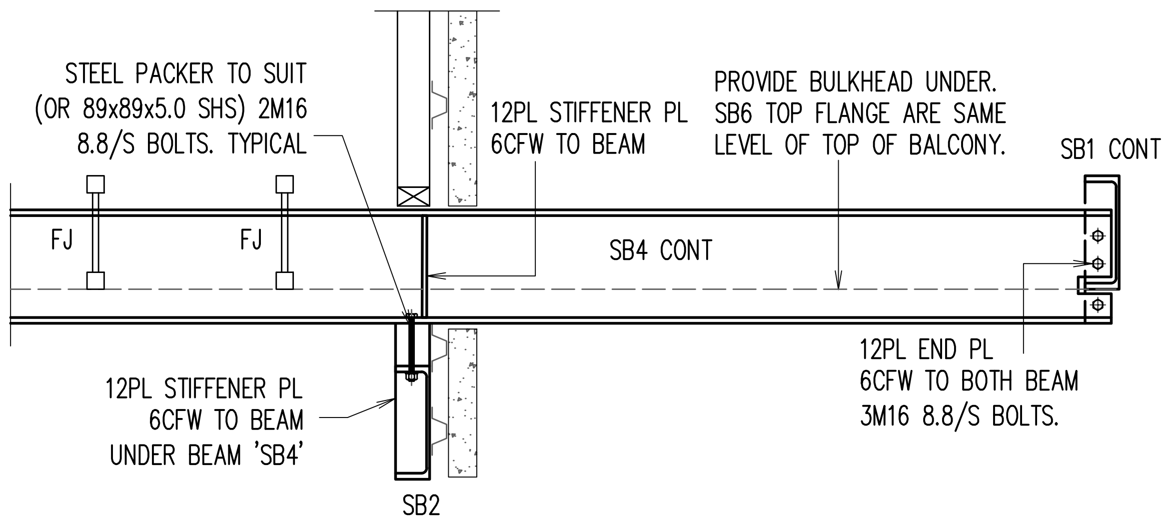


DETAIL – O

SCALE 1:20



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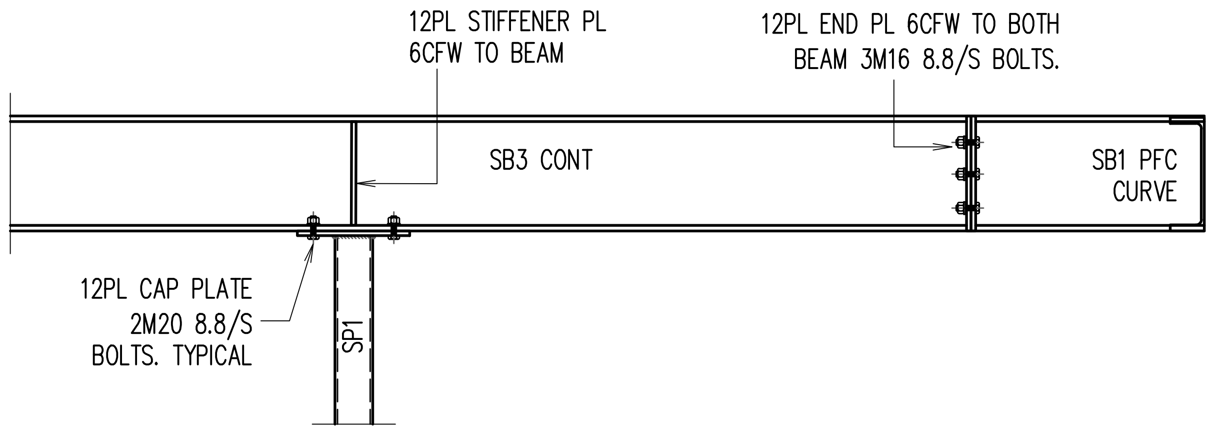
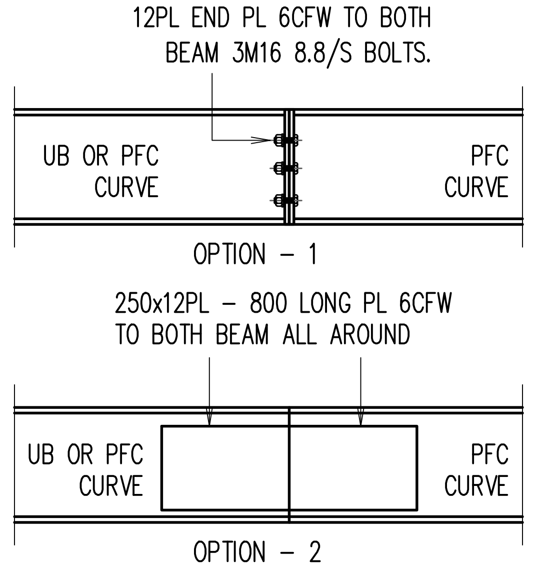
SCALE 1:20



DETAIL – Q

SCALE 1:20



TYPICAL CURVE STEEL BEAM DETAIL

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DESIGN BY:	DRAWN BY:	CHECK BY:	PAGE:
R.H.	C.V.	R.H.	13 OF 16

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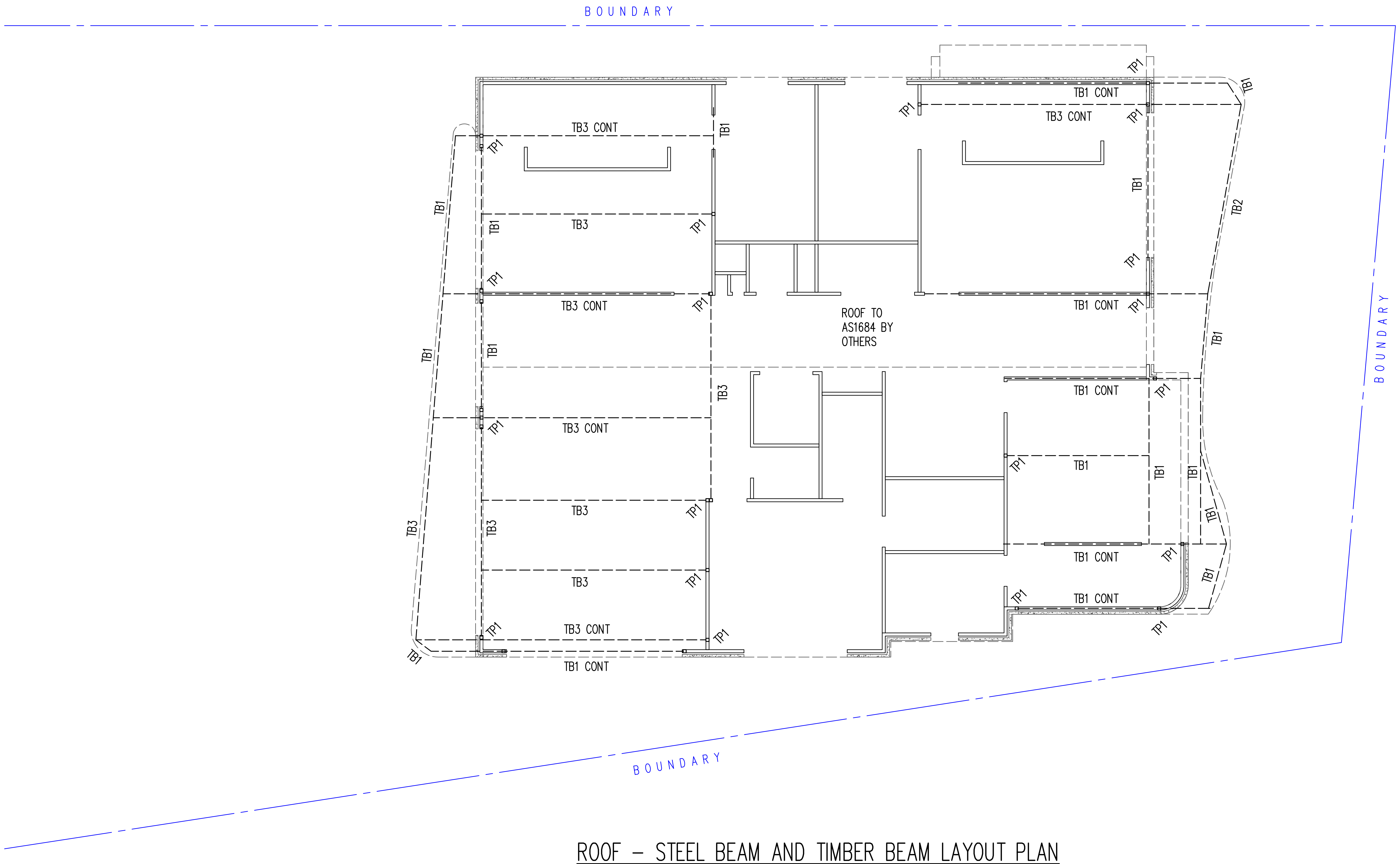

MEMBER SCHEDULE

MARK	SIZE	COMMENTS
TB1,HB1	300 x 63 HYSPAN	TIMBER BEAM
TB2	360 x 63 HYSPAN	TIMBER BEAM
TB3	400 x 63 HYSPAN	TIMBER BEAM
TB,HB	STRUCTURAL TIMBER BEAM	TO MANUFACTURES DETAILS
TP1	STRUCTURAL TIMBER POST	TO MANUFACTURES DETAILS

NOTE:

1. IF IN DOUBT FOR LINTEL SIZES OVER REMAINING WINDOW/DOOR OPENINGS CONTACT STRUCTURAL ENGINEER.
2. FIRST FLOOR PARAPET WALLS AND SUPPORT DETAILS BY OTHERS.
3. FLOOR JOIST ARE NOT DESIGN TO TAKE ROOF LOAD, FOR CONVENTIONAL ROOF, CARPENTER NEED TO PROVIDE TIMBER BEAM AS PER AS1684.
4. TIMBER TRUSS ROOF AND WALLS BRACING AND TIE DOWN DETAILS BY OTHERS & COMPLY WITH RELEVANT SECTIONS OF AS1684.

LEGEND

- DENOTES BRICK WALL UNDER
- DENOTES HEBEL WALL OVER
- DENOTES TIMBER WALL UNDER
- ⟷ DENOTES ROOF RAFTER DIRECTION

1. INSTALL HEBEL (OR SIMILAR PRODUCT) & STUD WALLS AS PER MANUFACTURERS DETAILS AND SPECIFICATIONS.
2. PROVIDE ARTICULATION JOINTS – LOCATION & DETAIL AS PER HEBEL MANUFACTURE DETAILS.

ROOF – STEEL BEAM AND TIMBER BEAM LAYOUT PLAN

SCALE 1:100

1. DRAWINGS TO BE READ IN CONJUNCTION WITH ARCHITECTURAL.
2. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETOUT, LEVELS, FALLS ETC.
3. TB1 OR HEADER BEAM TO BE PROVIDED ON OVER ALL OPENINGS
4. PROVIDE DOUBLE STUD UNDER ALL TIMBER AND HEADER BEAM.

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A2 0 1 2 3 4 5 6 7 8 9 10

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PROPOSED RESIDENCE
LOT 515 (NO. 18) THORPE WAY
BOX HILL, NSW
FOR –

BUILDER
DEZIRE HOMES

ROOF BEAM PLAN			
SCALE: 1:100, 1:20 U.N.O.			
DESIGN BY: R.H.	DRAWN BY: C.V.	CHECK BY: R.H.	PAGE: 14 OF 16
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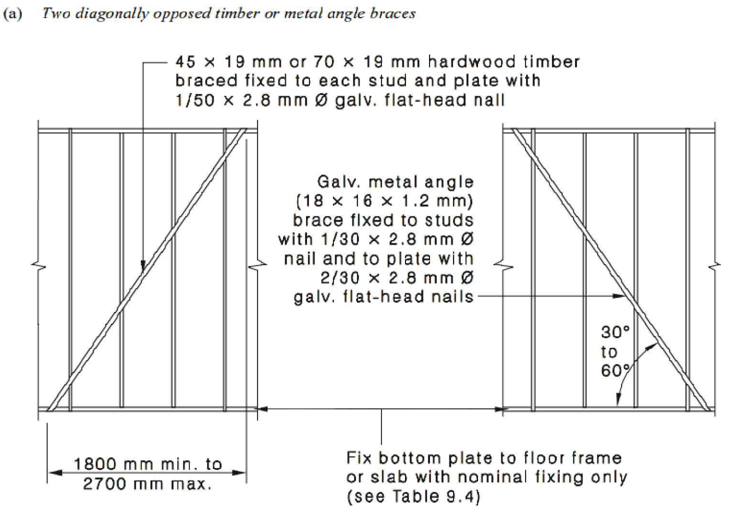
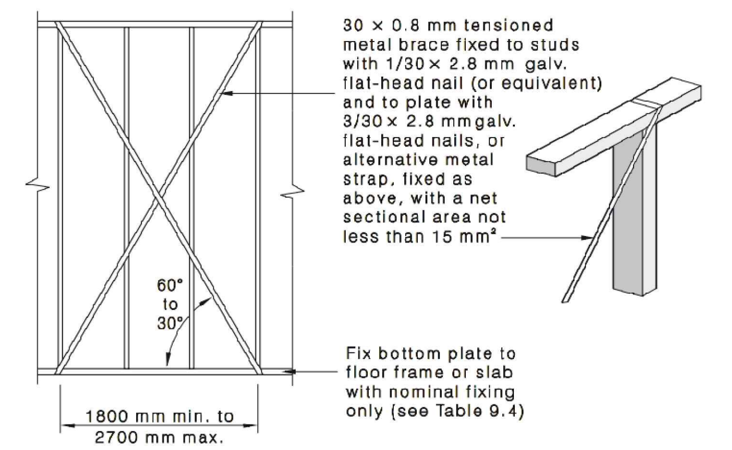
Type of bracing	Bracing capacity kN/m
(a) *Two diagonally opposed timber or metal angle braces*	0.8
(b) *Metal straps—Tensioned*	1.5
(c) *Timber and metal angle braces* The maximum depth of a notch or saw-cut shall not exceed 20 mm. Saw-cuts studs shall be designed as notched.	1.5
(d) *Metal straps—Tensioned—With stud straps*	3.0
(g) *Plywood* Plywood shall be nailed to frame using 30 mm × 2.8 mm Ø galvanized flat-head nails or equivalent.	3.4

(a) *Two diagonally opposed timber or metal angle braces*

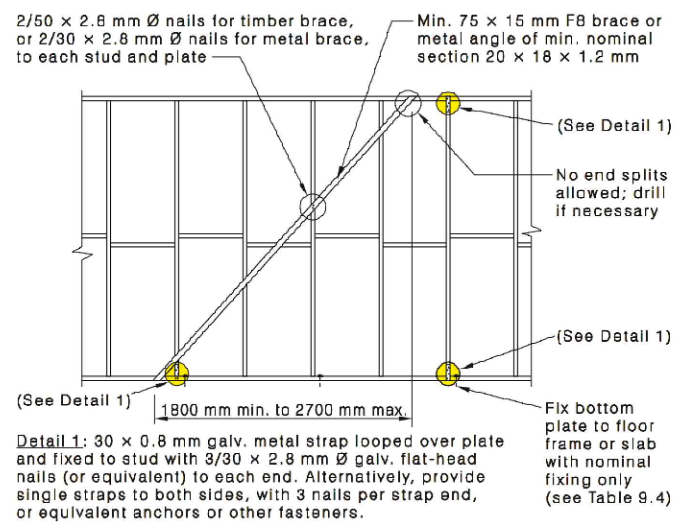


NOTE: All flat-head nails shall be galvanized or equivalent.

(b) *Metal straps—Tensioned*

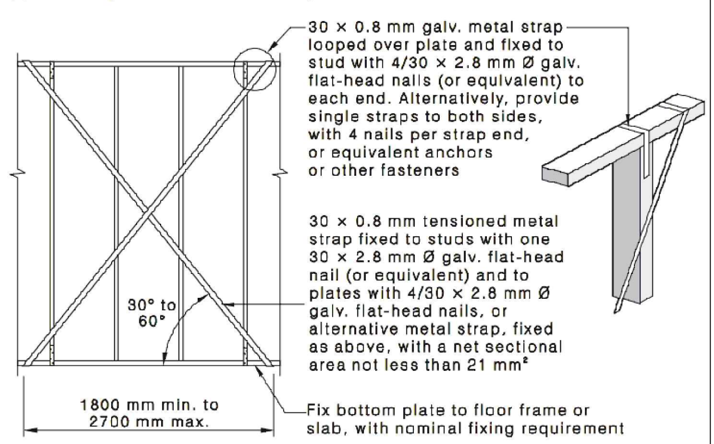


(c) *Timber and metal angle braces* The maximum depth of a notch or saw-cut shall not exceed 20 mm. Saw-cuts studs shall be designed as notched.

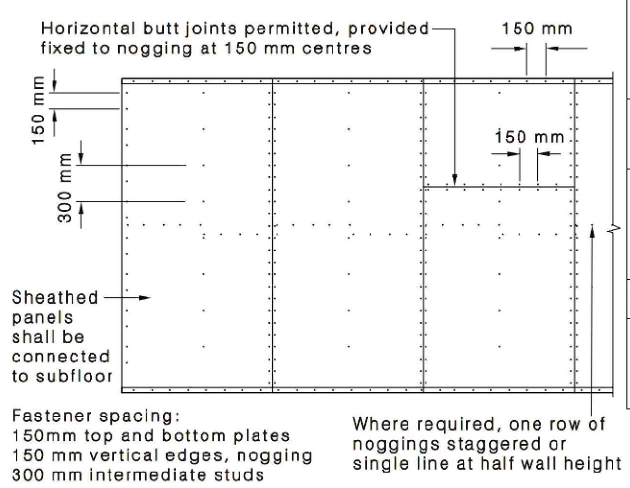


Detail 1: 30 × 0.8 mm galv. metal strap looped over plate and fixed to stud with 3/30 × 2.8 mm Ø galv. flat-head nails (or equivalent) to each end. Alternatively, provide single straps to both sides, with 3 nails per strap end, or equivalent anchors or other fasteners.

(d) *Metal straps—Tensioned—With stud straps*



(g) *Plywood* Plywood shall be nailed to frame using 30 mm × 2.8 mm Ø galvanized flat-head nails or equivalent.



Minimum plywood thickness, mm		
Stress grade	Stud spacing mm	
	450	600
No nogging (except horizontal butt joints)		
F8	7	9
F11	4.5	7
F14	4	6
F27	3	4.5
One row of nogging		
F8	7	7
F11	4.5	4.5
F14	4	4
F27	3	3

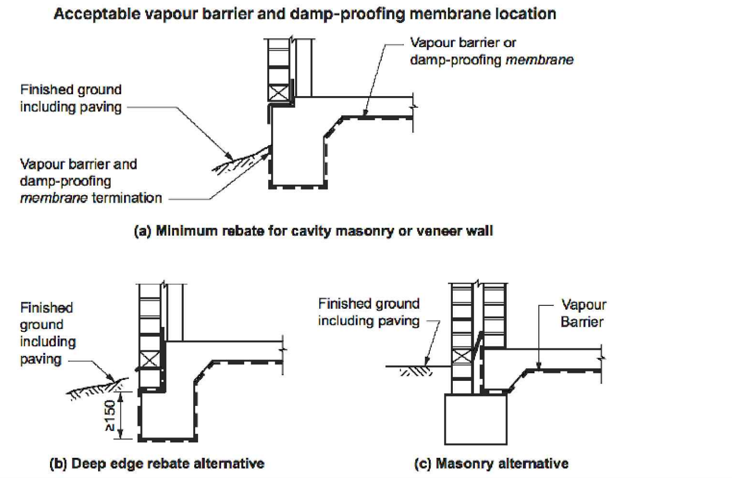
NOTES:
1 For plywood fixed to both sides of the wall, see Clauses 8.3.6.5 and 8.3.6.10.
2 No other rods or straps are required between top or bottom plate.
3 Fix bottom plate to floor frame or slab with nominal fixing only (see Table 9.4).

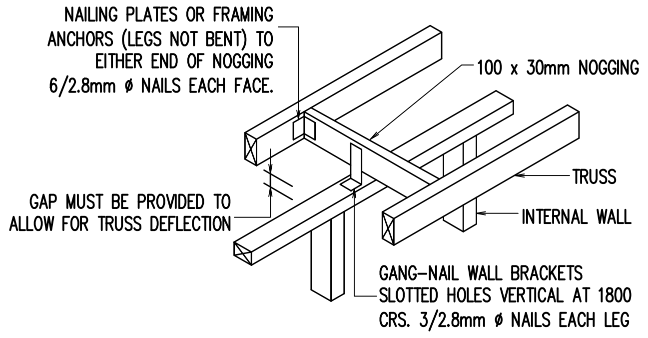
TYPICAL WALL BRACING DETAIL

NOTE: IN ACCORDANCE WITH REQUIREMENTS OF AS1684 NATIONAL FRAMING CODE.

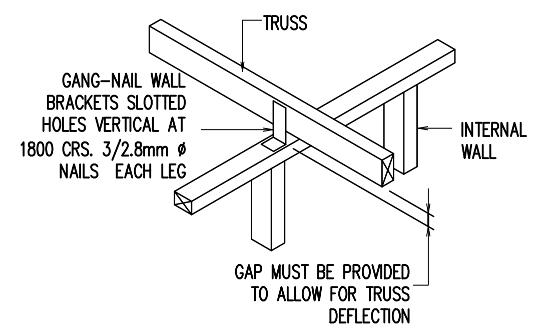
NOTE :
ALL BRACING TO COMPLY WITH RELEVANT SECTIONS OF AS1684 RESIDENTIAL TIMBER FRAMED CONSTRUCTION, FOR FURTHER INFORMATION REFER TO CLAUSE 8.3.6 OF PART 2 AS1684.2-2010

Figure 4.2.8: **Acceptable vapour barrier and damp-proofing membrane location**

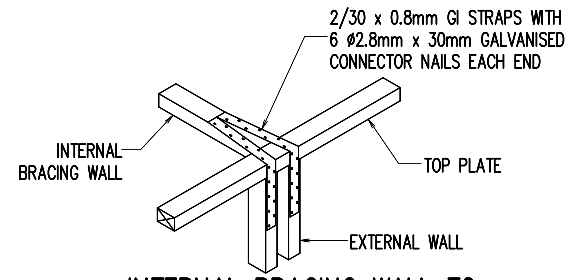




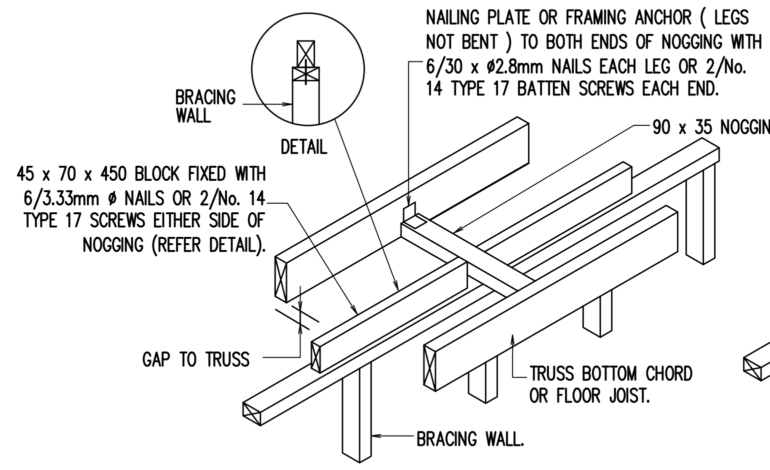
NON LOAD BEARING WALL PARALLEL TO TRUSSES



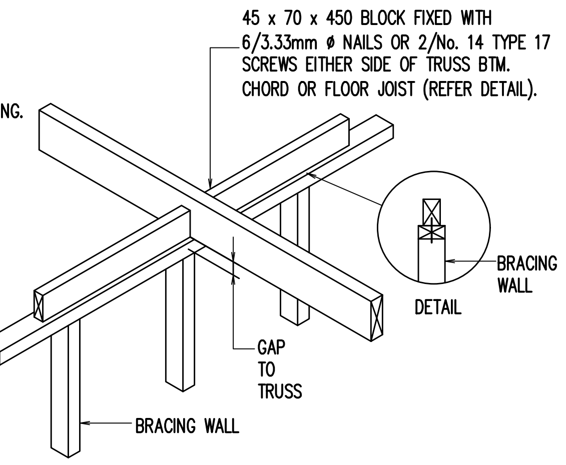
NON LOAD BEARING WALL PERPENDICULAR TO TRUSSES
SCALE NTS



INTERNAL BRACING WALL TO EXTERNAL WALL CONNECTION
SCALE NTS



PARALLEL TO TRUSS



PERPENDICULAR TO TRUSS

TIMBER BRACING WALL TOP CONNECTION DETAIL

A2 0 1 2 3 4 5 6 7 8 9 10

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TYPICAL BRACING DETAILS

SCALE: 1:100, 1:20 U.N.O.			
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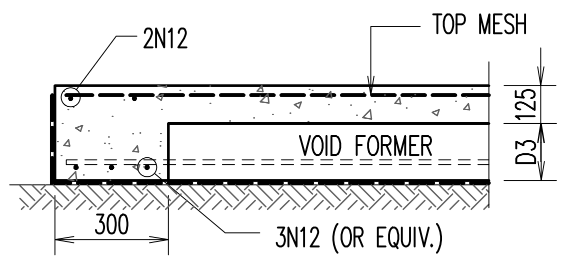
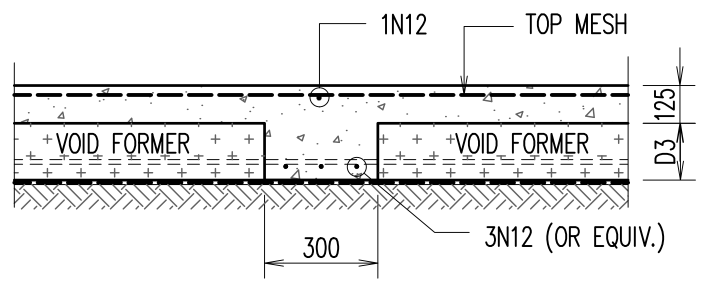
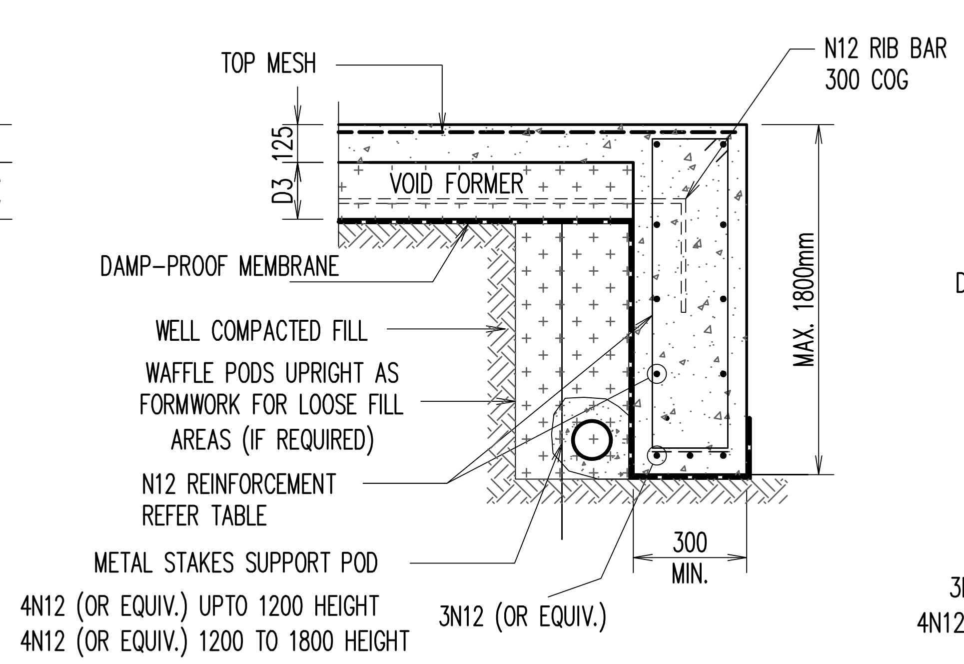


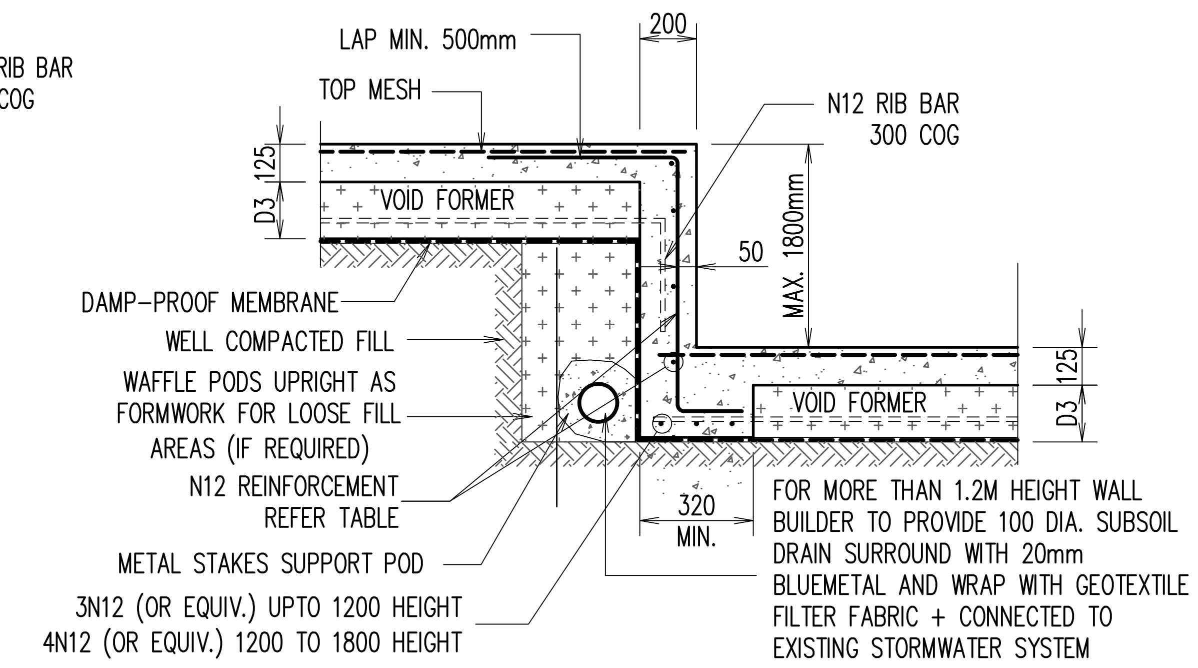
TYPICAL EDGE BEAM – 'EB10'



INTERNAL BEAM – 'IB10'



'EXTERNAL' DROP BEAM – 'DB10'



'INTERNAL' DROP BEAM – 'DB11'

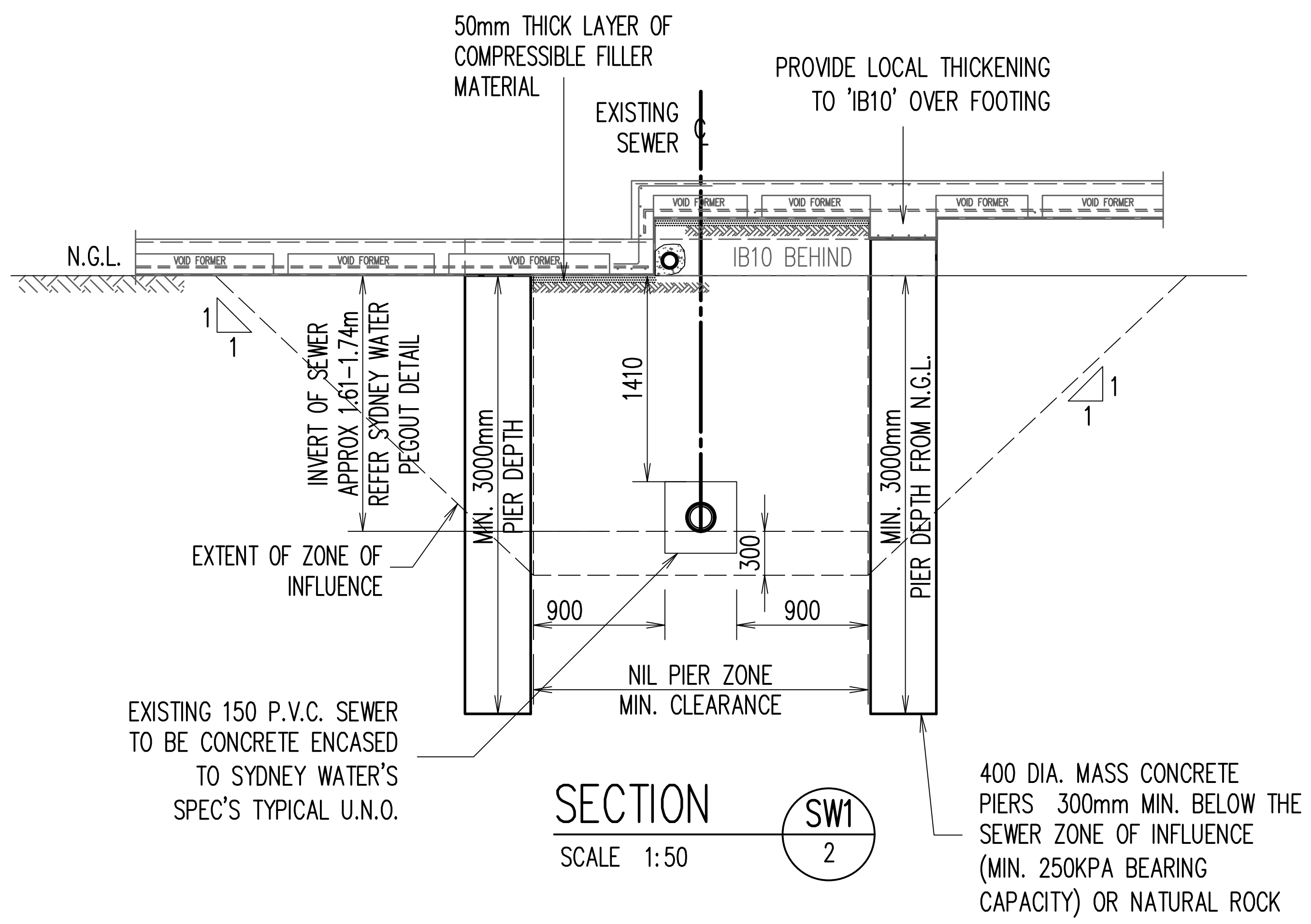
JOB SPECIFICATION	
POD DEPTH (D3)	150mm
SLAB THICKNESS	125mm
TOP MESH	SL92
BEAM REINFORCEMENT: 3N12 (ALTERNATIVELY 3–L11TM)	

REINFORCEMENT FOR EXTERNAL BEAMS
WHERE WIDTH EXCEEDS 300mm

WIDTH	TOP STEEL	BTM STEEL EXTRA
301–330	1N12	1N12
331–440	1N12	2N12
441–550	2N12	3N12
551–660	3N12	4N12

DROP EDGE BEAM REINFORCEMENT TABLE

DROP EDGE BEAM DEPTH	HORIZONTAL REO	VERTICAL REO
UP TO 400mm	1N12	N12–900 CTS
410 – 900mm	N12–400 CTS	N12–400 CTS
910 – 1500mm	N12–300 CTS	N12–300 CTS
1510 – 1800	N12–200 CTS	N12–200 CTS



SECTION SW1/2
SCALE 1:50

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SLAB & SEWER DETAILS			
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